OECD ECONOMIC SURVEYS

1999

SPECIAL FEATURES

Structural policies

Enhancing environmentally sustainable growth

FINLAND

© OECD, 1999.
© Software: 1987-1996, Acrobat is a trademark of ADOBE.
All rights reserved. OECD grants you the right to use one copy of this Program for your personal use only. Unauthorised reproduction, lending, hiring, transmission or distribution of any data or software is prohibited. You must treat the Program and associated materials and any elements thereof like any other copyrighted material.

All requests should be made to:

Head of Publications Service,
OECD Publications Service,
2, rue André-Pascal, 75775 Paris
Cedex 16, France.

OECD ECONOMIC SURVEYS

1998-1999

FINLAND

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

Pursuant to Article 1 of the Convention signed in Paris on 14th December 1960, and which came into force on 30th September 1961, the Organisation for Economic Co-operation and Development (OECD) shall promote policies designed:

- to achieve the highest sustainable economic growth and employment and a rising standard of living in Member countries, while maintaining financial stability, and thus to contribute to the development of the world economy;
- to contribute to sound economic expansion in Member as well as non-member countries in the process of economic development; and
- to contribute to the expansion of world trade on a multilateral, non-discriminatory basis in accordance with international obligations.

The original Member countries of the OECD are Austria, Belgium, Canada, Denmark, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, the United Kingdom and the United States. The following countries became Members subsequently through accession at the dates indicated hereafter: Japan (28th April 1964), Finland (28th January 1969), Australia (7th June 1971), New Zealand (29th May 1973), Mexico (18th May 1994), the Czech Republic (21st December 1995), Hungary (7th May 1996), Poland (22nd November 1996) and Korea (12th December 1996). The Commission of the European Communities takes part in the work of the OECD (Article 13 of the OECD Convention).

Publié également en français.

© OECD 1999
Permission to reproduce a portion of this work for non-commercial purposes or classroom use should be obtained through the Centre français d'exploitation du droit de copie (CFC), 20, rue des Grands-Augustins, 75006 Paris, France, Tel. (33-1) 44 07 47 70, Fax (33-1) 46 34 67 19, for every country except the United States. In the United States permission should be obtained through the Copyright Clearance Center, Customer Service, (508)750-8400, 222 Rosewood Drive, Danvers, MA 01923 USA, or CCC Online: http://www.copyright.com/. All other applications for permission to reproduce or translate all or part of this book should be made to OECD Publications, 2, rue André-Pascal, 75775 Paris Cedex 16, France.

Table of contents

Assessment and recommendations 7

I. Macroeconomic conditions 17

Admirable performance, but unemployment is still very high 17
Macroeconomic policy: a new era begins with the advent of the euro 32
Short-term outlook 37

II. Improving performance by stepping up structural reform 41

How to improve labour market performance 42
Product markets: competition increased substantially 51
Financial markets: banks complete payback of emergency loans 60
Medium-term public sector issues 64

III. Enhancing environmentally sustainable growth 73

The major issues: an overview 73
Institutional framework 75
Forestry policy aims at sustainable management of forests 77
Climate change policy 82
Transborder and local pollution problems 97
Assessment and scope for further action 107

Notes 111
Glossary of acronyms and terms 120
Bibliography 122

Annexes

I. Labour market reform measures in the 1990s 126
II. Calendar of main economic events 130

• • • • •

Boxes

1. The programme of the new government 44
2. High effective tax rates affect many Finns 49
3. The Stability and Growth Pact 67
4. The forest sector 78
5. Energy policy 84
6. Competitiveness: a valid reason for suboptimal environmental taxation? 89
7. Transport is emission-intensive 94
8. Transborder pollution problems and co-operation with neighbouring countries 98
9. Finnish agriculture and the environment 105

OECD 1999

Tables

1. Demand and output 19
2. Exports by industry 21
3. Household appropriation account 23
4. Labour market developments 26
5. Wages, costs and prices 29
6. Budgetary plans and outcomes 34
7. Short-term projections 39
8. Labour market indicators 46
9. Threshold wages 47
10. Domestic and foreign-owned companies 58
11. Public finances 66
12. Government employment 66
13. Recommendations for further structural reform 69
14. The forest sector in the economy 79
15. Forest land ownership 79
16. Spending on nature conservation 81
17. Energy taxation 88
18. Cost recovery in road and rail transport 95
19. Finnish policy targets for transborder environmental problems 99

Figures

1. Unemployment in Finland and the euro area 18
2. Monthly output developments 19
3. Industrial production 20
4. Cost competitiveness and export performance 22
5. Household spending 24
6. Business investment 24
7. Labour market 27
8. GDP and expenditures 31
9. Interest and exchange rate developments 36
10. Leading indicators 38
11. Structure of the labour force 42
12. GDP per capita 43
13. Tax rates in Finland and other OECD countries 48
14. Effective marginal tax rates for salary earners and self employed 50
15. Disposable income and income-dependent social benefits of households 50
16. Privatisation proceeds 52
17. State aid 55
18. Foreign investment in Finland 57
19. Development of the banking sector 62
20. Bank profitability and credit behaviour 63
21. Fiscal consolidation 1993-98 65
22. Use of forest resources 80
23. CO_2 emission intensities 83
24. Projections of greenhouse gas emissions 86
25. Greenhouse gas emissions by gas and sector 87
26. Prices and taxes for fuel oil and gasoline 92
27. Petrol taxes in international comparison 93
28. Emissions of acidifying gases 101
29. Sources of nutrient loading 103

BASIC STATISTICS OF FINLAND (1998)

THE LAND

Area (1 000 km^2, 1995):	338.1
of which:	
Cultivated land	27.5
Forests	230.0
Lakes	33.6

Major cities (thousand inhabitants, 31.12.97):	
Helsinki	539.4
Espoo	200.8
Tampere	188.7
Vantaa	171.3

THE PEOPLE

Population (thousands, 31.12.98)	5 159
Number of inhabitants per km^2	15.3
Net natural increase (thousands)	7.8
Net migration (thousands)	3.4

Labour force (thousands)	2 508
Employment (thousands)	2 222
Employment (% of total):	
Agriculture, forestry and fishing	6.5
Industry and construction	21.4
Services	72.2

PARLIAMENT AND GOVERNMENT (1999)

Composition of Parliament (number of seats):	
Social Democratic Party	51
Centre Party	48
National Coalition (Conservatives)	46
Left Alliance	20
Green League	11
Swedish People's Party	11
Christian League	10
Other	3
Total	200

Government, number of ministers from:	
Social Democratic Party	6
National Coalition (Conservatives)	6
Swedish People's Party	2
Left Alliance	2
Green League	1
Independent	1
Total	18

Last general elections: 21st March 1999

PRODUCTION AND PUBLIC SECTOR

Gross domestic product (billion FIM)	676.1
Gross domestic product per head (US$)	24 547
Gross fixed capital investment:	
% of GDP	19.1
Per head (US$)	4 693

Public consumption (% of GDP)	21.5
General government (% of GDP):	
Current and capital expenditure	49.1
Current revenue	50.6

FOREIGN TRADE

Exports of goods and services (% of GDP)	39.5
Main exports (% of total):	
Wood, pulp and paper	29.9
Metals, machinery and transport equipment	27.0
Electrical and optical equipment	25.6
Other goods	17.6

Imports of goods and services (% of GDP)	30.8
Main imports (% of total):	
Intermediate goods	41.5
Capital goods	27.2
Consumer goods	23.5
Energy	7.8

THE CURRENCY

Monetary unit: Markka
Fixed conversion rate to euro:

€ 1 = FIM 5.94573

Currency units per US$, average of daily figures:	
Year 1998	5.35
May 1999	5.59

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Finland by the Economic and Development Review Committee on 10 June 1999.

•

After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 24 June 1999.

•

The previous Survey of Finland was issued in August 1998.

Assessment and recommendations

Despite an impressive economic performance since 1994, unemployment is still very high

Since the deep recession of the early 1990s, Finland's economic performance has been impressive in most respects. Annual gross domestic product (GDP) growth of more than 4 per cent between 1993 and 1998 was among the strongest in the OECD area, while the drop in the unemployment rate – from a peak of 18 per cent in February 1994 to below 11 per cent in early 1999 – was the second fastest sustained drop ever registered in the OECD area. These remarkable achievements were partly the result of skilful macroeconomic policy management, underpinned by a favourable international economic climate. Tight budgetary policy led to a shift from a sizeable deficit of 7 per cent of GDP in 1993 to a surplus of 1½ per cent of GDP in 1998. This helped restore financial market confidence. Moderate wage agreements created favourable conditions for the pursuit of a relaxed monetary policy stance that stimulated investment and employment growth. In addition, the economy underwent rapid restructuring which covered most industrial sectors and was especially remarkable in the electronic equipment industry. However, while the Finnish unemployment rate has fallen to the euro area average, it is still higher than before the recession started, and double the rate in countries with a good labour market performance. Moreover, the tax burden has remained among the highest in the OECD, implying large distortions and negative incentive effects. Labour market and tax reforms appear to be crucial to create the conditions for a continuation of the strong growth performance and a further significant decline in the unemployment rate.

Growth is projected to slow but the economy should continue to do well

GDP growth edged down to 4¾ per cent in 1998, from 5½ per cent in 1997, due to weaker external demand following the financial crisis in emerging markets, but the slowdown was cushioned by a significant pick-up in private consumption. The Finnish economy is projected to continue to do well in 1999, although with a less favourable international situation, output growth is likely to decelerate to slightly above 3 per cent, and domestic demand will drive the economy even more than in 1998. The projected international pick-up is likely to lead to somewhat higher economic growth in 2000.

Monetary conditions may be too easy in 2000 and overheating is a major risk...

Inflation has been low in recent years. However, there was increasing concern during the first half of 1998 that a combination of overheating of the Finnish economy and easy monetary conditions in the euro area could develop. With the Finnish economy slowing down and its cyclical situation becoming somewhat more in tune with the rest of the euro area, this fear was dissipating. Changes in monetary policy influence economic activity with a considerable lag. Given the significant euro area monetary policy easing since late 1998, the projected pick-up in activity and the further labour market tightening, monetary conditions could be too easy in 2000 for Finland.

... especially in the labour market

In the context of monetary union and strong integration with other euro area countries, inflationary pressures concerning traded goods are likely to remain low. However, inflation could be strong in asset markets and the sheltered sector. More importantly, the further drop in unemployment over the next two years risks ending the prolonged period of moderate wage agreements. The strong profitability of the Finnish enterprises increases this risk. The labour market could become fairly tight in a number of rapidly growing regions. Wage inflation already started to edge up somewhat in 1998, with the number of vacancies rising sharply. Therefore, bold labour and product market reforms are needed to raise potential and prolong the period of strong non-inflationary growth. Concerning labour market policies, measures to raise labour supply and to reduce professional and regional mismatch should be pursued vigorously. Raising the effective retirement age by reducing the access to

early retirement would be especially helpful in this context. The negotiations of the 2000-01 collective wage agreement will start soon. The government intends to reward a low collective wage agreement with tax cuts. However, it should be borne in mind that in the current circumstance of a rapidly tightening labour market, a low collective wage agreement may still imply fairly high wage increases due to increasing wage drift. With the euro in place, excessive wage increases could lead to heavy employment losses, as the option to devalue is no longer available. In the current context, fiscal policy should stand ready to tighten significantly if signs of overheating re-emerge and, especially, if comprehensive labour market reform measures are not undertaken.

The government budget moved into surplus...

In 1998, the general government balance moved into a surplus of 1½ per cent of GDP – the first surplus since the beginning of the 1990-93 recession – and a turnaround of 8½ percentage points from the record deficit of 1993. Since then, the budget consolidation progressed rapidly, restoring credibility in financial markets, and inflation and real interest rates fell significantly. Fiscal policy is somewhat restrictive in 1999, though substantially less so than in 1998 and the previous years of budgetary austerity. The medium-term surplus target of 2¼ per cent of GDP in 2002 is more ambitious than the Stability and Growth Pact objective of "close to balance" over the medium-term, partly reflecting the greater sensitivity of the Finnish budget to economic fluctuations. The target is probably easily sufficient to prevent pro-cyclical budgetary measures during "normal" recessions without risking a breach of the 3 per cent deficit ceiling. The government aims at a structural surplus in the central government budget. This implies an increase in the actual surplus in the context of continuing favourable cyclical conditions. In phasing in the tax cuts, the cyclical situation should be taken into account.

... but taxation has remained high

Over the medium-term, strong budgetary measures are required because of the adverse expenditure and revenue consequences of ageing. Such measures need to focus on expenditure because the tax burden has remained high by international comparison and should be reduced. As noted

above, partly as an incentive to reach moderate wage agreements, the new government intends to cut income taxes by FIM 10 billion (1½ per cent of GDP) between now and 2003, financed by a freeze of central government expenditures in real terms. This is a step in the right direction and could be taken further: increased efficiency by local government, for instance by outsourcing, and a greater use of charges and reductions in national agricultural subsidies could make room for somewhat deeper tax cuts. To reduce the tax burden significantly, however, would require cuts in other outlays, especially social transfers.

Bolder labour market reforms are necessary to improve labour market performance

Despite strong employment growth in recent years, unemployment is still very high. This partly reflects the late start of labour market reforms, the time it takes before results emerge and the extraordinary character of the 1990-93 slump, by far the deepest post-war recession in the OECD area. Previous Finnish governments already undertook reforms to improve incentive structures. However, measures taken probably did not go far enough and should be followed-up by further and bolder initiatives. Experience of some other OECD countries show that comprehensive reforms eventually lead to a significant decline in structural unemployment. Though the specific content of the reform strategy has differed across these countries, the policies have been broadly in line with the *OECD Jobs Strategy* recommendations to increase societies' adaptability to change. The new government has identified the need to proceed with labour and product market reforms to keep output and economic growth high and to reach a satisfactory labour market situation. It has, however, not yet announced many concrete measures. The following paragraphs suggest where measures would be most appropriate.

Further income tax cuts are required, especially for low-income earners

Despite some decline in income taxation in recent years, the marginal effective tax rate still exceeds 50 per cent for one-third of the employed. High income taxes reduce incentives to raise work effort with the participation rate being the lowest among the Nordic countries. The new government's intention to cut income taxes further is a step in the right direction. Currently, 25 per cent of low-paid employees face a marginal effective tax rate of 50 per cent

or more. The increase in the earned-income deduction in recent years has reduced the unemployment trap, but the adverse effects of such changes on the marginal tax rate for low-income earners should be taken into account in designing new tax cuts, as is envisaged in the new government's programme. Adjustments in social transfers would also appear necessary to sharpen incentives. For instance, phasing down unemployment benefits over time would strengthen job search intensity. Currently, benefits are constant during the first 500 days of unemployment. Before 1989, benefits were lowered by 12.5 per cent after 200 days which provided a better incentive to take up work quickly.

Measures should be taken to raise the effective retirement age

Efforts should be made to reverse the fall in the effective retirement age. This is a matter of urgency because the first cohort of the "baby boom" generation will reach the age of 55 years in 2000. Several options exist to reach the new government's goal to raise the effective retirement age by 2 to 3 years. There is scope to further tighten the link between individual pension contributions and benefit entitlements with a view to reduce distortions in the retirement decision. Impediments to hire older workers, such as the high disability and unemployment pension costs borne by the final employer, should be eradicated and initiatives should be taken to increase job and training opportunities for older workers. The unemployed above the age of 55 years are effectively exempt from actively searching for a job. This could be lifted and an increase of the minimum age for unemployment pensions, currently 60 years, could be considered.

Eligibility criteria for unemployment benefits should be strictly enforced

Eligibility criteria for unemployment benefits should be strictly enforced. In 1998, the length of the period during which benefits are suspended in case an unemployed person stops looking for work or refuses to accept a suitable job or training offer was increased. At the same time, suspensions by employment offices have remained rare. Improved job counselling and the shift in active labour market policies from subsidised jobs to training should be pursued. There should be a general obligation for the

unemployed to regularly review job and training prospects in the employment offices.

Collective bargaining should allow greater scope for wages to adjust to market conditions

The two most recent central wage agreements have served the Finnish economy well as moderate wage rises kept cost pressures low and competitiveness strong. However, the income policy also had its drawbacks as it included higher wage increases for the low-skilled in the context of an already compressed wage structure and limited differentiation across industries. Greater scope for relative wages to respond to market conditions would enhance both the employability of low-skilled workers and work incentives for higher-skilled persons.

Privatisation should be continued to enhance efficiency

The accession to the European Union (EU) in 1995 and government policy initiatives have spurred structural change in product and financial markets, contributing to the strong recovery since 1994. Finnish GDP per capita is now above the EU average but there remains ample scope for raising productivity and living standards. The new government announced that it will continue the policy of privatisation. This is welcome as it enhances the efficiency of the Finnish business sector and increases the possibilities for these enterprises to forge alliances or consider mergers. With the euro in place and financial markets integrating rapidly, a greater volume of share offerings could easily be absorbed.

State aid should be evaluated carefully

The rise in total research and development (R&D) spending to among the highest in the OECD, partly due to the increased R&D support, has facilitated the phenomenal growth of the electronic equipment industry. However, the authorities should continue to assess and evaluate state aid carefully. Within the OECD area there are many examples of aid that led to the delay of inevitable restructuring. They should not hesitate to stop projects with poor results and should continue to streamline support programmes. Opportunity costs, for instance the positive effects of lower taxes, should be included in these assessments. Also regional policy initiatives should be assessed carefully.

The competition authorities need sufficient power and resources

The Finnish government started relatively early with regulatory reform and progressed rapidly, especially in liberalising the telecommunication and electricity sectors, with sizeable positive effects for consumers. The authorities should be vigilant that the dominance of the major company in network sectors does not hamper competition and limit the benefits of deregulation. Furthermore, a reform of the competition authorities could strengthen competitive forces in general. Properly staffed and empowered competition authorities are crucial to this end.

Sustainable development has become a policy priority

Sustainable development issues concerning environmental management are discussed in Chapter III of this *Survey*. The issues influenced policy making in the early 1990s, leading to the development of an extensive institutional framework for environmental management and an important extension of legislation. In many fields, progress has been made and pollution has been reduced. Finland is in a very good position to meet its international commitments regarding sulphur dioxide (SO_2) and nitrogen oxide (NO_x) emissions. Some further measures are necessary to reduce nutrient discharges in the Baltic Sea. Meeting the Kyoto targets for greenhouse gases (GHGs) is the most challenging current issue. Further emission reductions could imply a rapid increase in abatement costs. This increases the importance of conducting economic analysis of the various abatement options systematically and of applying cost-effective measures, as set out in the following paragraphs.

A more systematic economic assessment of sustainable development issues would be desirable

Finland has developed an extensive institutional framework for environmental management. The National Commission for Sustainable Development, chaired by the Prime Minister, has played a role as a forum of debate and in increasing the awareness of sustainable development issues. However, an important weakness in the government's approach is the lack of systematic economic assessment of policy measures. Such assessments would be beneficial for the implementation of a set of comprehensive sustainable policies, especially where traditional economic instruments, like taxes or subsidies, are, or could be, involved. Moreover, environmental matters will need to be further

integrated in sectoral policies and in the overall budgetary framework.

Pollution has been reduced but much remains to be done

Emissions from several sources have been reduced substantially, but further reductions in acid depositions in Finnish lakes and in nutrient discharges into the Baltic Sea are required. Given that a substantial part of harmful substances are imported from neighbouring eastern European countries, measures to reduce emissions of acidifying and eutrophying substances abroad would be cheaper than emission reductions in Finland. Measures are also needed to restrain domestic emissions. Regarding NO_x emissions, a charge on emissions from power plants would be a cost-effective instrument, as experience elsewhere shows. Charges reflecting the environmental costs associated with nutrient release should also be applied to industry and municipal waste water discharges.

New policy initiatives are required to meet the Kyoto target

Combating climate change by reducing GHGs as agreed in the Kyoto Protocol is one of the most challenging current issues. New policy initiatives will have to be made for the target to be met. Marginal abatement costs are relatively high in Finland. It should, therefore, consider using the two flexibility mechanisms allowed for in the Kyoto Protocol: making investments elsewhere reducing greenhouse gas emissions and buying emission quotas abroad, as soon as this becomes operational. The authorities could also consider creating a national trading system for emission quotas that would replace, at least partly, energy taxation. It would make it easier to participate in an international (EU or broader) trading system as soon as it becomes operational.

The current energy tax structure increases abatement costs

Finland was the first country to introduce a carbon dioxide (CO_2) tax, and initially with few exemptions. However, after many changes, the current structure of energy taxes increases the overall costs of CO_2 abatement, since it partially excludes the most energy-intensive industries and, therefore, obliges others to pay higher taxes to achieve the emission reductions. Refocusing the CO_2 tax so as to equalise marginal abatement costs would greatly enhance cost-effectiveness of abatement. However, EU regulations limit

possibilities to tax imported electricity. A unilateral introduction of a pure CO_2 tax could imply sizeable stranded costs if other countries were to start introducing CO_2 taxation later on. Harmonisation of energy taxes at the EU level would help in this respect. At the domestic level, eliminating the favourable treatment of diesel compared to gasoline, as well as shifting from car to fuel taxes, could also enhance cost-effectiveness. Finally, efforts to comply with the Kyoto Protocol should not solely focus on CO_2 emissions as abatement costs of other GHGs, especially methane (CH_4), are estimated to be lower.

Environmental concerns should be better integrated in sectoral policies

Initiatives should not be restricted to realising cost-effective environmental policies. It is also very important to improve the consistency of the various sectoral policies with respect to their environmental consequences. In many instances, it is difficult to reconcile the various objectives of agricultural, transport and energy policy with environmental goals. Peat production can be seen as an example where regional policy objectives are achieved at an environmental cost. The view of the Finnish authorities, however, is that utilising peat is beneficial to secure the energy supply as there are no indigenous fossil fuel resources. Agricultural subsidies aim to promote agricultural activity, but have become increasingly directed towards inducing sustainable agricultural production. Nevertheless, agriculture remains a significant source of environmentally harmful emissions. Concerning energy policy, greater use of gas through the interconnection to the European gas grid is under discussion, but it requires various decisions in neighbouring countries as well. The building of a fifth nuclear power plant would help in containing CO_2 emissions, but has the drawback of raising other environmental issues. Finally, transport policy should aim at bringing prices closer to marginal social cost by raising, for instance, the taxation of diesel.

Evaluate carefully the various instruments to safeguard biodiversity

Concerning the safeguarding of biodiversity, the recently introduced certification and labelling system is a tool that could lead to the internalisation of customers' preferences for "sound" ecological products through the market. To start up such a system, some financial support from the government may be appropriate. In any case, establishing a clear

list of priorities and a careful evaluation of the cost-effectiveness of alternative instruments (certification, subsidies, purchases of forests, direct regulations and taxes) should help in devising an adequate policy mix. Such certifications and labelling schemes should also be reviewed in an international context.

Summing up

Finland's recent economic performance has been admirable in most respects and the economy is projected to do well in 1999 and 2000. The monetary conditions may become too easy in 2000, however, and excessive wage rises due to labour market bottlenecks are a major risk, despite the still high unemployment. Sound macroeconomic fundamentals, a strong competitive position and the existence of a pole of excellence in the electronic equipment sector may not be sufficient to achieve a satisfactory labour market performance in the coming years. The new government has identified the need to reduce taxes and to proceed with labour and product market reforms, but the envisaged income tax cuts are fairly timid while most reform measures, such as the reform of the early retirement scheme, still have to be specified. Wide-ranging and timely reform measures would help to alleviate obstacles to the continuation of strong, non-inflationary growth in the medium term and increase the chances of further reducing unemployment significantly. Some labour market reforms, especially of the early retirement scheme, could have an immediate effect on labour supply. In the absence of such reforms, a tight fiscal policy stance will be all the more needed to prevent overheating. Concerning sustainable development, both a more systematic economic assessment of environmental policy measures and a further integration of environmental matters in sectoral policies are needed.

I. Macroeconomic conditions

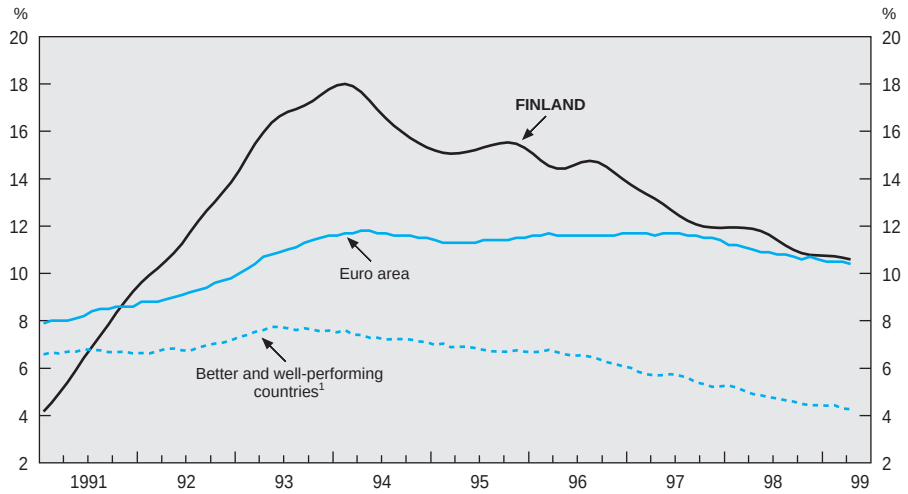
Admirable performance, but unemployment is still very high

The Finnish economy performed well in 1998. Real GDP increased by 4.7 per cent, employment expanded rapidly, unemployment dropped further, inflation was virtually absent, the general government budget was in surplus for the first time since the start of the 1990-93 recession and the government debt continued to fall. These remarkable achievements were mainly the result of years of skilful macroeconomic policy management and moderate wage rises, helped by a favourable international economic climate. Both fiscal and monetary policy contributed to the improvement in macroeconomic fundamentals. Tight budgetary policy led to a shift from a sizeable deficit of 7 per cent of GDP in 1993 to a 1½ per cent surplus in 1998, resulting in a drop of government debt by 8½ percentage points of GDP from its record level in 1993, and helped restore financial market confidence. Meanwhile, moderate wage agreements created favourable conditions for the pursuit of a relaxed monetary policy stance, which stimulated investment and employment growth.

In 1998, Finland outperformed the euro area and the EU as a whole, in virtually all macroeconomic aspects. Economic growth was higher, inflation lower, the government budget in better shape and employment growth was stronger. At the end of the year, even the unemployment rate, which had been much higher in the early 1990s, fell to the euro area average. However, the euro area average does not provide a strong benchmark given its rather modest macroeconomic achievements. Yet, performance has been also strong compared with the best performing countries apart from unemployment. While the Finnish unemployment rate has fallen to the euro area average, it is still twice as high as in European countries with a good labour market performance or in the United States (Figure 1).

The Finnish economy is projected to continue to do well in 1999 although, with a less favourable international situation on a year-to-year basis, output growth will decelerate somewhat further and economic growth will be driven even more by domestic demand than in 1998. The output gap is estimated by the

Figure 1. **Unemployment in Finland and the euro area**



1. Weighted average of the standardised unemployment rates for Austria, Denmark, Ireland and Netherlands.
Source: Eurostat; OECD, *Main Economic Indicators* and OECD Secretariat.

OECD Secretariat to be slightly positive, indicating some excess demand. This also contrasts with the euro area as a whole for which a negative output gap is estimated. With economic growth picking up in 2000, bottlenecks in some industries may re-emerge. Concerning the medium-term, a sound fiscal policy, including a reduction in the high tax burden and further labour market reform will be needed to reduce the risk of overheating and to keep output and employment growth high.

Growth has decelerated, but remains buoyant

Real GDP increased by 4.7 per cent in 1998, down from 5.5 per cent in the previous year (Table 1).[1] The slowdown was more pronounced during the year, with real GDP in December 1998 less than 2 per cent higher than a year earlier, having peaked at 8 per cent in March, but it picked up again in early 1999 (Figure 2).[2] Exports, government consumption, investment and stockbuilding all contributed to the slowdown. Export growth decelerated sharply due to weaker international demand initiated by the financial crisis in Asia and induced a weakening in machinery and equipment investment. The slowdown was cushioned by a sizeable pick-up in private consumption, stimulated by a surge in disposable income. The impact of the Russian crisis on the Finnish economy was rather

Table 1. **Demand and output**

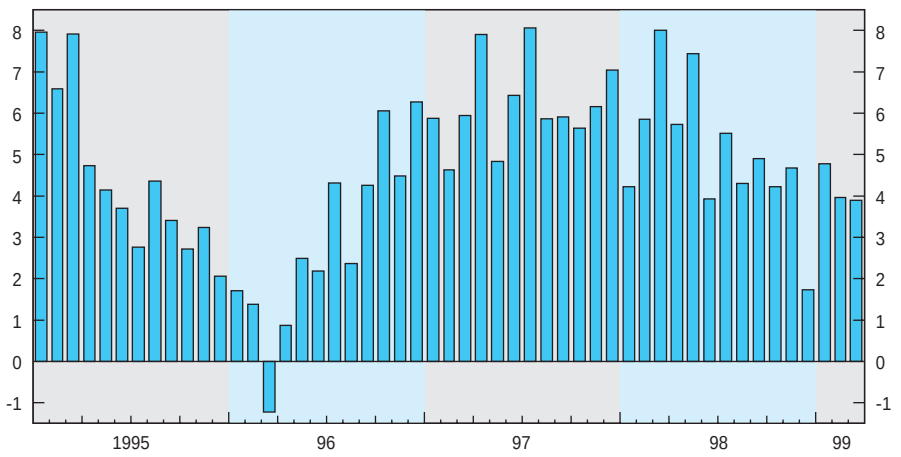
Percentage change, volume

	1995 Current prices, billion FIM	1996	1997	1998	Seasonally adjusted, annual rate		
					1997 2nd half	1998 1st half	1998 2nd half
Private consumption	291.8	3.5	2.6	4.5	3.1	4.3	6.5
of which: Durable goods	25.5	15.5	7.3	14.4	14.1	13.0	17.4
Government consumption	128.2	2.5	2.9	0.4	2.8	–1.6	2.0
Gross fixed capital formation	90.5	8.4	14.4	8.3	21.1	–1.8	18.0
Public sector	15.5	10.6	11.8	–1.6	4.7	–6.2	2.0
Private sector	75.0	8.0	14.9	10.3	24.7	–0.9	21.2
Residential	19.4	2.6	29.7	9.4	19.2	6.5	6.2
Non-residential	55.7	9.8	10.1	10.7	26.9	–3.7	27.4
Final domestic demand	510.4	4.1	4.9	4.3	6.4	1.6	7.7
Stockbuilding [1]	7.2	–1.5	0.8	0.0	0.6	–0.2	–0.1
Exports of goods and services	209.3	6.1	14.2	8.3	21.0	7.0	–0.1
Imports of goods and services	164.6	6.3	11.4	8.9	13.8	10.2	2.0
Foreign balance [1]	44.7	0.4	2.0	0.6	3.9	–0.3	–0.7
Statistical discrepancy [1]	–1.0	1.4	–1.7	0.2	..	..	..
GDP	561.4	4.1	5.5	4.7	7.9	3.2	4.5
of which: Manufacturing	127.6	3.0	9.6	8.5	13.9	6.7	7.1

1. Contribution to GDP growth.
Source: Statistics Finland and OECD Secretariat.

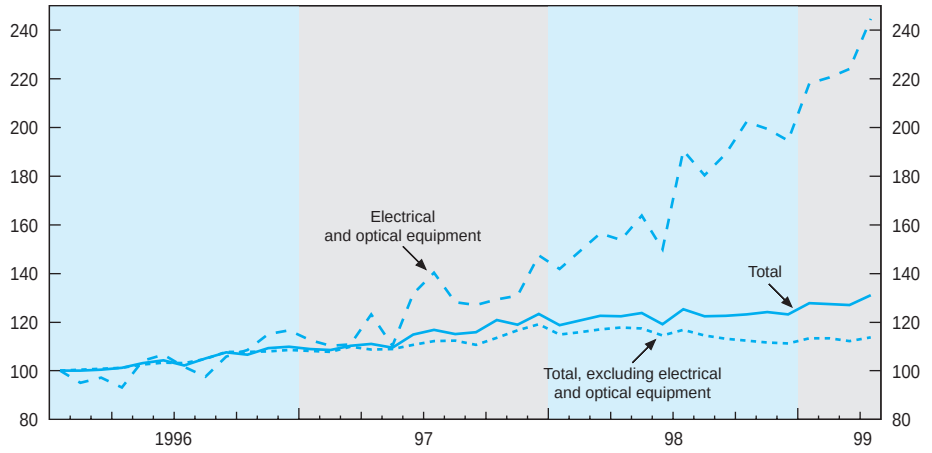
Figure 2. **Monthly output developments**

Per cent change over 12 months



Source: Statistics Finland.

Figure 3. **Industrial production**
Seasonally adjusted, January 1996 = 100



Source: Statistics Finland.

limited, mainly affecting food and transport industries, and not comparable with the substantial impact of the collapse of the Soviet Union at the end of the 1980s. Exports to Russia after the Russian debt moratorium dropped by more than a third compared to the first half of 1998, but year-on-year the drop in 1998 was only 11 per cent, while the share of exports to Russia has fallen to 7 per cent.

Industrial production excluding electronic equipment has been falling since mid-1998 and rose by only 3 per cent in 1998 on a year-to-year basis, compared with 7 per cent in 1997 (Figure 3). In the pulp and paper industry, factories were temporarily shut down in reaction to low world prices. In late 1998, the pulp benchmark price (Northern Softwood Bleached Kraft) hovered around US$460 per ton, compared with more than US$1 000 in November 1995. As a consequence, production of this still crucial industry started to fall in late 1998. On the other hand, the performance of the electronic equipment industry was again phenomenal in 1998 with production rising by 38 per cent due to strong world demand and positive supply factors (*i.e.* successful product innovations). The electronics equipment industry is estimated to have added 1 percentage point to annual real GDP growth during 1994-98 (ETLA, 1998). For 1998, the impact is estimated to exceed 1.5 percentage points, Nokia[3] alone accounting for more than two-thirds of this growth.

Weaker exports

In 1998, the double-digit export volume growth of 1997 was not repeated due to the slow-down in world demand. Exports of all product groups, except electronic equipment, softened. Growth in weighted world demand for Finnish goods exports fell to 6¼ per cent, from 8½ per cent in 1997 and decelerated further to 3½ per cent (saar) in the second half of 1998 and only ½ per cent in the first half of 1999.[4] In value terms, exports of goods dropped strongly in 1998 to Japan (–16 per cent), to the dynamic Asian economies (–14 per cent) and to Russia (–11 per cent).[5] However, the deceleration in exports in the second half of 1998 was mainly caused by weaker euro area imports, especially by Germany and Italy. The export performance remained good due to the favourable competitive position and the positive supply effects in the electronic equipment sector. A market gain of 1.1 per cent was made in 1998 after a substantial market gain of 2.5 per cent in the previous year (Figure 4). The main reasons for the smaller market gain were the significant weakening in the world demand for pulp and paper products and the substantial share of these products in Finnish exports. Pulp and paper exports increased by 3.6 per cent in real terms after rising by 18 per cent in 1997 (Table 2).

On the import side, the double-digit growth in 1997 was not repeated. Lower export growth had a dampening effect on the imports of raw materials and intermediate goods while weaker machinery and equipment investment showed up in the imports of investment goods. However, imports of consumption goods accelerated due to the surge in durable consumer goods' purchases. In value terms, imports from crisis-hit south-east Asia grew strongly in 1998, especially from Hong Kong China (+64 per cent), Korea (+58 per cent) and Thailand (+32 per cent).[6] In contrast to the previous year, import volume growth surpassed that of

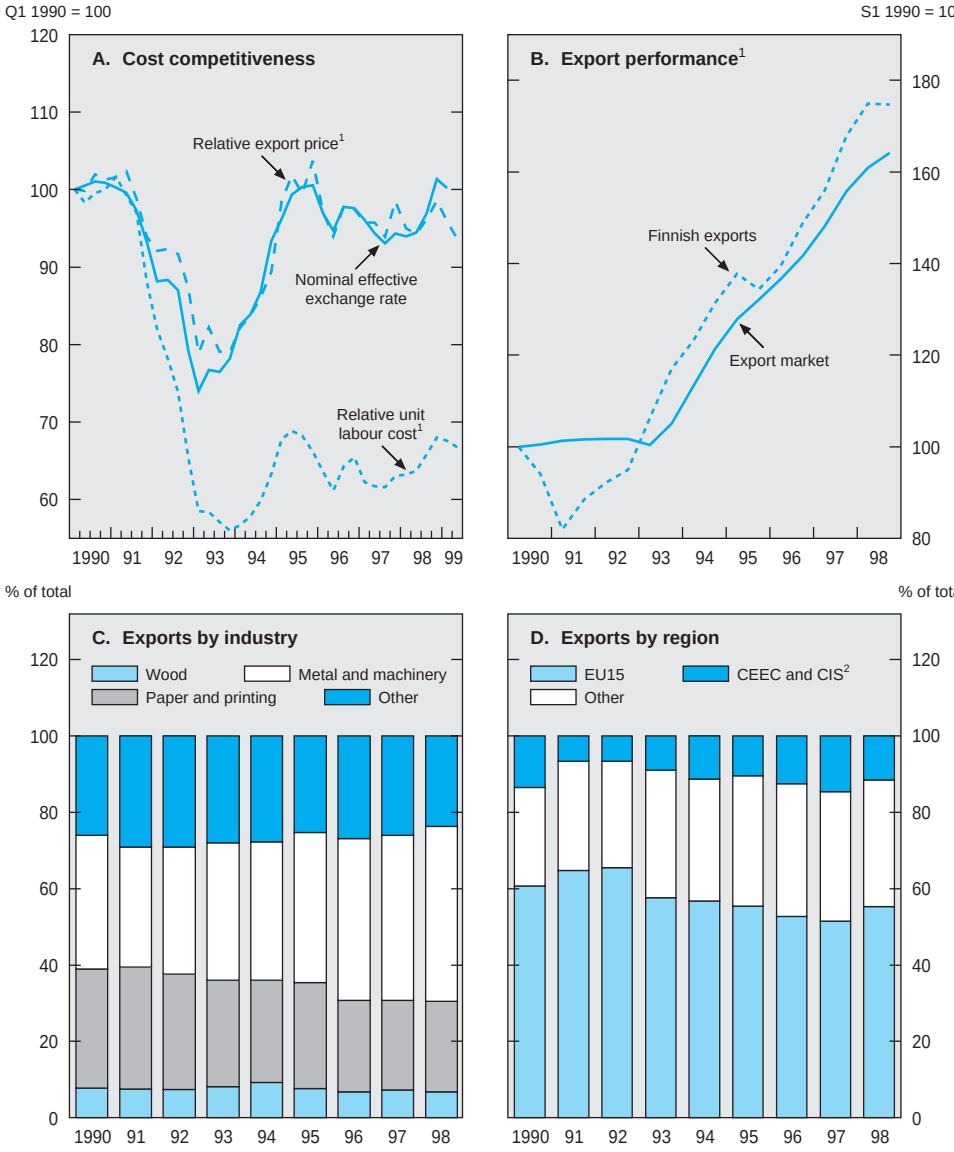
Table 2. **Exports by industry**

Percentage volume changes

	1990 Current prices, billion FIM	1991	1992	1993	1994	1995	1996	1997	1998
Manufacturing	100.0	–8.8	9.6	18.4	13.5	6.5	5.6	13.2	6.1
of which:									
Wood	7.8	–9.4	6.9	38.7	22.1	–2.9	0.0	8.8	6.3
Paper	31.7	–2.2	5.1	9.8	11.5	–2.3	–2.3	18.0	3.6
Metal products and machinery	35.5	–20.1	13.0	30.9	11.5	29.1	5.3	14.1	10.5
Total goods	101.3	–8.8	8.8	18.5	13.6	7.2	5.6	12.7	6.1

Source: Finnish Board of Customs and Statistics Finland.

Figure 4. **Cost competitiveness and export performance**



1. Manufacturing.
2. Central and eastern European countries and Community of Independant States (former USSR up to 1991).

Source: Statistics Finland; OECD, *Foreign Trade Statistics* and OECD Secretariat.

exports but with exports in level terms a fifth bigger than imports, net exports continued to contribute to output growth, albeit by much less than in previous years.

Private consumption surges, but the housing investment boom ends

Private consumption growth almost doubled in 1998, to 4½ per cent in volume terms, boosted by the strong rise in real disposable income (Table 3 and Figure 5). Consumption of durable goods was especially strong, rising by 14 per cent, buoyed by a jump in car sales of 20 per cent.[7] Strong employment growth, solid real wage rises, substantial growth of non-wage income and moderate tax cuts all boosted disposable income. Private consumption was also spurred by positive consumer sentiment, low interest rates and wealth effects from the strong house price rises. The wealth effect of the Finnish stock market surge was probably limited as households keep only a small part of their wealth in equities.[8]

The very strong rise in residential investment came to an end, despite positive consumer sentiment and the sharp rise in disposable income. Mainly due to strong carry-over, residential investment still increased by 9 per cent year-on-year, after increasing by 30 per cent in 1997. This exceptional rise in 1997 was

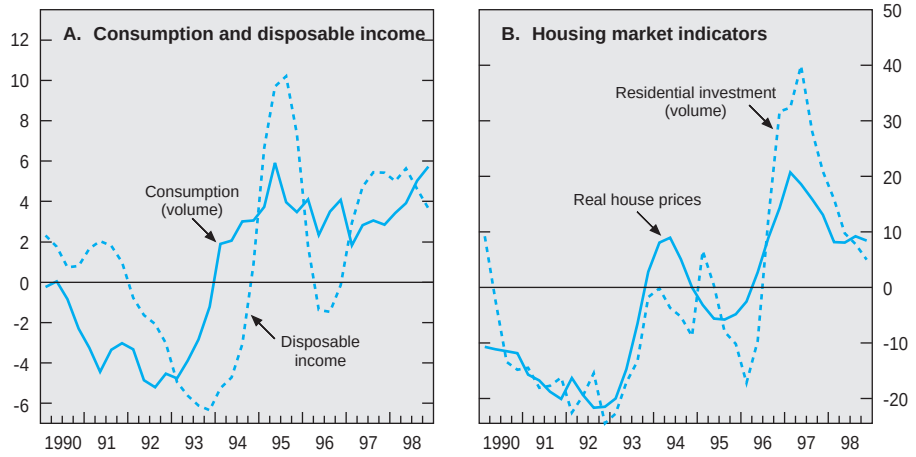
Table 3. **Household appropriation account**

	1992 Current prices, billion FIM	1993	1994	1995	1996	1997	1998
		Percentage changes					
Real disposable income	284.2	–5.8	–3.1	8.5	–0.3	4.6	4.7
Real consumption expenditure	267.3	–3.2	2.5	4.3	3.5	2.6	4.5
Saving ratio (per cent)[1]	10.2	7.5	2.6	6.2	2.8	4.7	4.8
		Percentage points					
Contribution to disposable income							
Compensation of employees	275.1	–5.4	1.7	6.5	4.3	4.7	7.2
Entrepreneurial and property income	78.7	–0.3	–0.7	1.7	–0.6	2.2	1.9
Transfers received	115.6	3.2	2.0	–0.1	0.1	0.0	–0.3
Less:							
Transfers paid	83.5	0.7	3.1	0.3	0.4	0.7	1.5
Direct taxes[2]	72.9	0.7	3.6	–0.6	3.3	0.1	1.7
Interest on consumer debt	28.7	–2.0	–1.8	–0.3	–1.1	–0.6	0.0
Disposable income	284.2	–1.9	–2.0	8.7	1.3	6.7	5.5

1. Ratio of household saving to disposable household income.
2. Direct tax payments in 1994 are heavily affected by a postponement of tax refunds to 1995.
Source: Statistics Finland and OECD Secretariat.

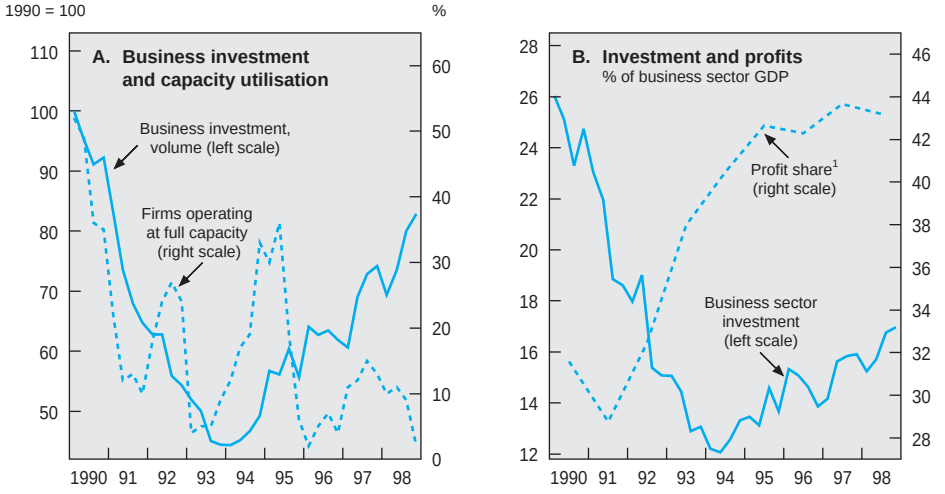
Figure 5. **Household spending**

Percentage change, annual rate



Source: Statistics Finland; OECD, *Quarterly National Accounts* and OECD Secretariat.

Figure 6. **Business investment**



1. Business sector GDP at factor cost less compensation.

Source: Statistics Finland; OECD, *Main Economic Indicators* and OECD Secretariat.

from a very low level and was supported by government subsidies. The slowdown during 1998 largely owes to government measures aimed at cooling down construction activity. Subsidised housing loans as well as subsidised housing construction were reduced. The number of subsidised dwellings built dropped strongly from 20 400 in 1997 to 13 000 in 1998. At the same time, the number of non-subsidised dwellings built increased from 10 500 to 18 000. The calmer situation resulted in smaller house price increases of 10 per cent compared with 18 per cent in 1997.[9] Housing shortages, and thus, overheating risks, still exist in the growth centres, mainly eight cities in southern Finland with strong employment and population growth. Moreover, high house prices relative to the cost of construction have made non-subsidised residential construction attractive.

Solid expansion of business investment

Business investment continued to expand solidly (by 10.7 per cent) despite a sharp deterioration in business sentiment during the year (Figure 6). Machinery and equipment investment growth slowed substantially due to the negative effect of the less favourable international economic conditions on capacity utilisation. Since mid-1998, capacity utilisation is falling and shortage of capacity is no longer a factor constraining production. The weaker machinery and equipment growth was partly offset by stronger investment in non-residential construction and civil engineering, with shopping centre construction being especially brisk. Investment in the business sector is being underpinned by good profitability, with the non-wage share in GDP remaining close to the record level reached in 1997.

Financial markets: vigorous growth in bank lending

Bank lending by households and non-financial corporations increased vigorously in 1998, for the first time in the 1990s. Strong borrowing by corporations was a new feature as investment in previous years was mainly financed by retained profits. Moreover, low interest rates and positive consumer sentiment made households much more willing to borrow for dwelling purchases. Demand for housing loans was also boosted by the abolition of stamp duties on loans in May 1998. It not only led to a spurt in new borrowing but also to substantial refinancing of existing housing loans at lower interest rates. Moreover, it heightened competition between banks as it made it easier for customers to switch from one bank to another and the margin added to prime rates narrowed somewhat in 1998. The strong rise in the purchase of durable consumer goods led to a substantial rise in consumer credit. However, as banks lost market share to non-bank financial companies, banks' consumer credit grew only modestly. Broad money (M3) increased by 2.4 per cent in 1998 – much less than the rise in bank credit due to the strong increase in financial liabilities of banks, especially deposits of

non-Economic and Monetary Union (EMU) residents and to the substantial increase in government holding of money market paper issued by banks.[10]

Strong employment growth

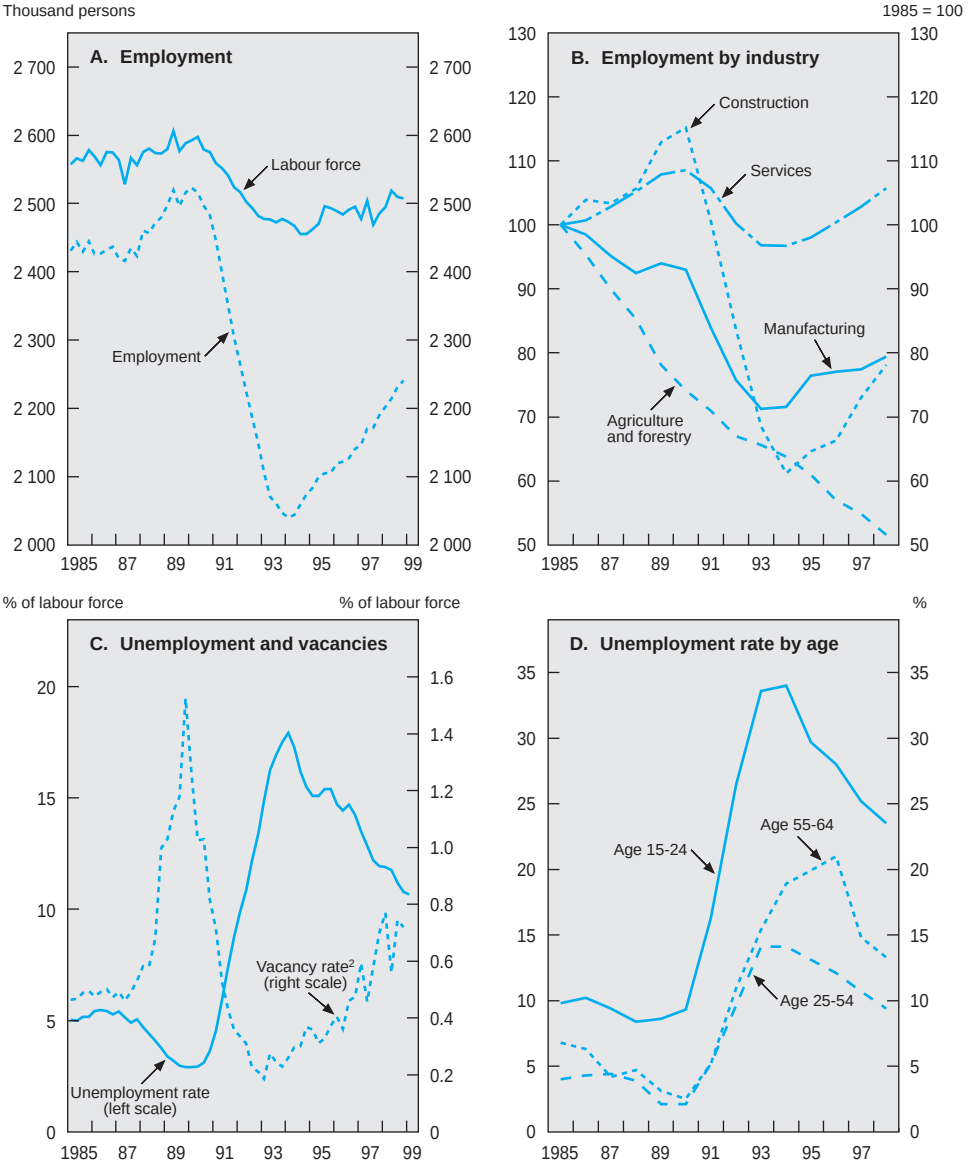
Despite a more subdued pace of output expansion, employment growth went up from 2.0 per cent in 1997 to 2.4 per cent in 1998, reflecting the lagged response of employment to output (Table 4 and Figure 7). In addition, output growth became more labour-intensive, with the number of employees in the private service sector increasing strongly and central government employment rising for the first time since 1991. On the other hand, weaker industrial production led to a fall in manufacturing employment in the fourth quarter of 1998. Differences in employment growth between regions were substantial. While employment increased by 4.2 per cent in the Uusimaa province, which includes Helsinki, there was practically no increase in the northern part of Finland, where unemployment is substantially higher.[11] This divergence was accompanied by stronger internal migration. More than a quarter of a million inhabitants (around

Table 4. **Labour market developments**

	1988-91 average [1]	1992	1993	1994	1995	1996	1997	1998
	Percentage changes							
Real GDP	–0.4	–3.2	–0.6	3.7	3.9	4.1	5.5	4.7
Real GDP per worker	0.9	4.2	5.9	4.6	1.7	2.7	3.4	2.2
Employment	–1.3	–7.1	–6.1	–0.8	2.2	1.4	2.0	2.4
Labour force	–0.4	–1.8	–0.9	–0.5	0.7	0.4	–0.2	0.9
	Per cent of labour force							
Unemployment rate	4.3	11.7	16.3	16.6	15.4	14.6	12.7	11.4
of which:								
Males	4.9	13.6	18.1	18.1	15.7	14.3	12.3	10.9
Females	3.7	9.6	14.4	14.8	15.1	14.9	13.0	12.0
Youth (aged 15-24)	10.7	26.4	33.6	34.0	29.7	28.0	25.2	23.5
Persons unemployed for more than 12 months (per cent of total)[2]	5.3	8.0	17.8	27.0	30.1	30.1	30.5	30.2
Active labour market programmes								
of which:								
Job creation	1.4	2.1	2.3	2.7	2.6	2.6	2.5	2.3
Labour market training	0.6	1.1	1.1	1.3	1.6	2.1	2.3	2.1

1. Annual rate.
2. Registered unemployed.
Source: Ministry of Labour, *Finnish Labour Review* and OECD Secretariat.

Figure 7. **Labour market**[1]



1. Break in series in 1988.
2. Change in legislation in 1988 forcing firms to declare vacant jobs.

Source: Ministry of Labour, *Finnish Labour Review* and OECD, *Main Economic Indicators.*

5 per cent of the population) changed communities, 7 per cent more than in the preceding year and the highest number since 1974. The stronger internal migration and the slowdown in output growth prevented serious labour shortages in the western and southern parts of the country. The improving employment situation brought discouraged workers back into the labour market. As a result, the labour force increased by 1 per cent, after falling slightly in 1997. It also led to an increase in the average hours worked per employee and in vacancies, which rose by 23 per cent to 0.7 per cent of the labour force. However, during the year, labour shortages eased and at year-end, they were no longer seen by enterprises as a constraint to further output growth.

In 1998, the unemployment rate dropped by 1.3 percentage points to 11.4 per cent. At the end of 1998, it fell to the euro area average, while it was 6.3 percentage points above it at the Finnish unemployment peak in February 1994. Less comfortable was the development of long-term unemployment (two years or more) as its share increased while the average length of an unemployed spell increased further.[12] Total labour market slack remained substantially above the labour market survey's measure of unemployment. Taking into account participants in training programmes, subsidised employment and early retirement schemes (unemployment pensions) gives a broad measure of the jobless rate amounting to some 20 per cent of the labour force.

Wage inflation picks up somewhat, while price inflation remains subdued

Wage inflation – as measured by the wage and salary earnings index[13] – picked-up in 1998 (Table 5). The two-year collective wage agreement for 1998-99 implied a wage rise of 2.7 per cent, with wage drift adding 1.0 percentage point. Wage drift has remained remarkably low given the rapid decline in unemployment. Wage rates rose more rapidly on a national accounts basis by 5.2 per cent due to a pick-up in overtime, bonus payments and the shift in employment to sectors with above average salaries. Non-wage labour costs (employers' contributions to social security and pension funds) grew as in the previous year by less than wages. Nevertheless, the rise in the compensation rate was significantly above the euro area average. The same holds for business sector unit labour costs, which rose by 2.7 per cent in 1998, compared with a drop of 0.7 per cent in the previous year, while for the euro area as a whole costs fell slightly in both 1997 and 1998. Last year was the first year since 1990 when unit labour costs rose above the euro area average.

Higher unit labour costs were reflected in a marginal pick-up in consumer price inflation (Table 5). On the other hand, lower raw material prices and the effective appreciation of the markka dampened cost pressures. As elsewhere, prices for services rose much more briskly than for goods. Lower import and higher labour content, lower productivity rises and more limited competition are

Table 5. **Wages, costs and prices**

Percentage changes

	1991	1992	1993	1994	1995	1996	1997	1998
Unit labour costs	**6.8**	**–2.7**	**–5.4**	**–2.6**	**2.6**	**0.0**	**–0.7**	**2.7**
Negotiated wages[1]	3.9	0.9	0.1	0.6	3.6	3.1	1.3	2.7
Wage drift[1]	2.5	1.1	0.7	1.4	1.0	0.8	0.9	1.0
Wage rate[2]	6.4	1.9	0.7	2.0	4.7	3.9	2.1	3.7
Wage rate[3]	1.4	–0.1	0.8	2.5	5.9	4.5	3.5	5.2
Compensation rate[3]	3.8	1.2	1.8	4.4	4.7	3.2	2.5	4.6
Consumer prices	**4.1**	**2.6**	**2.2**	**1.1**	**1.0**	**0.6**	**1.2**	**1.4**
At year end[4]	3.9	2.1	1.5	1.6	0.3	0.8	1.9	0.8
Contribution from:								
Labour cost and profit margins	1½	–2	0	0	½	–¼	½	¾
Import prices	½	3½	2	½	0	½	½	–½
Housing costs	–¼	–½	–¾	0	0	–¼	¾	½
Indirect taxes	2¼	1¼	¼	1	¾	½	¼	0
Food prices	0	0	0	0	–1¼	0	0	0

1. Annual average of quarterly data.
2. Total wage and salary earnings index (1964 = 100).
3. Private sector, national accounts data.
4. Rates of change from December to December.

Source: Statistics Finland, Ministry of Finance and Bank of Finland.

behind this phenomenon. Most of these factors also explain the relatively strong rise in the prices of other domestic expenditures. While the private consumption deflator increased by 0.7 per cent in 1998, the investment and government consumption deflators rose by 3.3 and 3.2 per cent, respectively. The downward trend in the consumer price index levelled off in the first quarter of 1999 due to the rebound in oil prices. For the first time in a year, producer prices increased in March from the previous month, but were still 4 per cent lower than in March 1998.

External position: new record current account surplus

With a rise in real net exports and the improvement in the terms of trade, the trade balance turned out a record surplus of 9.7 per cent of GDP on a balance of payments basis.[14] The current account balance posted a new record surplus of 5.8 per cent of GDP, the largest in the euro area. While the current account no longer poses a constraint on macroeconomic policy with the introduction of the euro, it remains a useful indicator of relative demand and price developments. Capital outflows exceeded inflows by FIM 19.4 billion. Net foreign direct investment increased sharply, from FIM 16.5 billion to FIM 46.5 billion, partly due to technical factors.[15] In recent years direct investment has been heavily

concentrated in the EU, with Sweden and the Netherlands being the major host countries. At the end of 1997, the EU accounted for 66 per cent of total direct investment.

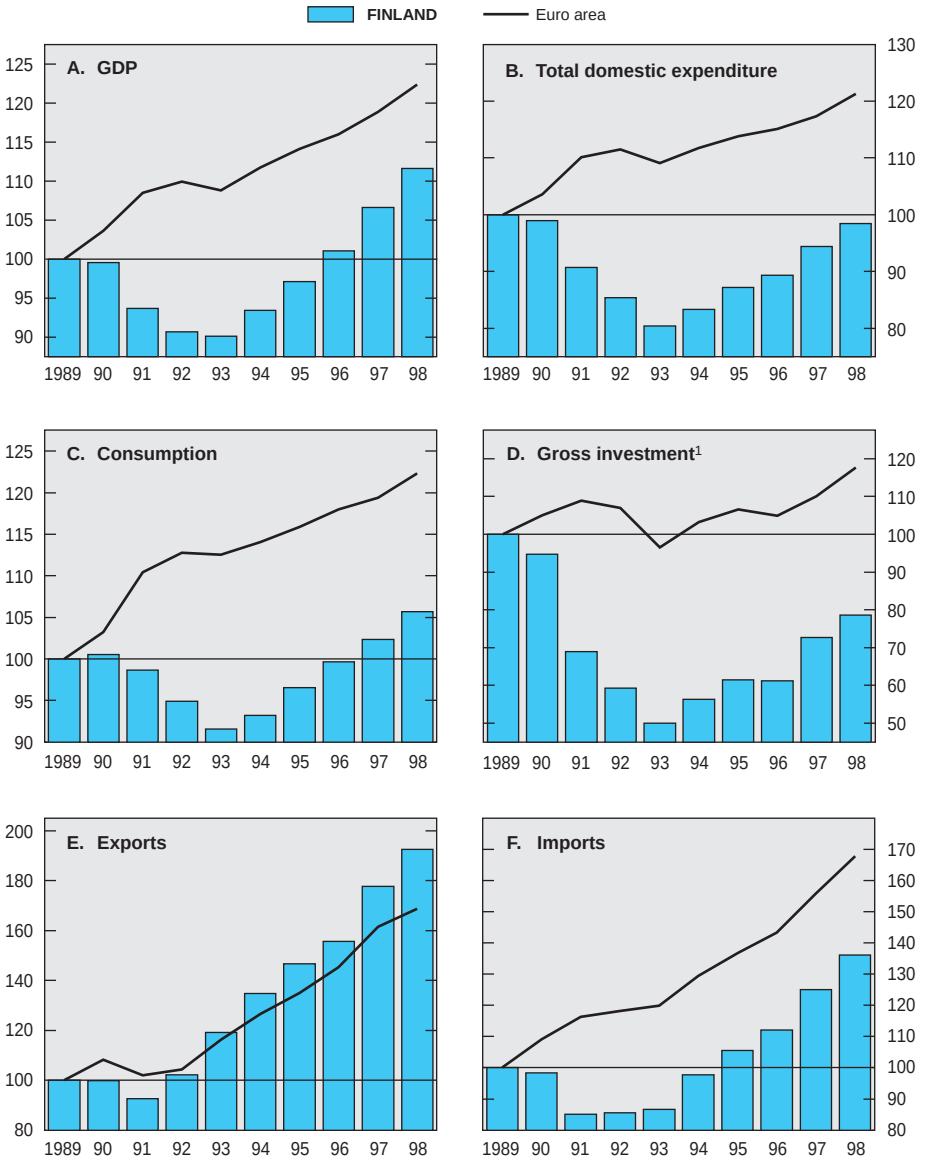
The strong rise in Finnish share prices led to a substantial rise in the net international investment position (net foreign liabilities), from FIM 271 billion at end-1997 to FIM 473 billion (70 per cent of GDP) at the end of 1998, despite the substantial current account surplus. The market value of shares held by non-residents surged by around 150 per cent in 1998, while the much smaller Finnish investment in foreign shares rose much less, by 64 per cent. However, due to the hefty current account surplus, the interest-bearing net external debt dropped by FIM 22 billion to FIM 138 billion (20 per cent of GDP). As a ratio of GDP, interest-bearing debt is back to its level at the end of the 1980s, but with a completely different composition. Currently, debt consists almost entirely of central government liabilities, while at the end of the 1980s, the central government had almost no foreign debt whereas the foreign debt obligations of the private sector were rising rapidly.

Finland in the 1990s: from deep recession to remarkable recovery

Since 1994, the Finnish economy has recovered rapidly from the 1990-93 recession, which had seen a drop in real GDP of 9.9 per cent, the deepest post-war recession in the OECD area[16] (Figure 8). This is a major achievement because deep recessions are often not followed by quick and complete recoveries. Real GDP surpassed the pre-recession level already in 1996 and, with an annual growth rate of 4½ per cent between 1993 and 1998, the expansion was the strongest in the OECD area after Ireland, Poland and Korea. This was accompanied by an outstanding productivity rise of 2.9 per cent per year, compared with 2.1 per cent and 1.7 per cent for the euro area and the OECD area, respectively. The recession and subsequent recovery led to a major shift in the composition of GDP, as the rise in real GDP was fully accounted for by a rise in net exports. In 1998, domestic demand was still, but only marginally, below the 1989 level while exports were 90 per cent higher. The rise in exports, related to a more general opening up to the global economy, also led to a strong rise in imports. Consumption in 1998 was higher than in 1989 but business investment and residential investment were still a quarter lower.[17] For two reasons, lower investment should not be seen as an indication of an incomplete recovery. First, the end of the 1980s were characterised by over-investment – an important reason for the deep 1990-93 slump. Second, the production structure shifted from physical capital to more human-capital-intensive activities, reducing business investment as measured in the national accounts.

The nominal effective depreciation in 1991-93 gave a major impetus to exports but depressed domestic demand and added to the difficulties

Figure 8. **GDP and expenditures**
Volume, 1989 = 100



1. Including stockbuilding.
Source: Statistics Finland and OECD Secretariat.

experienced by SMEs heavily indebted in foreign currency. Cost competitiveness improved by more than 30 per cent (Figure 4). Despite the strong effective appreciation of the currency in nominal terms since 1993, the competitiveness gain was broadly maintained, due to rapid productivity growth, moderate wage agreements and a monetary policy stance aimed at price stability. The tight fiscal policy was also crucial to the recovery as it led to a sustainable debt situation. The take-off of the electronic equipment industry was another important element. It was mainly attributable to entrepreneurial skills and the business strategy followed, but government R&D policy and deregulation of the telecommunications market probably played a role as well. Concerning the labour market, however, the recovery is far from complete. Employment has grown by 2 per cent per year since 1994, but in 1998 was still more than 10 per cent below the 1989 level.

Macroeconomic policy: a new era begins with the advent of the euro

Since the 1990-93 slump, macroeconomic policy has aimed at restoring macroeconomic fundamentals conducive to sustainable growth: low inflation and a government budget surplus has now been established reversing the previously adverse debt dynamics. These aims were also crucial for meeting the Maastricht criteria and for becoming a founding member of the EMU. From 1995 till the launch of the euro in early 1999, the monetary policy of the Bank of Finland was successful in keeping underlying inflation close to its official target of 2 per cent. Essential for achieving the inflation target was the appreciation of the markka from its excessively weak level reached soon after the 1992 decision to float the currency. Expectations of a markka appreciation helped the Bank of Finland to lower interest rates at a rapid rate, which was a precondition for a sustainable recovery. Fiscal consolidation succeeded in meeting the debt and deficit criteria of the Maastricht Treaty. Income policy also played an important role in the economic recovery with the government being involved in the negotiations of the 1996-97 and 1998-99 central wage agreements. Tax concessions were instrumental in reaching the agreements that led to moderate wage rises.

The launch of the euro in early 1999 is likely to have positive effects on living standards in the coming years but it also reduces the number of available macroeconomic instruments. The exchange rate and the short-term interest rate can no longer be used for demand management purposes. Getting fiscal policy right and an income policy that keeps labour cost pressures low are therefore even more important than in the past.

Fiscal policy: the government budget moves into surplus

In 1998, there was a general government surplus of 1½ per cent of GDP – the first since the beginning of the 1990-93 recession – and a turnaround of 8½ percentage points from the 1993 record deficit of 7 per cent of GDP. With interest payments rising as a result of the sizeable deficits during the slump and the re-capitalisation of the banking sector, the primary balance registered an even bigger swing, of 11 percentage points. Deficit reduction and economic recovery were clearly interrelated, with causality working both ways. Without the austerity measures, credibility in financial markets would not have been restored quickly and inflation and real interest rates would not have fallen to the low levels they did. Thus, the economic conditions would not have been there for rapid output and employment growth. At the same time, the economic recovery, through higher tax revenues and lower social security outlays, was essential for moving the budget balance into surplus.

The 1998 outcome, the 1999 budget and the 2000 budget deliberations

The 1998 government budget outcome was slightly better than expected in September 1998 when the draft 1999 budget of the central government was published and clearly better than expected a year earlier when the draft 1998 budget was presented to Parliament (Table 6). The better outcome is mainly the result of stronger than expected output growth leading to higher tax revenues, especially for corporate income tax and indirect taxes. The lower deficit, in combination with substantial privatisation receipts, led to a smaller than projected net borrowing requirement of the central government.

The draft 1999 budget included FIM 4 billion (0.6 per cent of GDP) additional spending cuts compared to the preliminary expenditure guideline aimed at limiting the risk of overheating. At the same time, it included income tax cuts of around FIM 4 billion. The general government surplus was officially projected to increase to 2.4 per cent of GDP, from 1.0 per cent of GDP in 1998, mainly as a result of a smaller central government deficit. The budget update of June 1999 foresaw a better outcome, with a general government surplus of 3.1 per cent of GDP. At the beginning of 1999, all income tax rates, except the highest one, were lowered by 0.5 percentage point while an inflation adjustment of 2 per cent was made to the tax scales. Furthermore, to increase work incentives, the earned-income deduction in municipal income tax and the maximum standard deduction for work-related expenses in central government income tax were raised. The OECD Secretariat's fiscal stance estimates indicate that fiscal policy in 1999 is somewhat restrictive but substantially less so than in 1998 and previous years.

At the beginning of 1999, buffer funds in the unemployment insurance and the employment pension funds were introduced. The aim of such funds is to

Table 6. **Budgetary plans and outcomes**

Billion FIM, national accounts basis[1]

	Outcome			1998		1999	
	1995	1996	1997	Budget[2]	Preliminary results	Budget[2]	Revised[3]
Central government							
Revenue	135.8	148.7	162.0	169.6	177.7	183.4	185.8
Expenditure[4]	188.7	190.2	187.8	190.5	188.5	189.5	187.0
Balance	–52.9	–41.5	–25.9	–20.9	–10.8	–6.1	–1.2
Per cent of GDP	*–9.4*	*–7.1*	*–4.1*	*–3.3*	*–1.6*	*–0.9*	*–0.2*
Net financing requirement (–)[5]	–58.9	–36.8	–13.4	–14.6	–0.4	–1.9	–2.6
Local government							
Revenue	103.6	102.6	99.6	98.7	104.2	103.5	107.0
Expenditure[4]	96.7	98.0	101.5	100.7	104.3	105.0	106.6
Balance	6.9	4.6	–1.9	–2.0	–0.1	–1.5	0.5
Per cent of GDP	*1.2*	*0.8*	*–0.3*	*–0.3*	*0.0*	*–0.2*	*0.1*
Social security							
Revenue	140.1	138.3	138.3	142.2	141.1	146.1	143.6
Expenditure[4]	118.6	119.1	120.4	120.1	120.2	121.6	120.8
Balance	21.5	19.2	17.9	22.1	20.9	24.5	22.8
Per cent of GDP	*3.8*	*3.3*	*2.8*	*3.4*	*3.1*	*3.4*	*3.2*
General government balance	–24.5	–17.7	–9.8	–0.8	10.0	16.9	22.1
Per cent of GDP	*–4.4*	*–3.0*	*–1.6*	*–0.1*	*1.5*	*2.4*	*3.1*
Underlying projections, per cent:							
Real GDP growth	4.0	4.1	5.6	3.5	4.7	4.0	3.8
Unemployment rate	15.4	14.6	12.7	13.5	11.4	9.0	10.3

1. The budget numbers are not fully comparable with outcomes, preliminary results and revised budget numbers which are based on the revised ESA95 national accounts.
2. September of preceding year.
3. June 1999.
4. Including consumption of fixed capital, capital formation and net capital transfers.
5. On a cash basis.

Source: Ministry of Finance.

smooth premiums over the cycle and to prevent pro-cyclical premium increases during downturns. Before the reform, unemployment insurance contributions had to match the projected outlays annually. The policy change is largely motivated by the introduction of the euro. The statutory authorised maximum size of the fund amounts to approximately FIM 3 billion (0.5 per cent of GDP). The buffer is twice as big, as the fund is authorised to borrow the same amount. The buffer will build up gradually, with a deposit of around FIM 0.5 billion in 1999. Contribution rates, and as a consequence changes in the size of the fund, will be proposed by

the board – formed by the social partners – but need approval by the Ministry for Social Affairs and Health.[18] At the same time, a supplementary buffer was introduced in the employment pension funds, with a maximum size of around FIM 3 billion. As for the unemployment buffer fund, it will build up gradually but in contrast, there will be no possibility to borrow.[19] Automatic stabilisers work through the social security and pension funds in most OECD countries and prevent pro-cyclical premium rises during cyclical downturns. However, if revenue shortfalls are first considered as purely cyclical but prove to be structural, this could delay necessary structural measures.

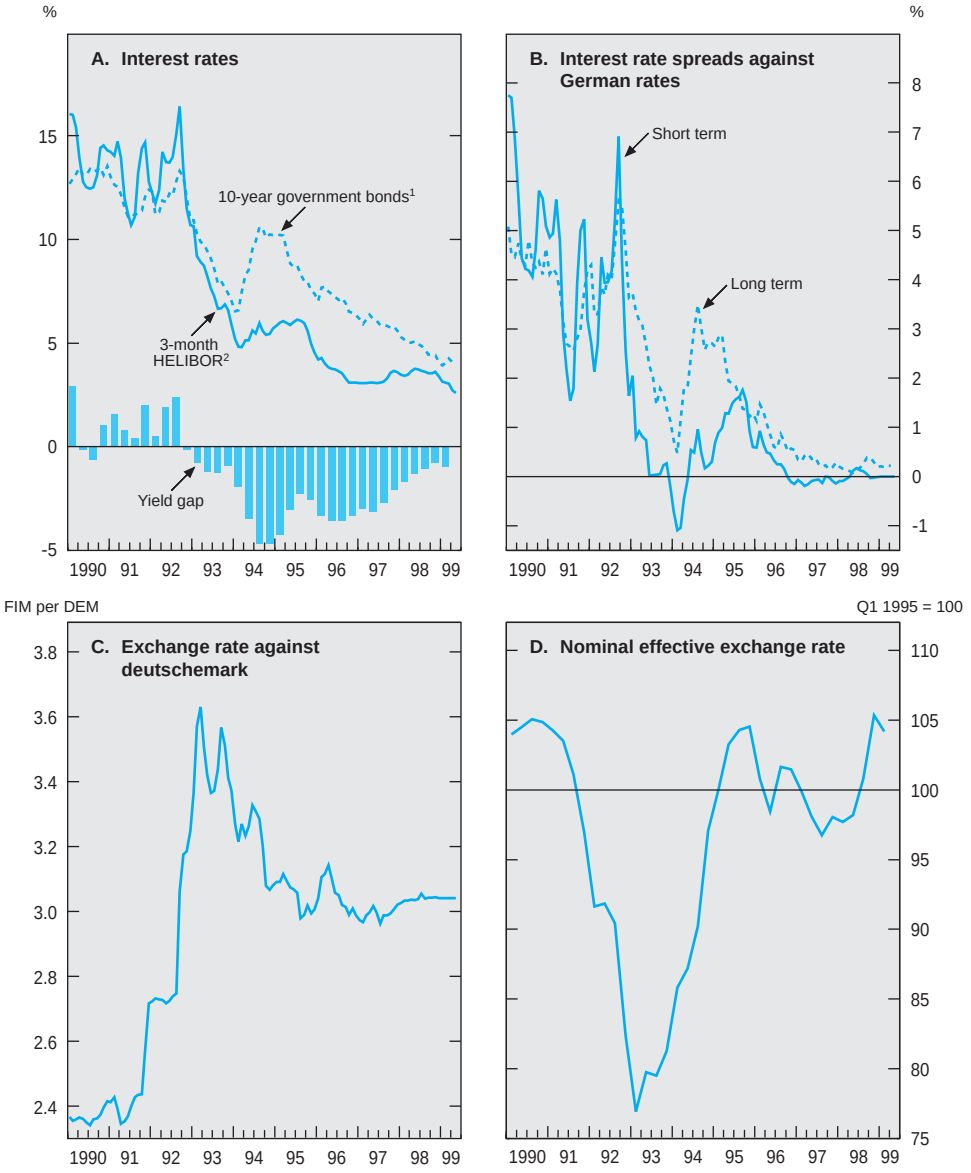
The new government formed after the March general elections announced in its programme that central government expenditures, including interest payments, will be kept constant in real terms over the medium-term. As interest payments will decrease, other expenditure could rise. The expenditure guidelines drawn up by the government in May mean a backloading of measures, as the ceiling for central government expenditure increases by 1¼ per cent in 2000.[20] The government will discuss the draft 2000 budget in July, but no detailed decisions on tax measures for 2000 will probably be taken as they depend on the wage negotiations starting later in the year.

Monetary policy: smooth run-up to the euro

The run up to the introduction of the euro has been smooth despite the turmoil in international financial markets. Before the decision on conversion rates was taken at the EU Summit of May 1998, the Bank of Finland had to intervene on the foreign exchange markets to prevent an appreciation of the markka *vis-à-vis* the Deutschemark. Market participants were speculating on a revaluation of the central rate for the markka because of the strong growth of the Finnish economy and signs of overheating. In order to signal its concerns about emerging inflationary pressures, the Bank of Finland raised the tender rate by 15 basis points to 3.40 per cent in March. Foreign exchange rate purchases of the central bank were some FIM 5 billion per month in March and April. However, after the decision to lock rates at the Exchange Rate Mechanism (ERM) central parities, upward pressure dissipated and the exchange rate *vis-à-vis* the Deutschemark was stable. At the same time, the money market rates converged to the German rates (Figure 9).

The Russian debt moratorium in August 1998 did not influence the markka *vis-à-vis* the Deutschemark or the Finnish money market. It had, however, a clear impact on the European bond markets with the interest rate differentials rising in most countries *vis-à-vis* the German rates. The differential between Finnish and German 10-year government bond rates doubled to around 40 basis points in September and October. Thereafter, the situation calmed down and the difference returned to around 20 basis points. This difference persisted in the first months of 1999 and reflects mainly liquidity differences between the two bond

Figure 9. **Interest and exchange rate developments**



1. Ten-year government bond rate from January 1993 onwards, five-year rate for earlier period.
2. EURIBOR as from January 1999.

Source: Bank of Finland and OECD, *Main Economic Indicators.*

markets. On 3 December, the Bank of Finland and nine other central banks of the prospective euro area lowered their official rates to 3 per cent.[21] In Finland, the decline in the tender rates was 40 basis points. The central banks took the measure in the light of weaker growth prospects and the favourable outlook for price stability in the euro area.

On 31 December 1998, the conversion rate of the markka was set at FIM 5.94573 per euro and the responsibility for monetary policy shifted to the European System of Central Banks (ESCB) with the Bank of Finland participating in the decision making of the eurosystem and implementing policy in Finland.[22] The changeover to the euro was smooth and did not lead to major technical problems due to the good preparation by the financial sector and the gradual adjustment of monetary policy instruments in preceding years.

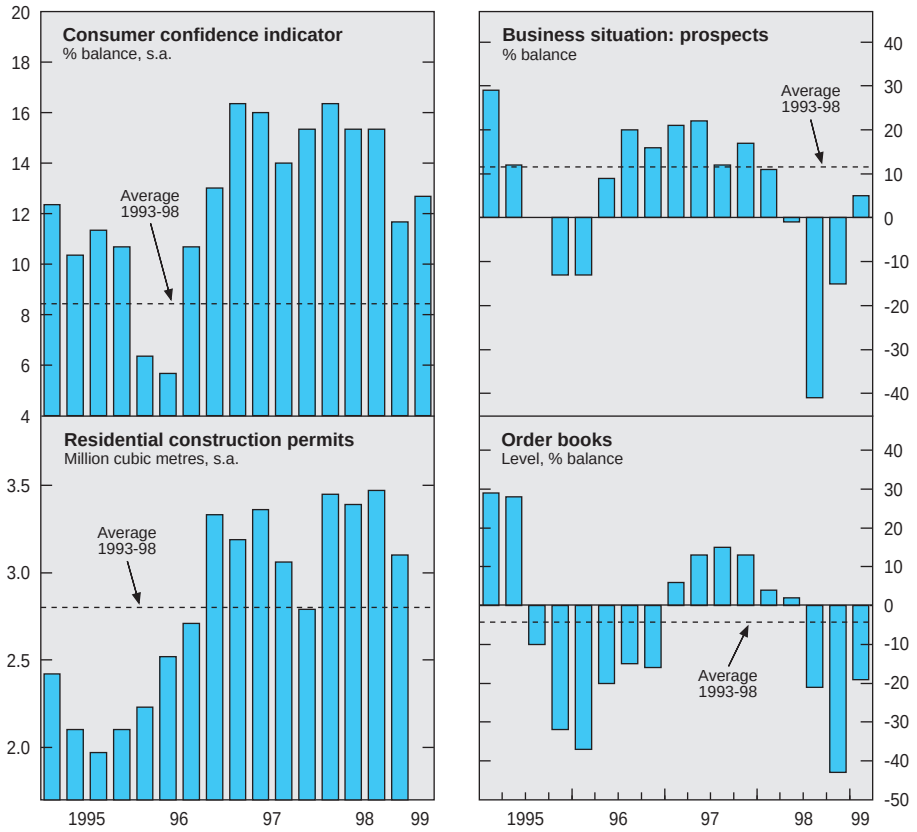
Monetary conditions appropriate in 1999, but probably too easy in 2000

During the first half of 1998, there was considerable concern that a combination of overheating of the Finnish economy and easy monetary conditions in the euro area could develop. With the Finnish economy slowing down and its cyclical situation becoming somewhat more in tune with the rest of the euro area, this fear was dissipating. However, given the reduction in the European Central Bank's (ECB) intervention rate in April, the weakness of the euro, the projected pick-up in activity in 2000 and the further labour market tightening, monetary policy of the ECB, though appropriate for the euro area as a whole, could become too easy for Finland, if not offset by a considerable further tightening of fiscal policy, especially if comprehensive labour market reforms are not undertaken.

Short-term outlook

Recent indicators for 1999 suggest that the slowdown in economic growth has not yet ended even though robust consumption provides a solid cushion. Industrial production excluding electronic equipment is still on a downward trend and export growth is weak, but retail sales are brisk. Total production picked up somewhat after December 1998, but with output in January-March 4 per cent higher than a year earlier, growth was still weaker than before mid-1998. Weak pulp and paper prices, albeit having picked up, have kept forest product exports low. Exports of goods to Russia remained extremely low. In the first three months of 1999 its value was 54 per cent less than a year earlier. The upward trend in employment continued in the first quarter, with a positive influence on private consumption, but unemployment did not drop substantially since December 1998. Building permit data indicate that construction activity will continue to expand at a moderate pace. The high level of house prices will give an impetus to residential investment. Leading indicators for the business sector suggest that a

Figure 10. **Leading indicators**



Source: OECD, *Main Economic Indicators.*

turning point will not occur in the coming three to six months (Figure 10). Order inflows, especially from abroad, are still falling and managers in industry remain pessimistic about economic developments in the near term. Consumer sentiment, on the other hand, remains strong. Domestic demand will also be supported by low interest rates. Fiscal policy is somewhat restrictive in 1999, but substantially less than in 1998 and the previous years of budgetary austerity. Furthermore, domestic demand should benefit from the solid financial position of households, the sound balance sheets of the business sector and the healthy

state of the banking industry. Another supporting factor is the positive supply-side shock generated by the electronic equipment industry. In 1998, the sector contributed 1½ percentage points to GDP growth. The contribution is likely to remain substantial over the next few years.

Nevertheless, due to weaker foreign demand, real GDP growth is projected to decelerate further to 3.3 per cent in 1999, which is still markedly above the euro area average[23] (Table 7). Private consumption is projected to rise

Table 7. **Short-term projections**

Percentage changes, volume

	1995 Current prices, billion FIM	1996	1997	1998	1999	2000
Private consumption	291.8	3.5	2.6	4.5	4.1	3.5
Government consumption	128.2	2.5	2.9	0.4	0.5	0.8
Gross fixed capital formation	90.5	8.4	14.4	8.3	5.4	6.6
Public sector	15.5	10.6	11.8	–1.6	–8.1	2.9
Private sector	75.0	8.0	14.9	10.3	7.9	7.1
Residential	19.4	2.6	29.7	9.4	9.1	8.9
Non-residential	55.7	9.8	10.1	10.7	7.5	6.5
Final domestic demand	510.4	4.1	4.9	4.3	3.6	3.6
Stockbuilding[1]	7.2	–1.5	0.8	0.0	0.3	–0.1
Total domestic demand	517.7	2.5	5.7	4.3	3.9	3.5
Exports of goods and services	209.3	6.1	14.2	8.3	3.6	5.6
Imports of goods and services	164.6	6.3	11.4	8.9	5.5	6.0
Foreign balance[1]	44.7	0.4	2.0	0.6	–0.3	0.4
GDP	561.4	4.1	5.5	4.7	3.3	3.6
GDP of the euro area	..	*1.6*	*2.5*	*2.9*	*2.1*	*2.6*
Employment	2 098.7[2]	1.4	2.0	2.4	1.4	1.2
Unemployment rate (per cent)	15.4	14.6	12.7	11.4	10.6	10.0
Household saving ratio[3]	6.2	2.8	4.7	4.8	3.8	3.3
Private consumption deflator	..	1.6	1.9	0.7	1.1	1.5
GDP deflator	..	0.6	1.7	2.4	1.3	1.8
Private sector wage rate	..	4.5	3.5	5.2	3.4	3.8
Current account (per cent of GDP)	..	4.1	5.6	5.8	5.8	6.1
Interest rates (per cent)						
Short-term	..	3.6	3.2	3.6	2.7	2.6
Long-term	..	7.1	6.0	4.8	4.3	4.4

1. Contribution to GDP growth.
2. Thousand persons.
3. Per cent of disposable income.

Source: Statistics Finland and OECD Secretariat.

strongly, by 4.1 per cent, due to positive consumer sentiment and the steady rise in disposable income. On the other hand, lower capacity utilisation rates should dampen investment growth. As with the other euro area countries, the depreciation of the euro in the first months of 1999 is likely to support exports, partly offsetting the lower pace of world demand. Employment growth will decelerate less than output as growth is likely to shift from capital-intensive exports to more labour-intensive domestic demand. The downward trend in unemployment is likely to continue with the unemployment rate falling to 10.6 per cent of the labour force, 0.6 percentage point below the euro area average. Inflation is projected to remain subdued, albeit picking-up somewhat due to the oil price rebound. With world demand picking up in 2000, output growth is projected to accelerate somewhat, to 3.6 per cent, while unemployment should decline further. The general government budget surplus will continue to rise, to 2.7 per cent of GDP in 2000, by far the strongest fiscal position in the euro area and the second strongest in the OECD, after Norway.

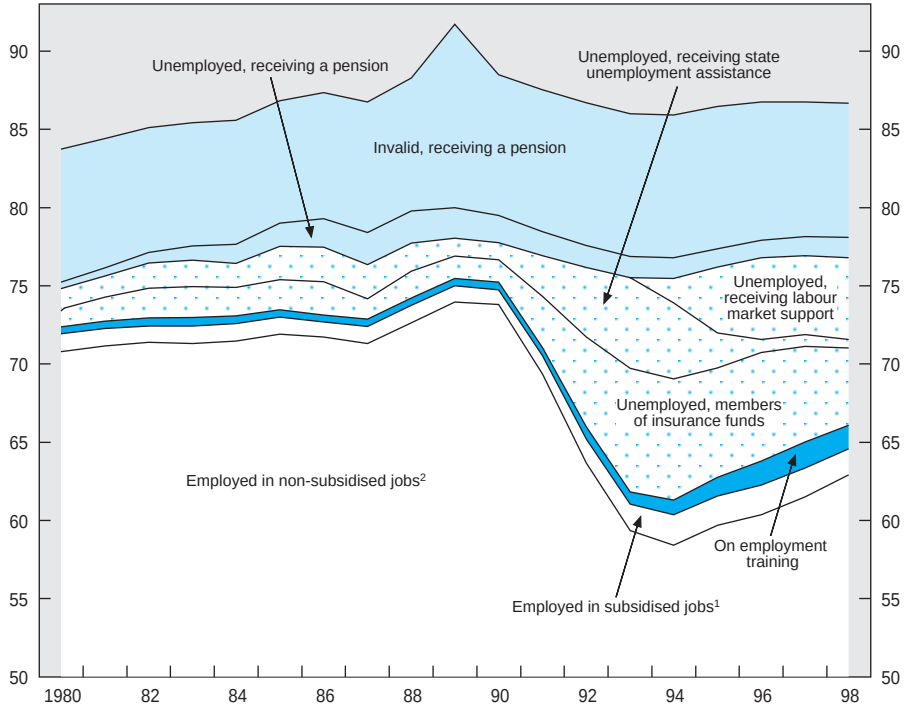
The year 2000 will be the third successive year with an output gap estimated to be positive by the OECD Secretariat. Therefore, the key domestic risk to the projection is that the further drop in unemployment will lead to an excessive wage agreement. With the euro in place, excessive wage increases could lead to heavy employment losses, as the option to devalue is no longer available. A stricter control of public expenditure would reduce the risk of overheating. The major external risk is that if economic growth in Western Europe and world trade in general does not materialise, Finnish export growth will suffer.

II. Improving performance by stepping up structural reform

The unemployment rate has dropped substantially since the recession of the early 1990s from a peak of 18 per cent in February 1994 to below 11 per cent in early 1999 – the second fastest drop ever registered in the OECD.[24] Nevertheless, the Finnish labour market performance has remained clearly inferior to countries like the Netherlands, Ireland, the United Kingdom and to a lesser extent Denmark.[25] Unemployment is still higher than at the beginning of the decade, while the employment rate is lower, and the large number of long-term unemployed is a major concern. Half of the long-term unemployed have been out of work for a period of two years or more. Moreover, the official unemployment figure underestimates the labour market problem as almost 20 per cent of the working-age population receives some form of assistance (Figure 11). The accession to the EU in 1995 and government policy initiatives have spurred structural change in product and financial markets, contributing to the strong recovery since 1994. Finnish GDP per capita passed the EU average in 1998 (Figure 12). However, living standards and productivity are still well below those of the United States and there remains ample scope for reforming product and labour markets. Such moves should be supported by a substantial lowering of the tax burden which would raise economic performance further.

Sound macroeconomic fundamentals, a strong competitive position and the existence of a pole of excellence in the electronic equipment sector may not be sufficient to achieve a satisfactory labour market performance in the coming years and to increase the living standards substantially. Indeed, the government projected in its 1998 Stability Programme only a meagre reduction of the unemployment rate, to 7.8 per cent in 2002, from 9.0 per cent in 1999.[26] The countries mentioned above, with a substantially improved labour market performance, all undertook structural reforms that were more radical and more comprehensive than the reforms made till now in Finland.[27] This chapter reviews the scope for structural reform in the labour, product and financial markets to improve the labour market situation and to provide the impetus for a continuation of high economic growth, while taking into account the specific institutional and social

Figure 11. **Structure of the labour force**
Per cent of population aged 15-64



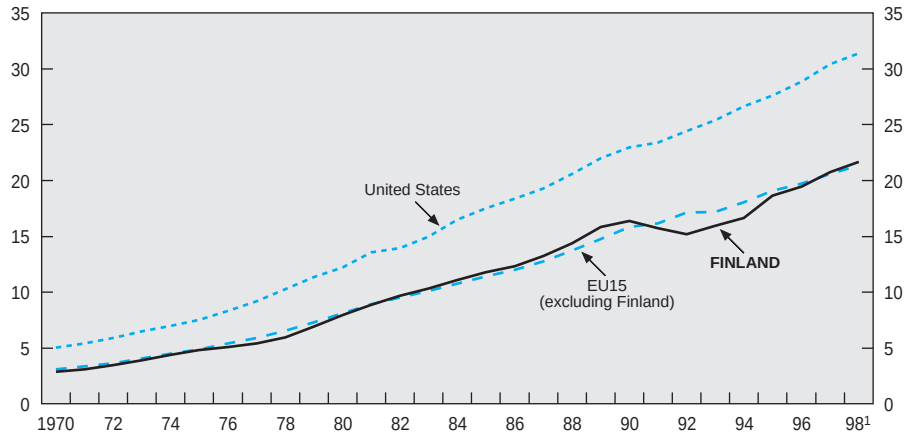
1. Persons employed with selective measures.
2. Total employment less persons employed in subsidised jobs.
Source: Ministry of Labour, *Finnish Labour Review* and OECD Secretariat.

characteristics in Finland. The recently announced policy initiatives of the new government are summarised in Box 1.

How to improve labour market performance

The labour market improvement since the deep 1990-93 recession is quite impressive (Table 8). In addition to sound monetary and fiscal policy, real wage flexibility at the aggregate level was crucial for this achievement. The limited wage rises of the central wage agreements since 1995 kept competitiveness strong and led to further market share gains. The structural reforms undertaken by

Figure 12. **GDP per capita**
In thousand US$, using current PPPs



1. Estimates.
Source: OECD, *National Accounts* and OECD Secretariat.

the Finnish government, especially following its 1995 Employment Programme, also played a role (see Annex I for a summary of measures taken in the 1990s). One of the more important measures taken was the increase in the earned-income deduction in the local government income tax, in combination with changes in housing support, basic welfare benefits (subsistence allowance) and children's day-care fees. As a consequence, the unemployment trap (the disincentive to take up work) has become less severe and increases in gross income lead to a rise in disposable income at a much lower level than before (Table 9). Another important measure was the marked reduction of benefit renewal through participation in active labour market programmes, effectively phasing down unemployment benefits over time and therefore enhancing the incentive to take up a job. Furthermore, the shift in focus of active labour market policies from subsidised jobs to training was positive. For instance, there were some successful conversion schemes, such as the training of building engineers in information technology. Finally, unemployment benefits became more conditional on labour market training. For instance, under 25-year-olds without vocational education receive an unemployment benefit only if they participate in labour market training.

Despite the substantial improvement in labour market conditions since 1993, all available indicators show that the situation is still worse than in

Box 1. The programme of the new government

Following the elections on 21 March 1999, a new coalition government was formed on 15 April. It includes the same parties as the previous one. The major policy goal of the new government is to raise the employment rate to close to 70 per cent by 2003 (from 65 per cent in 1998). To boost employment, a series of reforms to product, capital and labour markets will be implemented, including changes to the tax and social benefit systems. The government will pursue a consensual strategy with the social partners and tax reductions will be used as a carrot to encourage moderate wage agreements. It pledges to maintain the Nordic welfare society.

Budgetary and fiscal policy

The government is aiming at a structural central government surplus of an amount not specified in the programme. In 1998 there was a deficit of 1½ per cent of GDP. The government aims to reduce the central government debt to below 50 per cent of GDP in 2003 from 63 per cent in 1998. Central government spending, including interest rate payments, will be kept constant in real terms. As interest payments will decrease, other expenditure could rise in real terms. The state pension will be increased by FIM 50 a month (by approximately 2 per cent) from the beginning of 2002 in addition to the usual price index adjustment.

The government is committed to implementing a FIM 10-11 billion reduction in income tax and social security contributions (1½ per cent of GDP in 1998), partly through an increase in the earned income tax deduction in the local income tax, accompanied by lower central government tax rates so as to prevent a rise in marginal tax rates. The reduction of excise duties will be delayed until 2003, the real estate tax will be increased and the corporate and capital income tax rate will rise by 1 percentage point to 29 per cent in 2000. Charges for health care services will be increased from the beginning of 2000. Environmental and energy taxes will be increased, but with due regard to industrial competitiveness. The overall consequences of these measures for the tax to GDP ratio are not specified.

Labour market policy

The reform of social security benefits will be continued to raise the rewards from work. However, the incentives to find work for long-term unemployed persons will diminish due to a less stringent means test concerning the income of spouses. Labour market policy measures will be targeted primarily at the long-term unemployed. For youth after 6 months of unemployment, and for all other unemployed after 12 months, a workplace or traineeship will be offered or they will benefit from other labour market measures.

Labour legislation will be adjusted to realise a better balance between job security of employees and the flexibility required by enterprises, but no specific measures have been announced. The government will promote cyclical, sectoral and regional wage flexibility within the framework of binding collective agreements. As part of the measures designed to raise the average effective retirement age by 2 to 3 years in the long-term, tax relief on premiums for voluntary retirement pensions will be reduced.

(continued on next page)

(continued)

Product market policy

Decisions on sales of state assets will continue to be made case by case, with revenues primarily earmarked to repay state debt. In addition, revenues will be used to support R&D and venture capital. The main focus of industrial policy will be on improving the competitive environment as well as on lightening the regulatory regimes. Efforts will be made to maintain the overall spending on R&D at least at the current level, with greater emphasis on the less developed areas as an integral part of regional policy. In addition to R&D, the government will take measures to promote marketing and commercialisation of products, and to strengthen the educational system to support knowledge-based production.

The administrative burden on small and medium-sized enterprises (SMEs) will be alleviated, among other measures by improving the possibilities for small enterprises to do their business using electronic data transfer and combining the collection of data required by various authorities. Improved follow-up and control of support programmes will enhance the evaluation of the effectiveness of financial support. Subsidies to shipbuilding will be given as long as allowed under EU rules.

1990 – before the recession hit the Finnish economy (Table 8). There is the insidious threat that current unemployment becomes structural due to the erosion of professional skills, the increasing social stigma of unemployment and the negative attitude of employers towards the long-term unemployed. To some extent, the insufficient labour market improvement until now is due to the late start with structural reform, the time it takes before results emerge and the extraordinary character of the 1990-93 slump – by far the deepest post-war recession in the OECD area. More importantly, however, measures already taken did probably not go far enough and should be followed-up by further, bolder initiatives.

The previous *Survey* pointed to several areas where policy changes could improve labour market performance:

- Tax rates and unemployment benefit replacement rates are among the highest in the OECD, reducing incentives to raise work effort.
- The benefit level is constant for the first 500 days of unemployment, limiting the job search intensity.
- Labour market participation of older persons of working age is very low due to strong financial incentives to retire early.
- The eligibility criteria for unemployment benefits are not strictly enforced, reducing incentives to look for a job.
- The wage structure is among the most compressed in the OECD, reducing job opportunities for the low-skilled and efforts to upgrade human capital.

Table 8. **Labour market indicators**

	Units	1990	1993	1998	Situation in 1998 better or worse than in 1990?
Employment					
Employment rate[1]					
Total	%	74.1	60.6	64.6	↘
Men	%	76.4	61.2	67.5	↘
Women	%	70.8	59.1	61.6	↘
Total by age group:					
55-59	%	60.1	51.0	51.2	↘
60-64	%	25.0	18.8	19.6	↘
Subsidised employment	Thousand	31	57	57	↘
Rate	% of labour force	1.2	2.3	2.3	↘
Unemployment					
Total (labour force survey)	Thousand	82	405	285	↘
Rate	% of labour force	3.2	16.3	11.4	↘
Long-term unemployment[2, 3]	Thousand	3	86	113	↘
Rate	% of labour force	0.1	3.5	4.5	↘
Rate	% of registered unemployed	2.9	17.8	30.2	↘
Youth unemployment (age 15-24)[2]	Thousand	22	98	47	↘
Rate	% of youth labour force	6.1	34.9	15.7	↘
Rate	% of total youth	3.4	15.4	7.2	↘
Unemployment (age 55-59)[2]	Thousand	8	33	57	↘
Rate	% of labour force	3.2	17.3	15.4	↘
Average unemployment duration	Weeks	15	30	52	↘
Pensions and invalidity					
Recipients of:					
Old-age pensions	% of working-age population	19.7	20.4	21.6	↘
Unemployment pensions	% of population of age 60-64	23.3	18.2	19.5	↗
Invalidity pensions	% of working-age population	8.1	8.1	7.6	↗
Training					
Persons on employment training	Thousand	17	27	41	↗
Rate	% of labour force	0.7	1.1	1.7	↗
Flexibility indicators					
Fixed-term jobs	% of total	..	..	23.2	..
Part-time jobs	% of total	9.5	11.4	11.4	↗

1. Employment as a per cent of population of working age (15-64 years old).
2. Registered unemployed (*i.e.* unemployed job seekers).
3. Persons who have been unemployed for over a year.

Source: Ministry of Labour, *Finnish Labour Review* and Statistics Finland, *Bulletin of Statistics.*

Table 9. **Threshold wages**[1]

FIM per month

Household	1996	1998/99	% change
Single	4 300	2 800	–34.9
Married			
Childless	7 200	5 400	–25.0
1 child	7 600	6 700	–11.8
2 children	8 900	7 300	–18.0
3 children	9 900	7 600	–23.2
Single parent			
1 child	3 300	2 800	–15.2
2 children	3 100	2 600	–16.1
3 children	3 000	2 100	–30.0

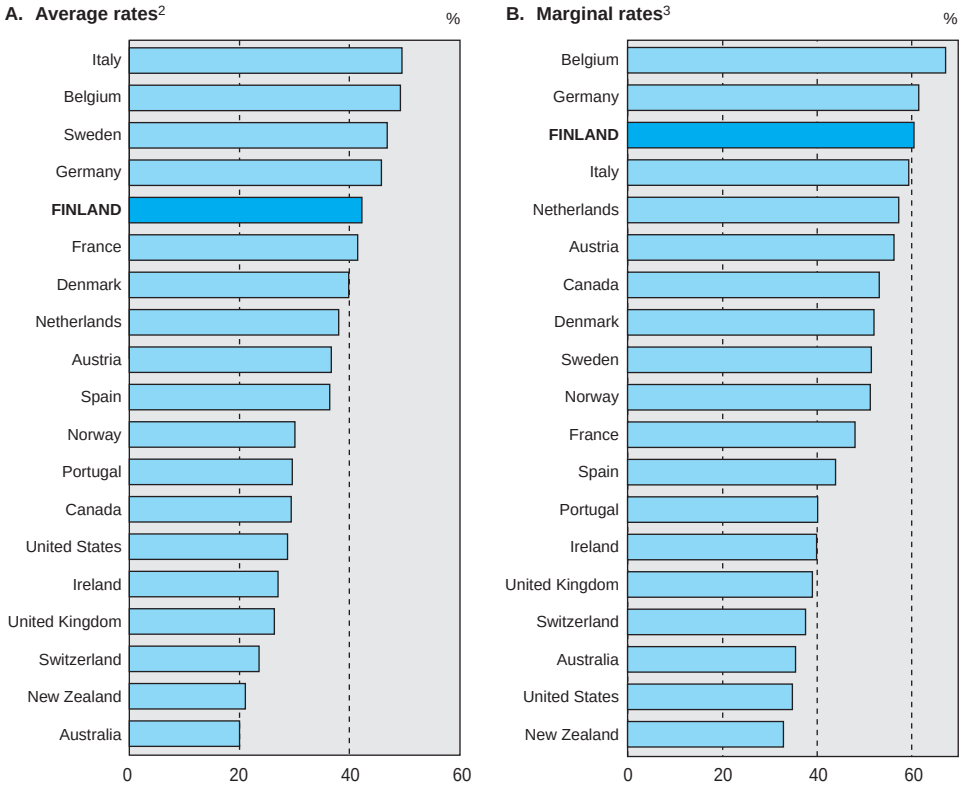
1. The threshold wage is the wage sum one has to earn to have a disposable income equal or above that of an unemployed with a basic/standard allowance, taking into account social benefits such as housing subsidises. In the case of a married couple, the combined wage is divided equally between the partners.
Due to the progressive income tax, threshold wages are higher if only one spouse works.

Source: Economic Council (1999).

High marginal tax rates have adverse effects on both low and high-income earners (Box 2). Due to means-tested social benefits such as rent subsidies, marginal tax rates are extremely high for low-income earners, creating a substantial poverty trap (disincentives to raise work effort). Tax rates have been lowered somewhat in recent years, but the Finnish average and marginal tax rates for the average worker remain among the highest in the OECD (Figure 13). In 1998, only three OECD countries had an "all-in" tax rate higher than in Finland (Kam and Bronchi, 1999). Incentives for high-income groups to increase hours worked and to make additional educational and vocational training efforts are distorted by a top income tax rate of more than 50 per cent.[28] Although the increase in the earned-income deduction in recent years was welcome as it reduced the unemployment trap, in designing new tax cuts, the adverse effects of such adjustments on the marginal tax rate for low-income earners should be taken into account, as is envisaged in the programme of the new government.

In 1998, only 40 per cent of the 55-59 year olds and 17 per cent of the 60-64 year olds were employed. Of the long-term unemployed, 55 per cent were aged 50 or older. A vicious circle exists: because the incentives to retire early are strong for employees, firms are hesitant to train older workers; this lack of training increases their risk of job loss, which seems to justify the existence of the early withdrawal schemes. There are also substantial disincentives to hire older

Figure 13. **Tax rates in Finland and other OECD countries**
In 1997 for a married couple with two children[1]



1. Where the principal earner makes 100% and the spouse 67% of the income of an average production worker.
2. Total tax wedge including employer's social security contributions.
3. Total tax wedge, principal earner.

Source: OECD (1998), *The tax/benefit position of employees,* 1997.

workers as the last employer has to pay a part of the pension in case of disability or unemployment.[29] Policy changes are needed not only because incentives are distorted and costs are currently high, but also because, in the next decade, costs will explode as the baby boom generation begins to reach retirement age. By 2010, the number of 50-64 year olds will have increased by 275 000 (bringing the total to 30 per cent above the 1998 level), while the number of persons of working age under 50 years will have fallen by more than 200 000 (9 per cent)

Box 2. High effective tax rates affect many Finns

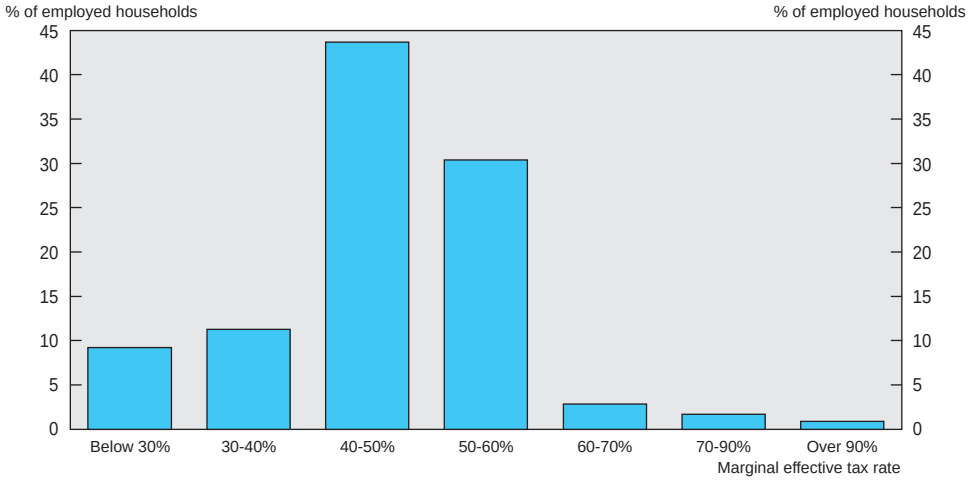
Disincentives due to the high tax burden affect a considerable part of wage earners and self-employed (Keso, 1999). Marginal effective tax rates, taking into account income taxes, social security contributions and income-dependent social benefits, exceed 50 per cent for one-third of the employed (Figure 14). The marginal effective tax rate for an average production worker who is unmarried is already around 50 per cent, while persons earning twice as much face marginal tax rates approaching 60 per cent.* Effective tax rates can be even higher for low-income households due to means-tested social benefits such as the housing allowance and the basic subsistence allowance. The lowest-paid workers can even experience a 100 per cent effective tax rate over a considerable income range (Figure 15). In some cases (1 per cent of the total), the effective tax rate can even surpass 100 per cent. Effective tax rates of 100 per cent are more frequent for people who were unemployed and take up (part-time) work.

* In 1999, the rate in the top scale of the central government income tax is 38 per cent. The municipal income tax is levied at flat rates. In 1999, this local tax rate varies between 15.5 and 19.75 per cent, the average rate being 17.6 per cent. The church tax, imposed on members of State Churches, is on average 1.3 per cent.

(Economic Council, 1998). The effective retirement age has dropped from 65 years in 1960 to 59 years in 1998 for males and from 63 to 59 years for females, while at the same time life expectancy and the health status of older workers have improved. Only a part of the drop can be explained by the increased preference for leisure at old age due to higher incomes. There is robust empirical evidence that the drop in the effective retirement age is strongly influenced by changes in the social security system which encourages retirement at a relatively young age by making continued work financially unattractive (Blöndal and Scarpetta, 1998 and OECD, 1998*b*). The implicit tax rate on continuing work after the age of 55 years is more than 40 per cent in Finland – comparable with the rate for the euro area as a whole, but substantially higher than in the United States and Japan where people tend to work much longer. Such a rate provides a substantial disincentive to continue work.

The length of the period, during which benefits are suspended in the case where an unemployed person stops looking for work or refuses to accept a suitable job or training offer, was increased in 1998. However, eligibility criteria are still not strictly enforced and suspensions remain rare. The change in the suspension period was part of a programme aimed at strengthening job-counselling and the brokerage activity of the employment offices and to re-emphasise the rights and obligations of unemployment benefit recipients to stimulate their job-search activity. The number of job counsellors at employment

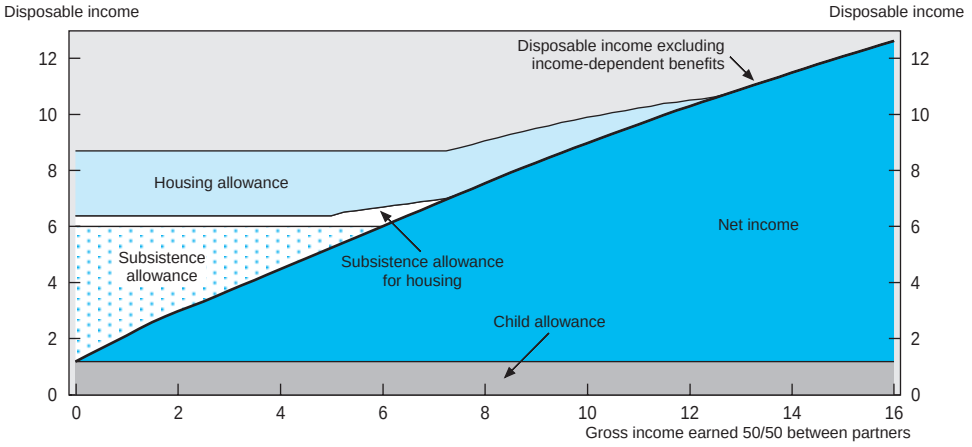
Figure 14. **Effective marginal tax rates for salary earners and self employed**[1]



1. Including child day-care fees.
Source: Ministry of Social Affairs and Health.

Figure 15. **Disposable income and income-dependent social benefits of households**[1]

Thousand FIM per month, 1999



1. For a couple with two children in public day-care, renting an apartment in an urban area.
Source: Ministry of Finance (1999).

offices was increased substantially in 1998 to intensify counselling and a target was introduced of a monthly contact with every unemployed person. However, the employment offices have not yet met this goal.

The two most recent central wage agreements have served the Finnish economy well as moderate wage rises kept competitiveness strong and led to further market share gains. However, the income policy also had its drawbacks, as it included higher wage increases for the low-skilled in the context of an already compressed wage structure and limited differentiation across industries. Fairly high wages for low-skilled workers can reduce their job opportunities. In addition to reducing labour demand, the compressed wage structure, in combination with the tax and social security system, lowers incentives to upgrade human capital.

The Finnish economic situation has improved substantially since the 1990-93 slump thanks to appropriate macroeconomic policy, wage moderation, and labour and product market reforms. However, much remains to be done to reach a satisfactory labour market outcome. The Finnish authorities could draw lessons from the low unemployment achieved in some other OECD Member countries, which is integrally linked to their labour-market reforms. Though the specific content of the reform strategy has differed across these countries, they have been broadly in line with recommendations of the *OECD Jobs Strategy* to increase societies' adaptability to change. Comparable reforms in Finland are likely to be a *sine qua non* to realise a further period of solid non-inflationary growth.

Product markets: competition increased substantially

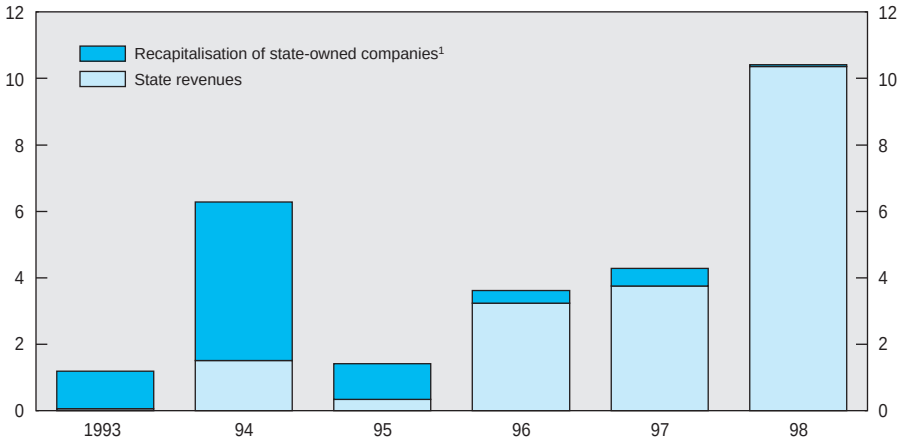
Competition in Finnish product markets has increased substantially during this decade strengthening the dynamism of the Finnish economy. The implementation of the Single Market measures following EU entry gave a major impetus but was certainly not the only reason for increased competition. For instance, the deregulation of the telecommunications and electricity markets started earlier than in most EU countries. The electricity market has been fully liberalised since 1997 for all customers, including households, while EU regulations only required liberalisation for major industrial and commercial customers from February 1999. In addition, markets are less distorted by state aid to the manufacturing sector than in most EU countries. In contrast, reinforcing dynamism and efficiency by privatisation of state-owned companies has been limited, although the pace increased in 1998. Furthermore, market forces are very weak in the agricultural sector due to the EU's Common Agricultural Policy and the "transitional" national support programme (described in Box 9 in Chapter III).

Privatisation: higher proceeds in 1998

In 1998, privatisation proceeds reached a record of FIM 10 billion (1½ per cent of GDP) (Figure 16). The government sold 22.2 per cent of its stake in Sonera (the dominant telecommunications company) through the largest ever privatisation by public offering, followed by 19.9 per cent in Fortum (the energy company formed by the merger earlier in 1998 of Imatran Voima [IVO], the biggest electricity supplier, and Neste, the dominant oil and gas supplier). Moreover, two mergers watered down the state share holding. The merger of the state-owned Enso and the Swedish company Stora, creating the worlds' second biggest forest industry company, reduced the Finnish public holding in the company from 44 to 18 per cent. The merger between the partly state-owned company, Valmet, and the privately-owned Rauma, both in engineering and machinery, resulted in a drop in state ownership from 20 to 12 per cent.

The privatisations in 1998 were part of the pragmatic and gradual privatisation process that started in 1993 – a process that the new government wants to continue. The original 1992 authorisation by Parliament did not allow for the complete privatisation of any state-owned company, but the mandate was broadened gradually and currently includes mandates for several complete

Figure 16. **Privatisation proceeds**
Billion FIM



1. New share issues to strengthen the capital base of the companies.
Source: Ministry of Trade and Industry.

privatisations (Ministry of Trade and Industry, 1998). The privatisation programme sets minimum stakes ranging from controlling power (at least two-thirds of the shares and votes) or a simple majority in companies operating in sectors considered of vital importance, to a substantial minority stake (at least one-third of the shares and votes) in other companies, but also allows complete privatisations. The privatisation process was influenced by the specific situation of the companies concerned, such as the need for additional equity capital. Moreover, the stock market situation influenced its pace. For instance, the public offering of Fortum was postponed in 1998 due to the turmoil on the international financial markets.[30] Since 1993, the sale of public holdings has generated gross privatisation proceeds in excess of FIM 27 billion (on average ½ per cent of GDP per annum). The state retained FIM 19 billion while equity capital of the companies involved was increased by FIM 8 billion. Most of the net revenues, FIM 11 billion, was used to reduce government debt with the rest mainly channelled into additional venture capital for SMEs (FIM 1 billion) and in R&D support (FIM 5 billion).

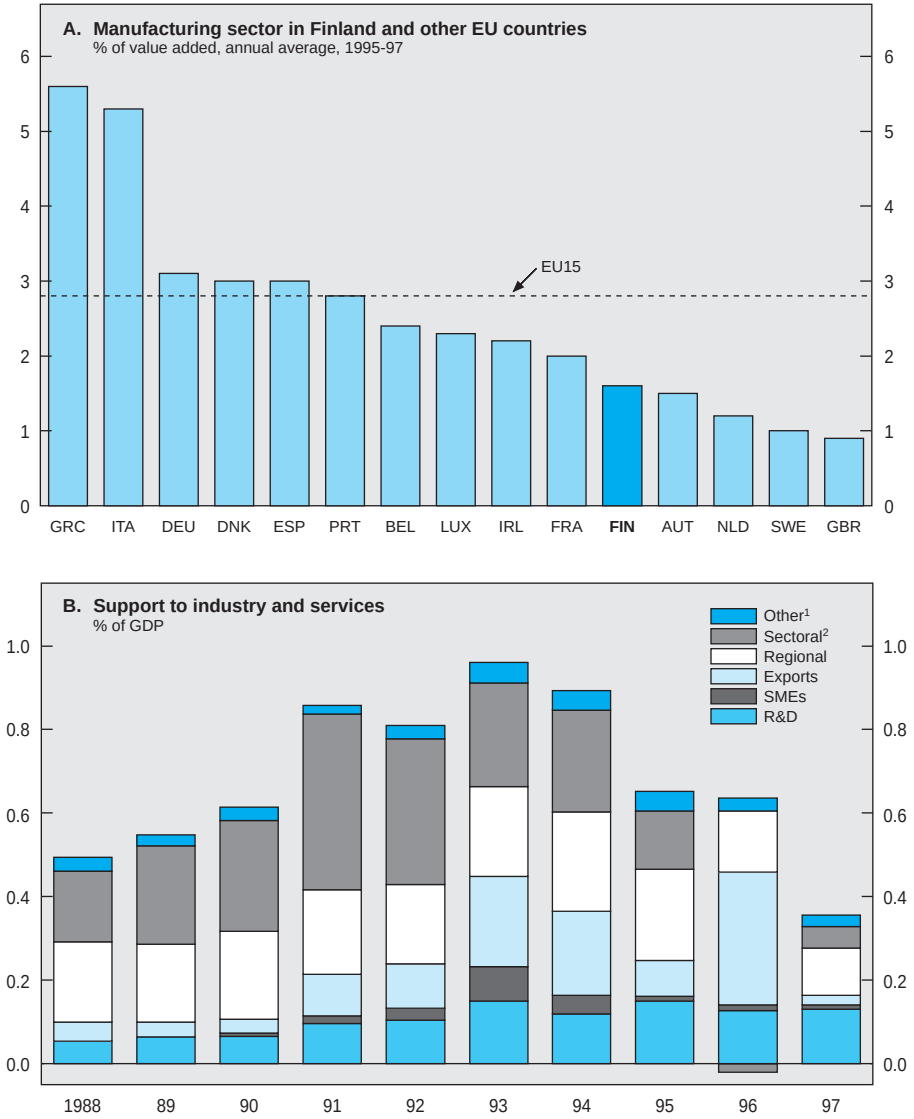
Up to now, privatisation has been only partial. The government privatised 13 companies completely, reduced its stake to less than one-third in four companies, but kept full control in 29 companies, a simple majority of shares in three major companies and a minority holding in three other important companies.[31] Only 20 per cent of the shares in stated-owned companies have been sold since 1992 while for over twenty companies, the government participation is still above the minimum stake authorised by Parliament. The potential proceeds of the unused privatisation mandates are estimated to be FIM 40 billion (6 per cent of GDP). Proceeds of complete privatisation of all public enterprises are estimated at FIM 90 billion (13 per cent of GDP). The new government announced that it will continue the policy of privatisation. This should enhance the efficiency of the Finnish business sector and improve corporate governance of the companies involved further. Full privatisation of currently partially privatised companies may make it easier for them to raise capital on favourable terms as domestic and foreign investors may be hesitant to buy shares in companies that have the government as major shareholder. Furthermore, mergers may be a necessary condition for reaching optimal scale. For instance, while the sale of a fifth of its stake in Sonera was a step in the right direction, it is relatively modest compared with the intended 49 per cent privatisation of Telenor/Telia, the company resulting from the merger of the Norwegian and Swedish state-owned telecommunication companies. The limited capacity of the Helsinki stock exchange to absorb share issues was in the past frequently mentioned by the Finnish authorities as a major reason for the slow pace of privatisation. However, with the euro now in place and the financial markets of the euro area countries further integrating, a much greater volume of share offerings should be possible. Moreover, there is probably still pent-up demand for Finnish shares as Finnish employment pension funds have been allowed to invest in shares only since 1997.

Subsidies to industry: below EU average

In 1997, state aid to the manufacturing sector amounted to 1.6 per cent of the value added of the sector, well below the EU average (Figure 17). Total aid to the industry and the service sectors declined gradually from the 1993 peak level of 1.0 per cent of GDP to slightly less than 0.4 per cent of GDP in 1997, dropping below the average level of 0.6 per cent of GDP in the latter half of the 1980s. Government support has not only dropped, but has also undergone a substantial change in orientation, from regional and sectoral objectives to horizontal objectives such as subsidising R&D, SMEs and export-oriented activities. In 1997, horizontal aid accounted for almost 55 per cent of total subsidies, compared with 30 per cent in 1991-92. The share of regional aid has hovered between 20 and 30 per cent over the 1990s, while the share of sectoral aid, the most distorting form of subsidisation, especially if it is given to specific companies, dropped from 45 per cent over the 1991-92 period to less than 15 per cent in 1997. It will probably drop considerably further in the coming years as shipbuilding subsidies, re-introduced in 1996 after the non-ratification of the OECD Shipbuilding Agreement, have to be phased out by 2001 under EU legislation. Currently, support to shipbuilding accounts for over 80 per cent of sectoral aid.

The rationale for horizontal state aid is to correct market failures affecting in particular small and medium-sized enterprises, especially in the area of technology diffusion. An important factor influencing technology diffusion is company spending on research and development (R&D). In recent years, the authorities took several initiatives to enhance R&D and to improve the interaction between science and industry. In September 1996, the government decided to use part of the privatisation proceeds to foster R&D and it set a target of R&D spending of 2.9 per cent of GDP by 1999 – one of the highest ratios in the OECD area. This goal was already reached in 1998. Total R&D spending increased annually by 15 per cent in nominal terms, with that of enterprises growing even more vigorously (18 per cent). In 1998, enterprises and government financed two-thirds and one-third of R&D outlays, respectively. Almost half of the public support came from privatisation proceeds. It is channelled through research councils of the Academy of Finland and through technology development centres (TEKES) of the Ministry of Trade and Industry. Evaluation, during and at the end of a programme, has been part of the process. In one case, an interim evaluation led to the premature halt of a programme. By the end of 1999, the performance of the research institutes will be evaluated. The excellent performance of the electronic equipment industry in recent years is mainly the result of successful business strategies and outstanding entrepreneurial skills, but the growing public support to R&D may have facilitated the growth of the sector. However, there is a question about what role the government should play in influencing technical progress. While it is commonly accepted that government can stimulate technical progress indirectly

Figure 17. **State aid**



1. Environment and energy.
2. Sectoral support in 1996 was FIM -123 million due to the withdrawal of State guarantees to shipbuilding.

Source: European Commission and Government Institute for Economic Research.

through the provision of a stable macroeconomic environment, non-distortive and low taxation, and good regulatory frameworks, there is less consensus about the positive effects of direct measures such as R&D subsidies (see also the previous *Survey*).

Financial support to SMEs totalled FIM 520 million (0.1 per cent of GDP) in 1998, of which more than 80 per cent were investment subsidies granted in particular to regional development and structural adjustment areas approved by the EU and partly co-funded by EU subsidies. In other parts of the country investment subsidies were limited to projects reinforcing the industrial base. Total aid to SMEs was somewhat larger than FIM 520 million as part of the R&D support is granted to SMEs. SMEs also benefited from the government support to venture capital and the cutting of red tape, using the possibilities of information technology.[32] For instance, the Ministry of Trade and Industry opened a so-called one-stop-shop Internet site, containing a wide range of application forms, allowing companies to complete and submit them in electronic form. Notification on company start-ups will be simplified in 2000 by combining the current separate notifications to the National Board of Patents and Registration and to the National Board of Taxation into one joint foundation notification. From 2000 onwards, businesses will be able to submit their auditors' reports to various authorities in a combined electronic form. Finally, SMEs benefit from the 15 Employment and Economic Development Centres operational since September 1997. The centres combine the regional units of three ministries: the Ministry of Trade and Industry, the Ministry of Agriculture and Forestry and the Ministry of Labour, thereby simplifying contact of companies with the various parts of the administration.

The authorities should continue to assess and evaluate state aid carefully. Within the OECD area, there are many examples of aid that led to the delay of inevitable restructuring. They should not hesitate to stop projects with poor results and should continue with streamlining support programmes. Opportunity costs such as the positive effects of lower taxes, should be included in these assessments. Also regional policy initiatives should be carefully evaluated. The Ministry of Trade and Industry has done some preliminary work on aid evaluation by making a benchmark study in co-operation with specialists from the UK and Ireland and by conducting some domestic studies on the general usefulness of state aids. In January 1998, the Act on the General Conditions of Business Subsidy entered into force. The Act stipulates that a business subsidy scheme shall primarily be aimed at the elimination of market failures, that business subsidy schemes should be of limited duration and that functioning of these schemes should be periodically assessed. The authorities should take care that the provisions of the Act are fully respected.

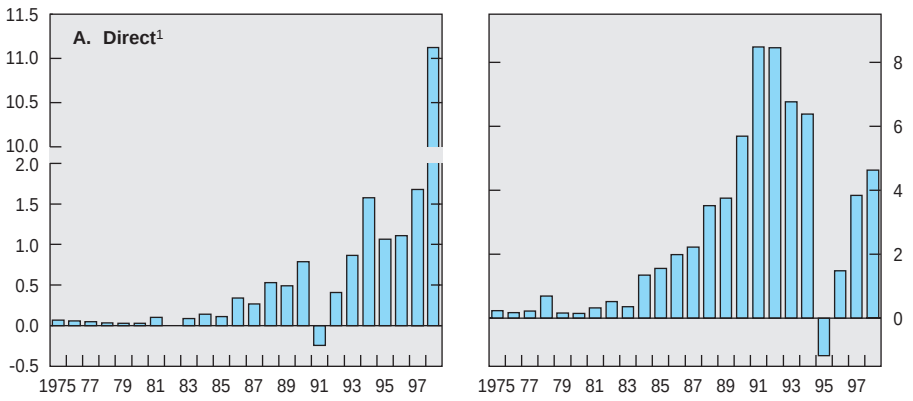
The Finnish economy has benefited from foreign ownership

In Finland, as in practically all other industrialised countries, company ownership has become more and more international. At present, one-third of the 500 largest Finnish companies is a foreign subsidiary or has at least 50 per cent of its shares owned by foreign investors (Ali-Yrkkö and Ylä-Anttila, 1999). The ten largest foreign subsidiaries employ around 6 per cent of the employees in the Finnish manufacturing sector. In 1997, foreign investment in the manufacturing sector increased by over 60 per cent and was equivalent to over half of all domestic manufacturing investment (Figure 18). Non-residential holdings of Finnish shares are around half of the total market capitalisation of the shares listed on the Helsinki stock exchange. For Nokia, the biggest listed company (representing slightly over 50 per cent of the total valuation of the Helsinki stock exchange), the share reached 77 per cent at the end of 1998. The economic results of foreign subsidiaries are better than those of domestically controlled companies (Table 10). Not only is the return on capital higher, but so is the ratio of value added to investment.

Finland's participation, first in the EU and later as the only Nordic country in the euro area, has probably had a positive impact on the inflow of foreign

Figure 18. **Foreign investment in Finland**

Billion FIM



1. The amount of foreign direct investment in 1998 is heavily influenced by two international mergers, Stora/Enso and Merita bank/Nordbanken, which had a substantial impact on gross direct investment flows.

Source: Bank of Finland and OECD Secretariat.

Table 10. **Domestic and foreign-owned companies**[1]

1996

	Domestic companies	Foreign-owned companies
Return on invested capital, %	16	23
As a per cent of value added		
Invested capital	140	100
Wage bill	50	55
Per employee, thousand FIM		
Value-added	394	377
Wage bill	195	210
As a per cent of turnover		
Investment	10	5
R&D	2	3
Debt	63	41
Ratio of exports to turnover (exporting companies)	32	42

1. The sample size of the research included 200 domestic and 108 foreign-owned companies.

Source: Pajarinen, M. and Ylä-Anttila P. (1998), "Ulkomaalaiset Yritykset Suomessa – Uhka vai uusi mahdollisuus?" (Foreign-owned companies in Finland – Threat or new possibility?), ETLA B:142, Taloustieto Oy, Helsinki.

capital. North American pension funds, for instance, probably would have invested less in Finnish stocks if Finland had not joined the third phase of the EMU. Another crucial factor was the change in product market policy, including regulatory reform, privatisation and deregulation – especially of the electricity and telecommunication markets. Moreover, during the past decade, virtually all restrictions on foreign direct investment have been abolished and replaced by simple notification procedures for administrative and statistical purposes.[33]

Foreign direct investment and foreign portfolio investment contributed to the recovery from the 1990-93 recession, and had a positive impact on productivity and employment. Moreover, there were positive spillover effects on the quality of management and on know-how. This investment also influenced positively the corporate governance of Finnish companies. Until the early 1990s, the main goal of Finnish corporations was robust turnover growth, while safeguarding existing jobs. Currently, more emphasis is put on raising shareholder value with positive effects on economic efficiency. Influenced by rate of return and pay-out ratios required by foreign investors, dividends paid by Finnish companies were over 40 per cent of total profits in 1998 compared with around 20 per cent in the mid-1990s.

In the Nordic countries, as elsewhere, industrial restructuring and consolidation will continue at a rapid pace in the coming years and will make ownership structures even more international. The Finnish authorities should keep their positive attitude towards this trend as it will enhance economic efficiency. Concerning state-owned companies, internationalisation should be pursued more actively.

Deregulation and competition policy

The Finnish government started relatively early with deregulation and progressed rapidly. However, alcoholic beverage distribution and railways are still heavily regulated, while postal services, although liberalised, have not yet attracted new entrants. Both the telecommunication and electricity markets are now fully liberalised. While electricity markets in the EU as a whole opened for competition for industrial and commercial customers using more than 40 000 MWh a year in February 1999, all Finnish households were already able to choose their electricity supplier in January 1997.[34] (The electricity market is also discussed in Box 5 in Chapter III.) Electricity prices, already low internationally, fell markedly (10-20 per cent) after the liberalisation. Since September 1998, small electricity consumers are no longer required to obtain an expensive hourly meter.[35] The liberalisation of the electricity market led to major restructuring, such as the creation at the end of 1996 of Fingrid, the national high-voltage transmission company, to remove possibilities for discrimination among market participants. In its recent review, the International Energy Agency praised the Finnish authorities on the very innovative way of introducing competition which should warrant a closer look by the international community (IEA, 1999). For instance, the design of the electricity transmission system gives proper price signals for grid expansion to the monopolist Fingrid.

The primary remaining problem concerning the energy and telecommunication markets is that the dominance of the major company could hamper a genuine level playing field. Both the 75 per cent state-owned Fortum and the 77 per cent state-owned Sonera dominate their market.[36] The European Commission, for instance, approved the Fortum merger only under the condition that it would reduce its stake in Gasum, Finland's only gas transportation company, to a minority stake. Also important were the amendments to the competition legislation in 1998 setting a maximum market share of 25 per cent for mergers of individual electricity suppliers. Moreover, as abuse of market power is probably no rarer for state-owned than for private companies, strong supervision is needed. In the recent past, Finnish competition authorities intervened in reaction to misuse in the wholesale electricity and wholesale gasoline markets. Recently, Fortum and the Swedish state-owned utility company, Vattenfal, have taken over several local distributors raising questions about the effects of increasing vertical

integration on competition. Investigative procedures into vertical integration of distribution and supply were initiated by the Finnish surveillance authorities.

Cross-border competition issues are under the supervision of the EU. In early 1999, the EU opened a legal proceeding concerning anti-competitive pricing of newsprint by major European companies that could lead to stiff fines for Finnish paper companies.[37] Furthermore, a reform of the Finnish competition authorities could strengthen market supervision. At the moment, the Finnish Competition Authority has the mandate to investigate infringements, but does not have the power to initiate proceedings or assess penalties. This power lies with the Competition Council whose functioning is undermined by the fact that it is mostly staffed by high-ranking public officials working on a part-time basis for the Council.[38] As already recommended in the 1998 *Economic Survey* on Finland (OECD, 1998*c*), it would be advisable, besides streamlining the organisational framework, to strengthen the staff and powers of the competition authorities.[39] Furthermore, the authorities could improve product market competition by making adequate information available. A register of all electricity retailers, accessible to all consumers, could, for instance, be introduced to provide greater price transparency.

Financial markets: banks complete payback of emergency loans

In 1998, all banks completed the payback of government support received (which started in 1996) during the banking crisis.[40] In the early 1990s, as in a few other OECD countries, banks' financial performance deteriorated to the point where the government had to support banks to preserve financial stability. The deterioration of bank balance sheets was particularly marked in Finland as bank loan losses surged due to the collapse in asset prices and the severe recession. The major factor behind the collapse was the financial deregulation which contributed to overheating of the economy, to excessive credit expansion and to the asset price bubble of the late 1980s. Delayed policy responses, the failure of banking supervision to enforce adequate internal risk-management by banks and moral hazard made the situation even worse. In contrast to the policy response in Norway, banks were not nationalised but were re-capitalised. The banking crisis was brought under control relatively quickly in Finland, with the total amount of public bank support escalating between 1991 and 1996 to almost FIM 57 billion, 10 per cent of GDP. In addition, public guarantees of FIM 32 billion were provided, bringing total support to around 16 per cent of GDP. Some of the bank support has already been recovered and more will be recovered in the coming years. According to calculations made by the Ministry of Finance, final costs of the banking crisis are expected to be FIM 45-48 billion (around 7 per cent of GDP). In reaction to the inadequate banking supervision, an independent

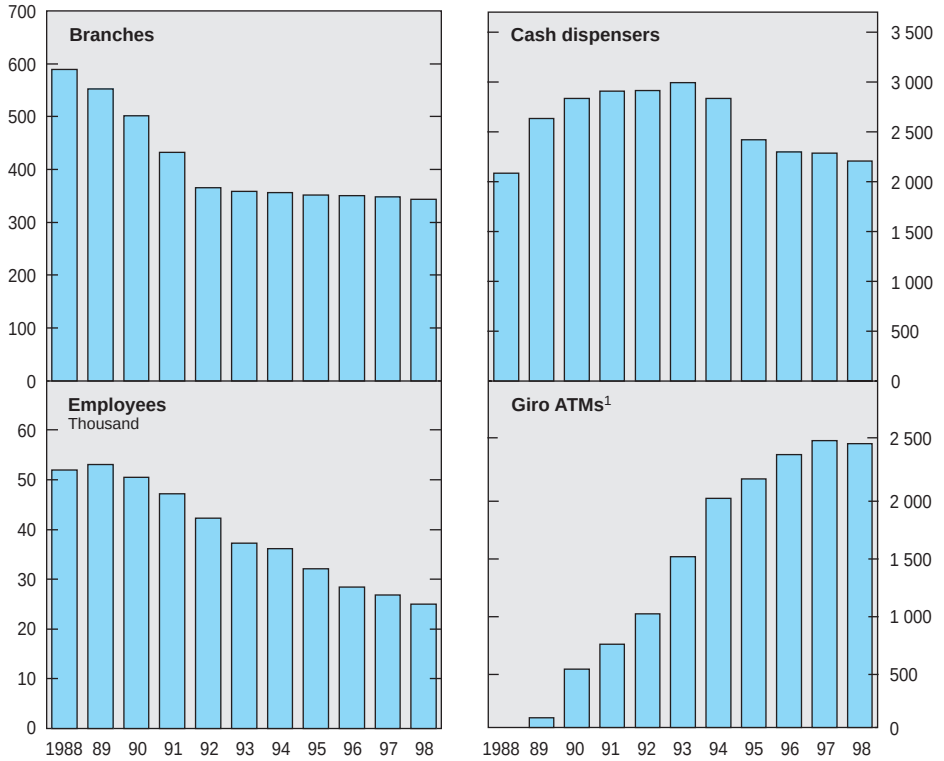
Financial Supervisory Authority was established in 1993 replacing the Banking Inspectorate within the Ministry of Finance which lacked resources and instruments to ensure sufficient banking supervision.[41]

Restructuring and mergers in the aftermath of the banking crisis led to significant changes in the banking sector in Finland with only three major domestic banking groups left.[42] The five major banking groups have a market share of 90 per cent. As a result, the scope for domestic mergers is practically exhausted, suggesting that future consolidation will be of a cross-border nature or between banks and insurance companies. The steady decline in lending margins indicates that concentration in the banking sector did not weaken competition. Market pressures forced banks to cut operating costs by reducing branch networks and staff numbers (Figure 19).[43] This was partly attributable to further financial market integration in the EU and a growing market penetration by foreign banks, although their market share has remained rather small. Due to cuts in operating costs, lower credit losses, profits from security trading and low deposit rates, total bank profits increased in 1998 by 26 per cent to FIM 8.2 billion (Figure 20).

In 1998, share prices on the Helsinki stock exchange rose by 69 per cent, the strongest rise in western Europe. Moreover, new issues doubled and turnover increased by 70 per cent to FIM 310 billion. Within the span of only a few years, the Helsinki stock exchange has become much more international, with half of the daily equity trading orders coming from abroad and around half of the shares of the listed companies in foreign hands (see also above). In November 1998, to strengthen its position, the Helsinki stock exchange signed a partnership deal with the Deutsche Börse AG and German-Swiss derivatives exchange, Eurex AG.[44]

The launch of the euro is likely to reinforce the existing trend of increased competition between banks in the EU, leading to narrower margins and a decline in fees (ECB, 1999*a* and 1999*b*). With profitability above the EU average and costs substantially reduced during the 1990s, Finnish banks are relatively well-positioned to react to the structural adjustment process in the EU banking system. In 1997, the return on equity was 15.2 per cent compared with 12.2 per cent in the 11 EU countries for which data are available. Recently, Finnish banks have benefited from the relatively low exposure to Asian and other crisis countries. Russian assets held were only 0.8 per cent of total gross foreign assets of Finnish banks in June 1998, while the share of Japan and southeast Asia was limited to 5.6 per cent (Bank of Finland, 1998). As with banks elsewhere, revenues are influenced negatively by the reduction in foreign exchange activity in currencies replaced by the euro and the costs of adjusting to the introduction of the euro. Currently, domestic loan losses are low due to solid balance sheets of Finnish enterprises and healthy financial positions of households. In 1997, loan loss provisions were 1 per cent of net income compared with 33 per cent in the 11 EU countries for which data are available (ECB, 1999*a*). In 1998, provisions were

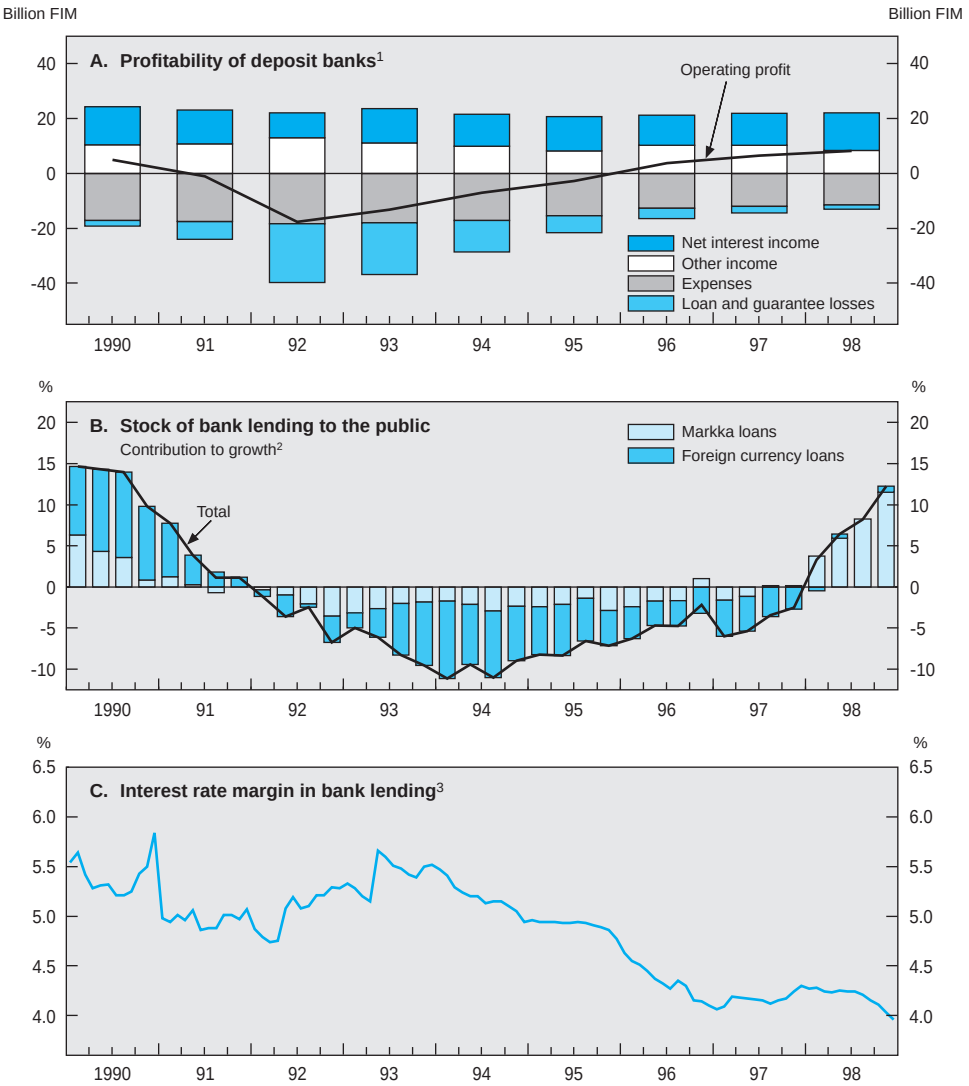
Figure 19. **Development of the banking sector**
Number of items



1. Automatic teller machines used for bank transactions such as transfers and payment of bills.
Source: Finnish Bankers' Association.

probably higher due to the Asian crisis. An advantage of Finnish banks that will probably be gradually eroded is the substantial level of low-cost resources. In 1997, interest income earned on low cost resources was 0.9 per cent of total assets, the highest ratio in the EU after Greece. The decision of the new government to abolish tax relief on bank accounts may accelerate the erosion.[45] A main challenge in the coming years for Finnish banks, as for many banks elsewhere, is to reach the scale necessary to compete in the international market place. Their competitive position will probably be eroded without greater geographical diversification.

Figure 20. **Bank profitability and credit behaviour**



1. Owing to the revision in 1995 of the accounting regulations the figures for 1995 onwards are not comparable with earlier years.
2. Contribution to changes in total bank lending to the public (as a per cent of total bank lending to the public in the same quarter of the previous year). Break in the first quarter of 1997.
3. Average rate on lending minus average deposit rate.

Source: Bank of Finland.

Medium-term public sector issues

The fiscal consolidation of the 1990s: mainly achieved through expenditure cuts

The fiscal consolidation since 1993 was almost completely achieved by lowering government outlays (Figure 21 and Table 11). Social transfers to households dropped most – mainly due to the fall in unemployment – but lower consumption and subsidies also contributed to the turnaround. Government investment was broadly unchanged, but had fallen sharply earlier during the recession. There was only a minor rise in the overall tax burden. Business taxes rose strongly, mainly as a consequence of the profit recovery and to a much lesser extent the increase in the corporate tax rate. However, higher business taxes were almost completely offset by lower income, indirect taxes and by lower social security contributions. Hefty cuts in central government transfers forced local governments to reduce outlays. As a consequence, consumption expenditure at the local level dropped more strongly than at the central level. Cuts were largest in health and education. However, local government employment increased by 6.7 per cent between 1993 and 1998, partly reflecting jobs created by municipal employment subsidies, while central government employment was unchanged (Table 12). Nevertheless, as substantial cuts were made during the recession, government employment in 1998 was 3.9 per cent below its 1991 peak level.

The OECD Secretariat's fiscal stance estimates suggest policy measures led to a drop in current outlays of 9 per cent of GDP between 1993 and 1998 while the economic recovery caused a drop of 1 per cent of GDP.[46] The recovery led to an increase in government receipts of 5.5 per cent of GDP – more than offset by policy measures reducing revenues by 7 per cent of GDP. However, these numbers still underestimate the role of fiscal consolidation, as part of the economic recovery is attributable to the budgetary measures taken.[47] Despite the substantial retrenchment measures and the impressive recovery, government outlays in 1998 were still significantly higher than before the 1990-93 recession and remained clearly above EU and OECD averages. Central government primary outlays, however, have returned to their pre-recession level.

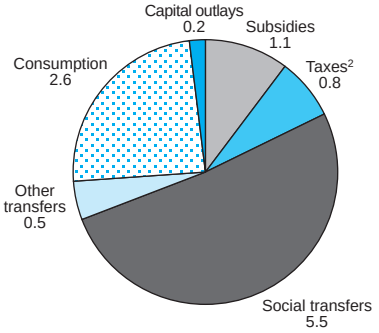
Medium-term policy challenges

A target of a general government surplus of 2.3 per cent of GDP in 2002 was set by the Finnish government in its first Stability Programme[48] published in September 1998.[49] In May 1998, the new government estimated the surplus in the range of 3 per cent of GDP for the period 1999-2003. The surplus target of the Stability Programme is the highest in the euro area,[50] but implies a broadly unchanged fiscal position from 1999 onwards. It is stricter than the

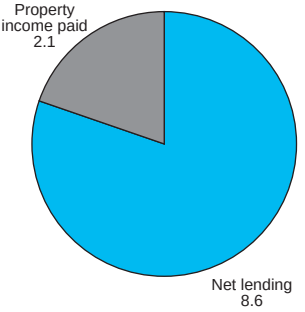
Figure 21. **Fiscal consolidation 1993-98**[1]
Change in percentage ratios to GDP

A. General government

Lower outlays and higher revenues led to...

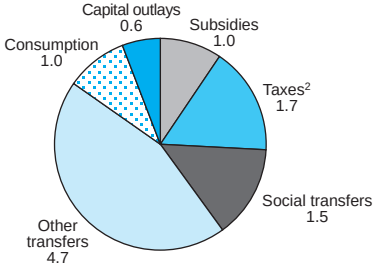


... an improvement in the budget balance despite higher interest payments

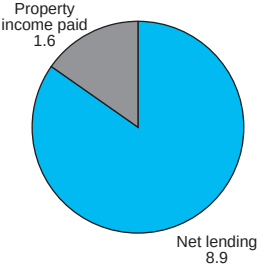


B. Central government

Lower outlays and higher revenues led to...

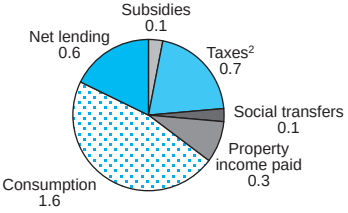


... an improvement in the budget balance despite higher interest payments

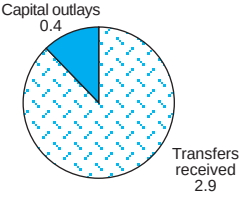


C. Local government

Lower outlays and higher revenues...



... offset drop in transfers received from central government



1. Fiscal consolidation by social security funds and pension funds is not shown. Both transfers from central government and social security premia received from employers dropped by 1.4 per cent of GDP. This was offset by lower transfers paid to households.
2. Including social security contributions.

Source: OECD Secretariat.

Table 11. **Public finances**

Per cent of GDP

	1991	1992	1993	1994	1995	1996	1997	1998
Expenditure	**52.8**	**57.6**	**59.0**	**57.6**	**55.1**	**54.5**	**51.9**	**49.1**
Revenue	**51.8**	**52.0**	**51.9**	**51.7**	**50.7**	**51.5**	**50.4**	**50.6**
Tax[1]	46.8	46.4	45.7	47.6	45.5	46.7	45.9	46.5
Other revenue[2]	4.9	5.7	6.2	4.1	5.1	4.8	4.4	4.1
Net lending	**–1.1**	**–5.5**	**–7.1**	**–5.8**	**–4.4**	**–3.0**	**–1.6**	**1.5**
of which:								
Central government	–4.5	–8.0	–10.5	–11.0	–9.4	–7.1	–4.1	–1.6
Local government	–0.4	–0.1	0.5	1.4	1.2	0.8	–0.3	0.0
Social security funds	3.8	2.5	2.8	3.7	3.8	3.3	2.8	3.1
Gross debt[3]	**23.0**	**41.5**	**58.0**	**59.6**	**58.1**	**57.8**	**54.9**	**49.6**
of which: Central government	16.9	34.0	51.9	59.0	64.0	67.3	66.3	62.2
Government bond yield,%[4]	11.9	12.1	8.8	9.1	8.8	7.1	6.0	4.8

1. Including social security contributions.
2. Property income and transfers received other than social security contributions.
3. EMU definition.
4. Ten-year government bond rate from 1993 onwards, five-year rate for earlier years.

Source: Statistics Finland; Bank of Finland and OECD Secretariat.

Table 12. **Government employment**

Thousand persons

	1990	1991	1992	1993	1994	1995	1996	1997	1998[1]
Government activities	575.2	581.5	565.4	533.7	536.1	541.8	546.2	556.4	558.6
Central government	149.3	152.2	151.9	146.2	144.9	144.7	141.3	139.4	146.2
Local government	417.1	420.5	404.7	378.8	382.4	388.4	396.2	408.3	404.2
Social security funds	8.8	8.8	8.8	8.7	8.8	8.7	8.7	8.7	..

1. Estimate.

Source: Statistics Finland (1999), *National Accounts 1990-1998* and OECD Secretariat.

minimum requirement of the Stability and Growth Pact (Box 3) of a medium-term budget objective of "close to balance", partly because the Finnish budget is more sensitive to economic fluctuations than the budgets of the other euro area countries.[51] Nevertheless, the target is probably sufficient to let automatic stabilisers work, and to prevent pro-cyclical budgetary measures during "normal" recessions without risking a breach of the 3 per cent deficit ceiling.[52] However, a medium-term objective of a somewhat bigger surplus could be considered, as it could facilitate adapting to the budgetary consequences of the ageing of the Finnish population in the coming decades.

Box 3. The Stability and Growth Pact

The Stability and Growth Pact clarifies the Maastricht Treaty's provision for dealing with "excessive deficits" and provides an institutional framework for its enforcement, in part through strengthened surveillance and co-ordination of economic policies *via* the annual review of national stability programmes. The Pact also calls on participants in the Economic and Monetary Union (EMU) and those countries with a derogation from initial participation in stage 3 of EMU – the actual introduction of a single monetary policy – to adopt budgetary balance (or even a surplus) as the medium-term objective.

The Pact considers a general government deficit above 3 per cent as excessive unless the European Council judges it to be temporary and exceptional when resulting from an unusual event or severe economic downturn. Temporariness implies that, according to the projections of the Commission, the deficit would fall beneath the 3 per cent threshold following the end of these events. In the case of an excessive budget, the Council will, on the recommendation of the Commission, propose a course of action for the country, which should be followed by effective measures within four months. The Council will monitor the measures and if they are found to be inappropriate it will make further more detailed proposals, which will be published. If corrective measures have not been implemented within ten months, sanctions will be imposed.* These would initially take the form of non-remunerated deposits, with a fixed component equal to 0.2 per cent of GDP and a variable component rising in line with the size of the excessive deficit. Such deposits are limited to a maximum of 0.5 per cent of GDP per year, but would accumulate each year until the excessive deficit is eliminated. Provided the excessive deficit is corrected within two years the deposits are returned to the country, otherwise the deposits will become non-refundable. In the event that the measures taken by a country are ineffective, the whole process will be restarted, but sanctions will be imposed within three months.

When a country is judged to be in recession, which is defined as an annual fall in real output (GDP) of at least 0.75 per cent, the Pact will be enforced in a differentiated manner. If economic output in a Member country declines by 2 per cent or more – and provided the deficit is temporary – exemption from the procedure is granted automatically. In the event GDP falls by between 0.75 per cent and 2 per cent, exemption can be granted in special circumstances by the Council. The country would need to convince the Council that the economic decline was "exceptional" in terms of its abruptness or in relation to past experience.

* Sanctions do not apply to those countries with an opt-out or derogation from participation in stage 3 of EMU.

For several reasons, bolder measures – especially on the expenditure side – than those incorporated in the Stability Programme for the period until 2003 are required. First, the income tax burden remains high in international comparison and constitutes an impediment to the continuation of strong employment growth. Moreover, although the support for the Nordic model of the welfare state is strong, there may be pressure from well-paid employees to cut income tax

rates to rates elsewhere. Second, the erosion of the indirect tax base, especially of excise taxes on alcohol and tobacco, may be stronger than expected. The derogation of EU regulations concerning the import of alcohol from other EU countries will expire in 2004.[53] In the Stability Programme, a negative EU-integration effect on indirect taxes of FIM 5 billion in 2002 is taken into account but the actual impact could be bigger.[54] Rising e-commerce could be one of the reasons for disappointing indirect tax revenues. In this respect, the Finnish Internet penetration (in 1998 the highest in the world) is important. Another reason for lower than currently projected indirect tax revenues could be the introduction within the EU of lower value added tax (VAT) rates on certain labour-intensive service sectors. Third, with the memory of the deep 1990-93 recession fading and the budget balance in hefty surplus, political pressures to increase government spending may become more intense. Fourth, the budget is insufficiently prepared for the expenditure and revenue consequences of ageing. Pension expenditures are projected to rise from 11.5 per cent of GDP currently to 15-19 per cent of GDP in 2050, depending on developments in productivity and labour force participation (Economic Council, 1998), while health expenditures could increase by 2 per cent of GDP to 6 per cent of GDP. Finally, the unemployment rate at the end of the Stability Programme projection period is still fairly high at 7.8 per cent in 2002 – only 1.2 percentage points lower than in 1999. Budgetary policy, in combination with structural policy initiatives, should aim at a better result.

The new government should review the policy options which would allow increased efficiency of the government sector by "doing more with less". There is considerable scope for efficiency improvements in local government as highlighted in a recent government report (Ministry of Interior Affairs, 1999). The report recommends, for instance, the outsourcing of services and the introduction of additional user charges to increase productivity. The new government should also explore possibilities to shift the tax burden from labour. In this context, it could consider higher property taxes, by raising the tax rate and reducing existing exemptions. The "greening" of taxation (Chapter III) may be another option. Both possibilities are envisaged in the programme of the new government. However, efficiency gains, higher user charges and tax shifts will not suffice and cuts in outlays, especially social transfers, will be needed to reduce the heavy tax burden (Table 13).

Table 13. **Recommandations for further structural reform**

Based on previous and current *Surveys* and action taken since early 1998

Previous *Survey*	Action taken	Current *Survey*
A. Reduce tax burden		
Reduce labour taxation to increase incentives to raise work effort.	Early 1999: tax scales except the top rate were lowered by 0.5 percentage point; earned-income deduction in the municipal income tax and the maximum standard deduction for work-related expenses in the central government income tax were raised.	Continue reducing income tax rates, especially marginal rates on low incomes.
B. Reform unemployment and related benefits		
a) Shorten maximum unemployment benefit duration to increase job search intensity.		Same as previous *Survey*.
b) Reduce benefit levels over time to reduce unemployment trap. Reduce the "income norm" to limit the means-tested cash benefits.	Early 1998: the indexation of benefits was scaled down from wage indexation to consumer price indexation.	Phase down unemployment benefits during the first 500 days of unemployment.
c) Enforce and tighten unemployment benefit eligibility criteria to increase incentives to look for a job.	Early 1998: eligibility for labour market support for young unskilled unemployed persons was made conditional on training; the waiting period was increased by 2 days to 7 working days.	Enforce strictly the eligibility criteria for unemployment benefits.
d)		Increase regional mobility of the unemployed.
e) Introduce medical re-examination of disability pension beneficiaries.		Same as previous *Survey*.
C. Reduce incentives for early retirement		
a) Reduce early withdrawal from labour force.	Early 1998: the minimum age for part-time retirement was lowered from 58 to 56 years.	Increase gradually the minimum age for unemployment pension.
b)		Reduce impediments for hiring older workers. Reduce, for instance, the contribution of the last employer to disability and unemployment pensions.

Table 13. **Recommandations for further structural reform** *(cont.)*

Based on previous and current *Surveys* and action taken since early 1998

Previous *Survey*	Action taken	Current *Survey*
D. Make active labour market programmes more effective		
a)	1998: number of job counsellors was increased; individual counselling of unemployed persons was improved; introduction of in-depth interviews with unemployed persons.	Pursue better job counselling, for instance, reach the target of one contact per month with every unemployed person.
b)	1998: active labour market policy shifted somewhat from subsidised jobs to training and job-search assistance.	Reduce the number of subsidised jobs offered by local government and continue to shift the focus from subsidised jobs to training and job counselling.
c)		Evaluate the effectiveness of all active labour market programmes and end programmes with poor results.
E. Increase wage and labour cost flexibility		
a) Reform employment protection and working time legislation needed to encourage job creation, especially by SMEs.		Same as previous *Survey.*
b) Review existing wage formation arrangements in order to enhance employability of low-skilled workers, in particular in private services.		Support greater wage differentiation in coming wage agreements, to promote the employability of low-skilled workers, while maintaining moderate average wage rises.
F. Enhance product market competition		
a) Examine scope for completing privatisation of partially privatised state enterprises to increase efficiency.	Record privatisation proceeds in 1998.	Same as previous *Survey.*
b) Streamline the number of support programmes in industry; continue efforts to reduce industrial support.		Assess state aid carefully and end programmes with poor results.
c) Continue efforts to reduce agriculture support.	Decrease in agricultural subsidies granted for the EU transition period.	Reduce agricultural subsidies further.
d) Strengthen competitive forces by reforming the Finnish competition authorities.		Same as previous *Survey.*
e)		Promote genuine level playing fields in the energy and telecommunication markets.

Table 13. **Recommandations for further structural reform** *(cont.)*

Based on previous and current *Surveys* and action taken since early 1998

Previous *Survey*	Action taken	Current *Survey*
G. Improve public sector efficiency		
a) Reduce government outlays to make income tax cuts possible.	Some additional spending restraint in 1998 and 1999.	Step up restraint to make sizeable income tax cuts possible.
b) Shift tax burden off labour.		Same as previous *Survey.*
c) Increase economic efficiency of local government services.		Same as previous *Survey.*
d)		Apply more user charges.
e) Move towards pensions based on career average earnings.		Same as previous *Survey.*

Source: OECD Secretariat.

III. Enhancing environmentally sustainable growth

Since the early 1990s, sustainable development issues have started to influence policy-making in Finland, as highlighted by the creation of the high-level National Commission for Sustainable Development in 1993. Recently, sustainable development has also been made a top priority area for the work of the OECD. The contribution of OECD Economic Surveys is focused on policies that have an impact on the environment and on natural resource use.[55] Environmental policies, and to a lesser extent resource management issues in forestry, and their interaction with sectoral policies are the major focus of this chapter. It builds on the work undertaken elsewhere in the Organisation.[56]

Finland has realised considerable progress in improving the environment in many areas. It is in a very good position to meet its international commitments for reducing sulphur dioxide (SO_2) and nitrogen oxide (NO_x) emissions. Some further measures are needed to reduce nutrient discharges in the Baltic Sea. Meeting the Kyoto targets for GHGs is the most challenging current issue. However, the decoupling of environmental pressures from output growth witnessed so far could become harder to achieve with continuing growth, putting a premium on ensuring cost-effectiveness of environmental and resource use policies. This will increase the importance of assessing the various abatement operations regularly and of making greater use of economic instruments. At the same time, a better design and integration of sectoral policies could often help to solve environmental and resource management problems at low, or negative, cost in terms of efficiency.

The major issues: an overview

A central focus of sustainable development is intergenerational equity: are future generations well-equipped to attain the same or a higher level of well being as the current one? The opportunities for future generations largely depend on the total stock of capital (wealth) they inherit. Here, the capital stock is

defined broadly, consisting of man-made physical, human and natural (resource and environmental) capital. "Social capital", the system of social values and institutions, is certainly also critical in this respect, and institutional aspects important for the setting of resource and environmental policy objectives and instruments are discussed in this chapter. A key issue concerning sustainable development is the degree of substitutability of the various forms of capital. If they were substitutes, a decline in natural capital, for instance, could be compensated by a rise in physical capital. However, some forms of natural capital can probably not be substituted beyond (often unknown) limits. Their depletion could lead to irreversibilities, or regeneration processes could be very slow (forest ecosystems, biodiversity, ozone layer, etc.). This could severely impair the welfare of future generations.

Unfortunately, total wealth is very difficult to measure, largely because many environmental resources are very difficult to value. It is possible, however, to trace the development of an important indicator of total wealth, potential output, that is driven by physical and human capital formation, and technical progress. Potential output in Finland has continued to grow, even though it has suffered from the fall in investment during the recession in the early 1990s and the upward drift in structural unemployment. Continued growth reflects, among other factors, that educational qualifications of the Finnish population have risen strongly and are among the highest in the world. Consistent official estimates of total wealth *per se* (human, physical and natural resources, such as forests) for the past and projections for the future, which would provide a better indicator on whether the economy follows a sustainable path, are not available so far.[57] Attempting to measure an even wider definition of wealth, in particular including environmental resources, runs into serious valuation problems. Thus, judgements concerning the sustainable use of environmental assets have to rely on skimming a large number of diverse indicators. Indicators of environmental pressures for Finland suggest that assets of this sort may have decreased until the 1970s, but are likely to have risen again (even though some important issues such as the eutrophication of the Baltic Sea and acidification of some lakes remain serious concerns for environmental policy).

Policies aiming at cost-effectiveness are vital to achieve sustainability as they allow faster wealth accumulation. The natural environment is largely a public good, available to all, despite its exhaustible character. This implies that policy needs to ensure that production and consumption decisions take into account their effects on the environment. Ideally, cost-effective policies would aim to correct market failures by properly pricing the use of natural capital while prompting cost minimisation by the individual economic actors. This would imply that policy decisions are based on cost-benefit analysis. In practice, it is often difficult to evaluate the benefits of environmental policies. There is often considerable uncertainty concerning the cost. Concerning the policy instruments, it is generally

agreed that economic instruments, such as taxes or permit trading schemes allow economic actors to search for the cheapest abatement method. However, they are little used in practice because of the transaction costs involved in establishing targeted tax or trading schemes. Of equal importance, economic policies in other domains can significantly affect the environment with agricultural, energy and transport policies being the most prominent examples. While policy goals often conflict, it is frequently possible to design better policies, which will reduce distortions, while keeping to the initial goals. For example, the aim of agricultural policy to raise the living standard of the farm community could be achieved in a way that reduces distortions in the use of natural resources by decoupling farm assistance from production.

Institutional framework

Finland has an extensive institutional framework for environmental management. The Ministry of the Environment is responsible for a large part of the environmental issues, while the Finnish Environment Institute, an agency of the ministry, monitors and assesses the implementation of environmental policies and provides information to the public. The Ministry of Agriculture and Forestry plays an important role in promoting sustainable forest management and deals with environmental problems linked to agriculture. Municipal environmental administrations were created in the late 1980s, and are taking increased responsibility in managing local problems, in particular regarding water.[58] They are also involved in land planning and environmental education, and since 1997, less commonly, in a campaign to reduce greenhouse gas (GHG) emissions in urban areas. Important regional bodies are the regional environment centres (supervised by the Ministry of the Environment), with a prominent role in the enforcement of environmental legislation (*e.g.* permit procedures) and in the collection of information on environmental issues, and the regional forestry centres (supervised by the Ministry of Agriculture and Forestry), with responsibility for biodiversity management among other roles.

Finnish environmental policy is not yet based on a comprehensive environmental law, but the legislative and regulatory framework for environmental protection has been considerably extended and upgraded throughout the 1990s, partly to harmonise with EU legislation.[59] The Waste Act, the Water Act, the Nature Conservation Act and several pieces of legislation on forestry address problems directly related to the environment. The Health Protection Act and the Building Act are also of relevance. A proposal has been made recently for a consolidated Environmental Protection Act which would replace the laws on air pollution control, noise abatement and environmental permit procedures. A system of integrated pollution prevention is in preparation that would rely on a case-

by-case approach and on emission standards. Such an integrated system would allow more cost-effective emission abatement if accompanied by systematic economic assessment.

Since 1993, a specific forum, the Finnish National Commission for Sustainable Development, exists for debate on policies that aim at sustainable development. Chaired by the Prime Minister, its composition is very broad, with participation of the Parliament, central public administration, local authorities, business representatives, labour unions, scientists and non-governmental organisations (NGOs). It is divided into five subcommittees[60] and operates through working groups (*i.e.* climate change, biodiversity, forestry, trade and the environment, development co-operation). It is also in charge of monitoring the implementation of the government's Programme for Sustainable Development which was approved in 1998. The National Commission plays a role mainly in raising awareness of, and giving political impetus to, sustainable development issues.

Specific programmes, such as the Action Programme on Transport, the Agri-Environmental Programme and the Environmental Programme for Forestry, have been prepared by interministerial working groups. The practical implementation of sustainability objectives lies with various administration bodies. The Ministry of Trade and Industry (concerning energy) and the Ministry of Transport and Communications deal with environmental matters in different departments, but they do not have specific environmental units. There is no environmental unit in the Ministry of Finance. A specific unit on sustainable development issues exists in the Ministry of the Environment, and several other departments carry out activities related to the integration of environmental concerns into agriculture, energy and transport policies. The integration between ministries is based mainly on expert level co-operation. However, economic assessments are not systematically pursued in the Ministry of the Environment. Such assessments, accompanied by co-ordination with other ministries, especially the Ministry of Finance, would be beneficial for the implementation of a set of comprehensive sustainable policies, especially where traditional economic instruments like taxes could be used, or where subsidies have adverse environmental side effects.

Efforts have been made to develop environmental and resource accounting. Forest accounts have been the pilot project, but data have also been collected on the various environmental issues and policies. The main focus has been to establish environmental accounts compatible with input-output accounts and analysis. The possibility of constructing a green national GDP was considered in the early 1990s, but has not been pursued mainly because of the strong uncertainty attached to valuation. A set of 99 sustainability indicators is currently being developed on various issues (*e.g.* loss of biodiversity, economic and social indicators). Based on the environmental statistics, an interministerial working group prepares a yearly review on "Finland's Natural Resources and the Environment"

which is published by Statistics Finland and the Ministry of Environment and is attached to the budget proposal. It provides a useful summary of sustainable development issues but does not yet play a strong role in the integration of sustainable development issues in the regular budget process. In 1998, the government decided to use this document as a tool to assess progress on sustainable development.[61]

Forestry policy aims at sustainable management of forests

Finland is endowed with very rich forest resources (Box 4). In recent years, the management of forests has evolved markedly from pursuing sustainable wood yield to a wider concept of sustainable development that includes long-term ecological concerns. Following the international conferences in Rio de Janeiro in 1992 and in Helsinki in 1993 (Second Ministerial Conference on the Protection of Forests in Europe), key new laws were approved: the Forest and Park Service Act (1994) which regulates the institution in charge of public forests; a new Forest Act (1997) that promotes "economically, ecologically and socially sustainable management" of forests; and the Act on the Financing of Sustainable Forestry (1997) which provides subsidies to forest projects that ensure the conservation of biodiversity. The commercial and conservation activities of the Finnish Forest and Park Service, which manages state-owned forests, were completely separated in 1998. At the same time, National Forest Programmes (in existence since the early 1960s) have broadened their focus to also include issues such as the recreational use of forests and environmental matters.[62]

During the two decades that followed the Second World War, a considerable effort was made to increase the production and export of wood and wood products, as this is an area in which Finland has a clear competitive advantage. Annual harvesting increased substantially during this period and at the end of the 1950s overtook annual growth so that the forest resources dwindled. At the end of the 1960s, there was a conscious effort to implement forest management methods that ensured rising wood production in the long run. Since then, forest growth has exceeded annual drain[63] and overharvesting is no longer an issue (Figure 22).

The pace of forest exploitation is affected by different incentives (subsidies and taxes on production, profits or land). Subsidies to forest activities have been reduced to very low levels, whereas the tax regime has become neutral with respect to that of other economic activities (Box 4). It might be argued, though, that forestry should be subject to an extra tax that recovers a scarcity rent accruing from the use of land. Such a tax is worth studying if rents are considered to be sizeable. However, its practical implementation could be difficult since it may not be easy to distinguish the scarcity rent from the return on capital. Indeed, owners

Box 4. The forest sector

A large share (86 per cent) of the Finnish land area is covered by forests – by far the highest proportion in the OECD. Forest activities (sylviculture, hunting, recreational activities, etc.) have always played a major role in the economy. The share of GDP of direct forest exploitation and the forest-related pulp and paper industries, though declining gradually, is at present still around 8 per cent (Table 14). Four to 5 per cent of the labour force works in the forest sector, and some of the most important Finnish multinationals are forest companies. The wood processing industry, which includes wood products and pulp and paper products, is an export-oriented industry. It exports 70 per cent of its production, mainly to European countries, and accounts for about 30 per cent of Finnish exports. At least 330 000 Finns own some forest land. In addition, the state owns about a third of the forestry surface, mainly in the less productive, northern part of Finland, which is partly dedicated to commercial use (Table 15). Following the old Nordic tradition of Every Man's Right, everyone has free access to all forests in the country. These two factors are reflected in a long-standing involvement of the population in forest matters.

The state does not intervene in the roundwood market. Until now, forest owner associations have negotiated a benchmark price for wood with the industry, which has been used as a reference by small owners in individual negotiations. However, the Finnish Competition Authority is reviewing the current system, and the reference price may be suppressed in the future. Since the share of income from wood production represents on average only 20 per cent of total income for forest owners, they may chose not to sell their production immediately, but can wait until they are satisfied with the price. Even though the paper and wood product sectors are highly concentrated,* there is little evidence for market power on the demand side. In fact, even if Finland is naturally endowed with very rich forest resources, the processing sector of the forest industry does not benefit from lower roundwood prices as compared with its closest international competitors. While the state does not intervene in the roundwood market, it promotes the use of wood as an energy source.

Taxation of forestry activities has been reformed recently and is now in line with most other types of economic activity, although forest land is exempted from real estate tax. Earlier, the forest sector was subject to a tax based on land area (a property tax), corrected by a coefficient which depended on the average productivity calculated for large regional areas (and not for individual slots), thus mimicking a production tax. In 1993, this system was replaced by a profit tax based on net income, with a voluntary transition period of 13 years. Under the old system, the tax was largely based on acreage instead of direct production. Furthermore, tax deductions were sizeable, which lowered the tax as compared to alternative uses of funds. The new method applies the standard corporate tax rate of 28 per cent and allows the same deductions as for the business sector in general.

* Three Finnish multinational groups control 80 per cent of sawn timber production and 98 per cent of pulp and paper production and own controlling participation in other plants in Europe. They are Stora Enso (partly state-owned), UPM-Kymmene and Metsäliitto. See OECD (1997*a*).

Table 14. **The forest sector in the economy**

As a per cent of total

	1990	1991	1992	1993	1994	1995	1996	1997
Value added[1]								
Forestry sector	7.5	5.8	6.5	7.4	8.2	9.2	7.4	7.8
Forestry and logging[2]	2.9	2.3	2.3	2.2	2.5	2.7	2.4	2.6
Manufactured products[3]	4.5	3.5	4.2	5.2	5.7	6.5	5.0	5.3
Exports								
Forestry sector[4]	37.7	38.0	36.3	34.5	34.2	33.7	29.0	29.0
Raw materials	8.5	7.7	7.4	7.2	8.2	7.3	5.7	6.3
Manufactured products	29.2	30.4	28.9	27.3	26.1	26.4	23.2	22.7
Employment								
Forestry sector	4.9	4.8	4.9	4.6	4.6	4.5	4.3	4.3
Forestry and logging[2]	1.5	1.5	1.4	1.4	1.2	1.2	1.2	1.1
Manufacturing	3.4	3.3	3.5	3.3	3.4	3.3	3.2	3.2

1. Gross at factor cost.
2. Includes related service activities.
3. Wood and wood products; pulp, paper and paper products.
4. SITC Revision 3 classification. Raw materials and manufactured products of cork, wood, pulp and waste paper.

Source: OECD (1998), *Foreign Trade by Commodities*; Statistics Finland (1999), *National Accounts 1990-1998* and *Bulletin of Statistics.*

Table 15. **Forest land ownership**

	Inventory period	Area 1 000 ha	% of area			
			Farmers and other private owners	Companies	State	Other
Whole country	1986-97	20 085	61.8	8.8	24.5	4.9
Southern Finland	1986-97	11 119	75.8	11.8	7.1	5.2
Northern Finland	1992-94	8 966	44.4	5.1	46.0	4.5

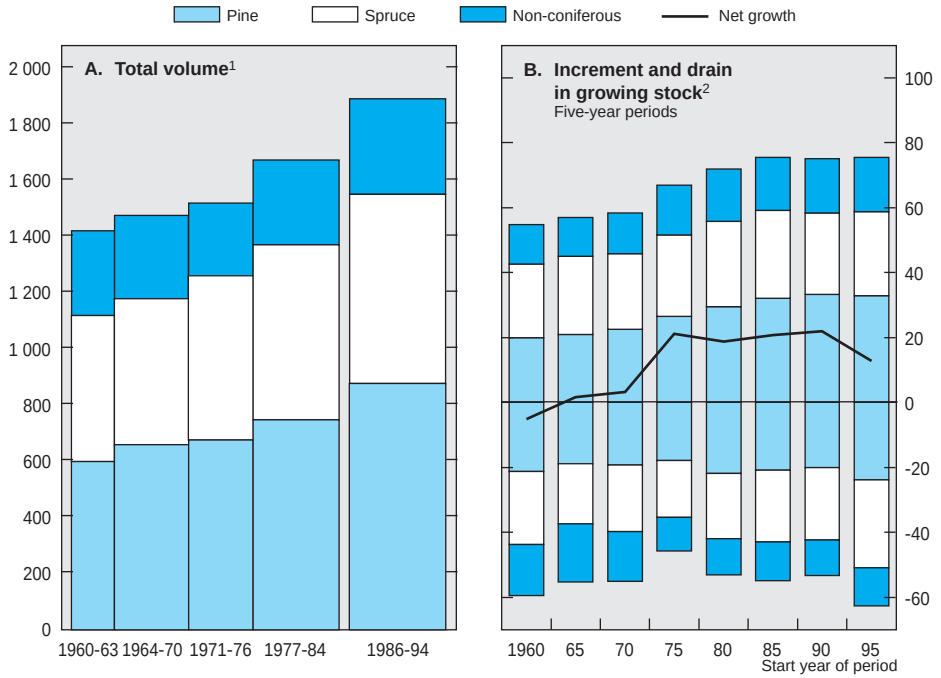
Source: Finnish Forest Research Institute, *Finnish Statistical Yearbook of Forestry 1998.*

have already paid a price for their properties that includes an estimated future stream of scarcity rents. In addition, forestry activities would be taxed more, while farming is heavily subsidised, with both activities usually carried out by the same entity.

While overharvesting is no longer an issue, there are important environmental issues relating to the past reduction in the area dedicated to old forests[64] and inadequate practices in commercial forests (especially in the south). Both elements directly affect biological diversity. Finnish forests lie mainly in the

Figure 22. **Use of forest resources**

Million cubic metres



1. The width of the bar indicates the period of the forest inventory.
2. Million cubic metres per year. Drain refers to the losses in growing stock due to fellings, sylvicultural measures and natural mortality. The data for 1995 covers the period 1995-97.

Source: Finnish Forest Research Institute, *Finnish Statistical Yearbook of Forestry 1998.*

boreal coniferous forest area. The genetic diversity of the species living there can be of particular importance. Four per cent of the species in Finland are classified either as endangered, threatened or in need of monitoring.[65] Lack of sufficient decaying wood, forest fires and fragmentation of large habitats are considered to be the three most important factors behind an increase in the number of endangered species and the deterioration of genetic diversity in wood areas. For instance, large investments in roundwood transport roads inside forests led to an excessive fragmentation of contiguous natural environments which are needed for many slow-adapting species that rely on large untouched habitats for their survival.

The rationale for some sort of public prevention to preserve biodiversity is quite compelling. From an economic point of view, there is a market failure since market prices for forest products do not reflect the risk of extinction and the "existence" and "option value"[66] of biological diversity. At the same time, the lack of a strong direct link between forestry practices and loss of biodiversity makes the use of taxes for the internalisation of such externalities rather difficult. For instance, the damage caused by building a road for wood transport may vary considerably from one area to another which makes a uniform tax on roads inefficient.

Several instruments are used to preserve biodiversity.[67] Purchases of forest land by the state are mainly directed at old forests, which are probably the most important biotopes from an international perspective.[68] The share of old forests in total forest land has decreased dramatically during the century and now accounts for less than 20 per cent. Since 1997, a considerable amount of private land has been purchased for nature conservation purposes. The funds for such purchases have almost doubled between 1996 and 1998, and most were dedicated to the acquisition of old forests (Table 16). It is expected that the area of protected old forests will increase from 9 000 ha in 1996 to 345 000 ha in 2007. Concerns about biodiversity have also surfaced in Finland with respect to roundwood imports from Russia, the source of around 13 per cent of the raw wood used by the industry. The sustainability of production of Russian providers, especially in the Carelia region, was cast into doubt during the 1980s. However, annual felling in Russia has decreased in the 1990s, and the total forest area is increasing again in the Carelia region. Concerns about the exploitation of old forests, and the lack of information on the origin of wood, have led the Finnish industry to establish a voluntary moratorium on imports from certain areas.

Recent government initiatives pay special attention to the regulation of private forests, accompanied by subsidies to cover part of the costs and information on better forestry practices. The Forest Act defines particularly significant

Table 16. **Spending on nature conservation**

Million FIM

	1994	1995	1996	1997	1998[1]
Purchase of land	161	183	184	321	362
Management of conservation areas	85	75	69	75	75
Compensation payments	20	19	19	43	47
Other	100	35	53	40	66

1. Budgeted.
Source: Statistics Finland.

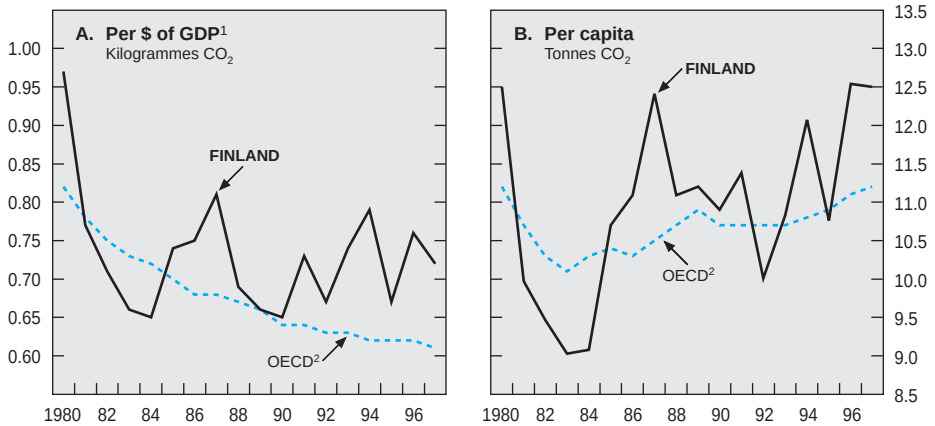
habitats in commercial forests where management has to be carried out in a way that retains their characteristics.[69] Regional forestry centres have a mandate to monitor the provisions of the Forest Act and courts can penalise the violation of the law. To compensate for income losses due to protection costs, the Act on Financing of Sustainable Forestry provides funding to private forest owners.[70] When land owners can demonstrate that environmental measures reduce the yield substantially, they may ask for compensation. Losses exceeding 4 per cent of the logging value or FIM 40 000 are fully compensated. No subsidy is granted for lower losses. Finally, information on environmental guidelines is disseminated by the 13 regional Forest Centres since it is considered that the spread of knowledge on ecological issues can be a cheap way to attain biodiversity goals.

In addition to government measures, a forest certification system is being implemented by all participants in the forest sector. Certification and eco-labelling is an information mechanism that ensures that preferences for "sound" ecological products are internalised. In principle, there should be no need for government intervention in the design of such a system, but given the high cost of implementing a narrowly-targeted certification procedure,[71] a direct financial contribution by the state may be appropriate. The Ministry of Agriculture and Forestry is indeed facilitating the set up of the system by partly funding its implementation.[72] A careful analysis of the implementation costs of different certification systems and the benefits from increased income and improved environmental efficiency is necessary to assess the appropriate level of government funding. This evaluation should be carried out in a comprehensive framework that takes into account all policy instruments available for attaining well-defined biodiversity goals. The cost-effectiveness of tools already in place (regulations and subsidises, land purchases, information and training) should be assessed and compared with that of more market-oriented tools (like certification and labelling, or taxes where they are feasible) to find an optimal policy mix.

Climate change policy

Climate change is the top environmental policy priority in Finland. According to the burden-sharing of the Kyoto target within the EU, Finland is committed to return to its 1990 national emission level between 2008 and 2012. Although this is less stringent than for the EU as a whole, the goal is nonetheless ambitious.[73] Energy intensity is high in Finland, due to a number of structural features. First, the cold climate requires significant energy consumption for heating. Second, despite the remarkable development of the electronic equipment industry in recent years, forest-based and metal industries and to a lesser extent the chemical industry (which are highly energy-intensive) still form the backbone of the Finnish industry, together accounting for more than 60 per cent of total

Figure 23. **CO_2 emission intensities**



1. Using 1991 prices and purchasing power parities.
2. Average excluding eastern Germany, Czech Republic, Hungary and Poland.

Source: IEA (1998), *CO_2 Emissions from Fuel Combustion;* OECD Environmental Indicators 1998 and OECD Secretariat.

manufacturing output. Third, transportation distances are long, due to the country's low population density. Hence, CO_2 emission intensities are high in Finland compared with the OECD average, and they show no decreasing trend in per capita terms while, in relation to GDP, the decrease has halted since the second half of the 1990s (Figure 23). At the same time, energy efficiency is already high in Finland, in large part due to a wide utilisation of co-generation in electricity production (Box 5). Abatement will, therefore, be more costly.

Indeed, the official baseline projection puts total GHG emissions in 2010 about 27 per cent higher than in 1990, the bulk of the increase coming from CO_2 emissions after 2000 (Figure 24). In 1997, despite the recession of the first half of the 1990s and a substantial decrease in methane (CH_4) emissions due to changes in waste management policies, Finnish GHG emissions were already 4 per cent higher than in 1990 (and CO_2 emissions 10 per cent higher). Man-made CO_2 emissions originate almost exclusively from energy production and from transport, while agriculture and waste are the major sources for CH_4 and N_2O emissions (Figure 25).[74]

Finland was the first country to introduce a CO_2 tax in 1990, initially with few exemptions for specific fuels or sectors.[75] Since then, however, energy taxation has been changed many times and substantially, from a low but "pure"

Box 5. Energy policy

The general objective of energy policy has been to ensure reliable supply of energy at competitive prices, in a country where energy intensity is high. The policy also aims at meeting environmental requirements. The energy supply is quite diversified and energy imports cover about 60 per cent of total use. During the past three decades, energy supply shifted away from oil towards coal, and more recently, peat, nuclear and natural gas. Finland is also importing a non-negligible share of its electricity supply. Industry accounts for about half of total energy use and its share is increasing over time, in contrast to the other OECD countries. The forest industry is the largest consumer, but produces a substantial share of its needs (more than 40 per cent) by using renewables (waste wood and other by-products). The share of wood and derived products in total energy supply is indeed the highest in Europe.

Direct government intervention to steer the choice of energy sources is scarce, apart from decisions regarding the use of nuclear power. In 1993, the Finnish Parliament overturned the Council of State's decision to construct a fifth nuclear power plant. However, economic instruments, both taxation and subsidies, have been used to improve energy efficiency and to favour the development of domestic energy sources such as peat and biomass.

Peat, in particular, as one of the few indigenous energy resources, was given substantial support in the form of R&D and investment subsidies, and tax exemptions. Finland's peat technology is highly advanced and allows the use of wood in the same boilers. Today, peat can compete with coal and sometimes oil in combined heat and power production in the sparsely populated central, eastern and northern parts of Finland which are remote from the gas grid and from ports for coal imports, but only due to the energy tax exemption (IEA, 1999). Peat production also plays a role in supporting employment in high-unemployment regions (harvesting involves 2 100 persons yearly). However, peat causes also a number of environmentally harmful emissions, including sulphur dioxide (SO_2), nitrogen oxides (NO_x), volatile organic compounds (VOCs) and, not least, carbon dioxide (CO_2). Peat emits the same amount of CO_2 as coal per unit of energy. Although peat fields also absorb greenhouse gases, CO_2 emissions from burning peat are included in the Kyoto Protocol.

In recent years, the electricity market has changed considerably. Following the United Kingdom and Norway, the Finnish electricity market has been gradually liberalised since 1995, and since 1997 Finland is effectively part of the Scandinavian electricity market with Denmark, Norway and Sweden. As a result of increased competition and improved overall efficiency, the price of electricity has been decreasing for large consumers and has started to do so for small consumers as well, for which competition was introduced in autumn 1998. This trend, which is expected to continue in the future, may result in an increase in electricity consumption and may have therefore important unintended side-effects on greenhouse gas emissions. In some cases, such as in the United Kingdom, electricity market liberalisation, combined with coal and gas supply reforms, led to a reduction in greenhouse gas emissions through a switching from coal to gas. In the case of countries relying to a non-negligible extent on non-fossil fuels, such as Finland, it may however, accelerate the shift towards a more fossil-fuel based electricity production. To avoid a surge in greenhouse gas emissions, it will probably become all the more important to have a policy in place, which internalises environ-

(continued on next page)

(continued)

mental externalities. There is already an electricity tax, but it is not differentiated by carbon content. On the other hand, more competitive markets, would have the advantage of responding more quickly to new environmental constraints (IEA, 1998).

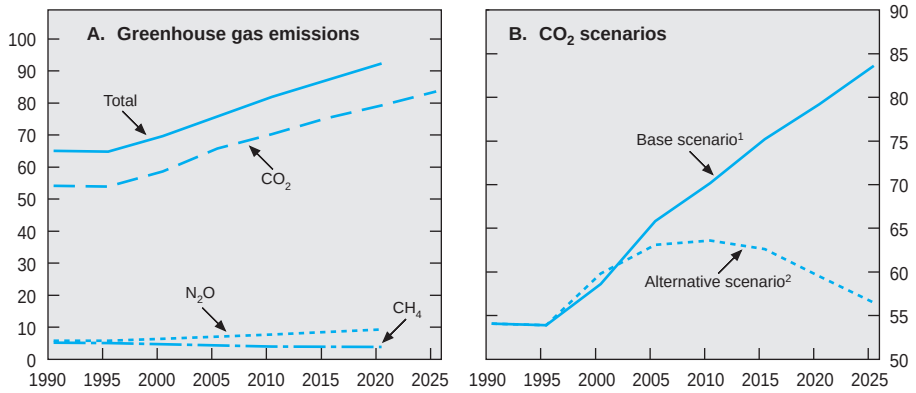
As noted in the Finnish Government Programme for Sustainable Development, the greatest challenge for Finland's national energy policy is to succeed in reducing trade-offs due to conflicting goals, in particular, the provision of energy at internationally competitive rates and the restriction of greenhouse gas emissions. Decisions have to be taken soon on future energy supply. The potential for increasing hydropower is very limited, and the development of renewable resources cannot be expected to meet the needs. Two main options are being considered by the government: the construction of a second gas pipeline, or the building of a fifth nuclear power plant. The nuclear option has the advantage of not increasing greenhouse gas emissions, but involves other important environmental issues. Moreover, there are questions about economic efficiency in the new competitive environment where large-scale, capital-intensive facilities are likely to lose market share.* One major concern related to the gas option is the reliability of supply. Today, Finland depends on one pipeline and a single supplier, the Russian Federation. The long-term objective is to create a more diverse gas market, with interconnection to the European grid, which requires major international decisions. One option, hooking on to the Nordic gas grid, would extend the gas grid to the west, while the other one would provide an alternative route for Russian gas to Western Europe. Which option is chosen (gas or nuclear), the additional capacity would not be operational before 2010.

* A 1998 study conducted jointly by the IEA and the OECD Nuclear Energy Agency concluded that, with a 10 per cent discount rate (which is a level generally used in the gas market), coal and gas are the cheapest option for a new plant to be built between 2005 and 2010. Nuclear was found to be the most costly option, even at a 5 per cent discount rate.

CO_2 tax to a much higher but much less CO_2-related tax. After a number of increases in the CO_2 tax rate, the first major change occurred in 1994, when an additional component based on the energy content of the fuels was introduced, as well as special taxes on nuclear and hydropower. Imported electricity was taxed at the average rate applied to domestically produced electricity. As noted by the Environmental Economics Committee (1994), which was set up to examine the efficacy of the various tax options to reduce emissions, the numerous changes were mainly motivated by fiscal rather than environmental concerns.

The second important revision of energy taxation took place in 1997, prompted by the opening of the Nordic electricity market. Domestic industries, and in particular the electricity sector, felt disadvantaged by the fact that energy-intensive sectors were exempted from the CO_2 tax in the other Nordic countries, some of which also have ample carbon-free power supply. In addition, electricity imports could not be taxed according to their carbon content. At the same time,

Figure 24. **Projections of greenhouse gas emissions**
Million tons CO_2-equivalents



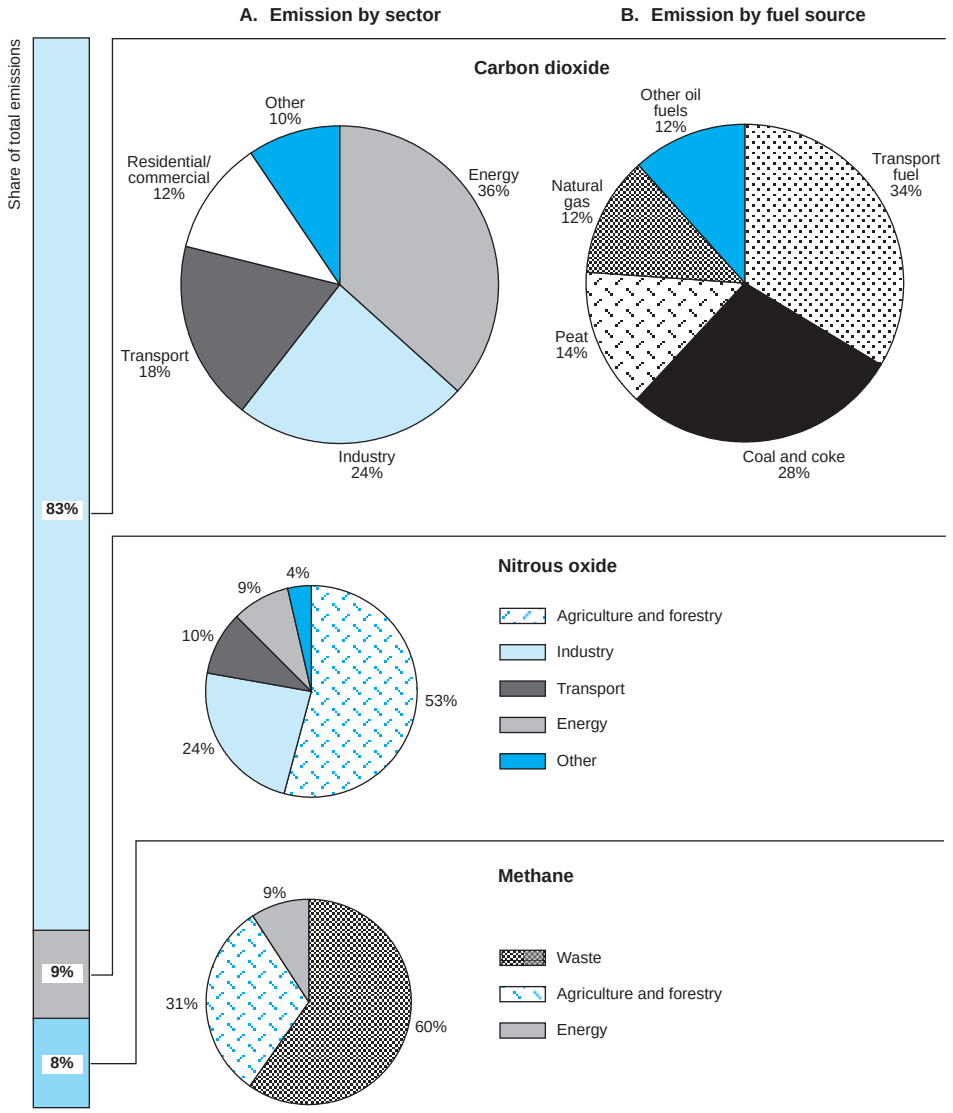
1. Scenario based on no change in current policy.
2. Policy scenario including increased use of gas and bioenergy for electricity generation combined with improvement in energy efficiency and higher taxation levels (see main text).

Source: Ministry of Trade and Industry.

the border tax on imported electricity was found to be out of line with the EU single market legislation. Therefore, to avoid harming the competitiveness of domestic industries, the carbon/energy tax based on fuel inputs in the electricity sector was scrapped and an electricity consumption tax was introduced, with a lower rate for industry and greenhouse cultivation (slightly above half the rate on households and service sectors). Source fuels for heating and transport continued to be taxed, but only on their carbon content, with a reduced rate for natural gas and peat. Since then, tax rates have been raised on several occasions and further exemptions added. In addition, energy excise duties paid by energy-intensive industries will be partly refunded on the condition of approval by the European Commission.[76]

The current system is rather complex and the link between the taxes and the carbon content in electricity generation is clearly looser than in the earlier system, thus providing little incentive for cost-effective abatement (Table 17). The electricity tax, geared to consumption, provides an incentive to save energy but not to reduce carbon emissions in electricity generation. Moreover, the incentive effect is much stronger for households than for industries, due to the differentiated rates for the electricity tax and the additional exemptions for industry – electricity represents more than 40 per cent of energy consumption. The link with

Figure 25. **Greenhouse gas emissions by gas and sector**
1995



Source : Ministry of the Environment (1999), "Finland's National Greenhouse Gas Inventory to the UN's Framework Convention on Climate Change", Helsinki; and Lehtilä and Tuhkanen (1999).

Table 17. **Energy taxation**

At 1 January 1999

	Unit[1]	Tax			Markka per tonne of CO_2	
		CO_2	Excise	Total	CO_2 tax	Total tax
Fuel						
Gasoline	Penni/litre	23.9	304	**328**	102	1 400
Diesel	Penni/litre	26.9	152	**179**	102	679
Light fuel oil	Penni/litre	27.0	10.9	**38**	102	144
Heavy fuel oil	Penni/kilogram	32.1	0	**32**	102	102
Natural gas	Markka/MWh	10.3	0	**10**	61	61
Coal	Markka/MWh	35.2	0	**35**	102	102
Peat	Markka/MWh	9.0	0	**9**	24	24
Electricity						
Industry	Penni/kWh			**2.5**		
Other	Penni/kWh			**4.1**		

1. MWh = Megawatt hour
kWh = Kilowatt hour
1 markka = 100 pennis

Source: Ministry of Trade and Industry.

the carbon content will also become loose for energy-intensive firms if the refunding scheme is approved by the Commission. Overall such a system, since it partially excludes the most energy-intensive industries and, therefore, obliges others to pay higher taxes to achieve the emission reductions, increases the overall costs of reducing CO_2 emissions. In addition, from a longer run perspective, the current CO_2 tax structure favours CO_2-intensive activities which conflicts with the objective of the tax (Box 6).

Apart from the CO_2 tax, a number of other measures affecting GHG emissions have been undertaken, although, as in other OECD countries, most of them were initially designed to achieve other objectives:

– To improve energy efficiency, a number of voluntary agreements have been signed recently with industries, covering now a substantial part of the sector. These agreements, which run until 2005, set fixed quantified targets for firms but not at the sectoral level. Further development of measurement methodology would be needed, however, to allow a proper monitoring of these agreements and to assess their effectiveness (IEA, 1999). Energy conservation is also promoted through subsidies to energy-saving investment (up to 30 per cent of the costs), and to energy auditing of buildings in the industry and the service sectors

Box 6. Competitiveness: a valid reason for suboptimal environmental taxation?

Competitiveness issues have always been a stumbling block to the implementation of a pure CO_2 tax. Indeed, in other countries applying a CO_2 tax (*e.g.* Sweden, Denmark, Norway) energy-intensive industries have been tax-exempt from the start. The arguments traditionally used against the unilateral implementation of a CO_2 tax applied to all sectors may appear even stronger in the case of Finland. Energy-intensive industries represent about two-thirds of manufacturing output and a substantial share of their output is exported to international markets characterised by strong price competition. In addition, compared with the main competitors on the electricity market, Norway and Sweden, the energy supply is much more carbon-intensive in Finland, and a high tax on coal-produced electricity would substantially reduce the competitiveness of Finnish electricity production. On the other hand, the tax would be very effective at low levels. Given the competitiveness concerns, the Finnish authorities have often underlined the need for harmonisation of energy taxes at the European level, favouring the introduction of energy taxes at a sufficiently high level to steer behaviour (Ministry of Trade and Industry, 1997).

It is unlikely that such a harmonisation will take place quickly. Two proposals have been made for an EU-wide energy taxation system, without agreement so far. The first one envisaged applying broad-based minimum tax rates on all fuels (except renewables) with a CO_2 component and an energy component. It was to be phased in progressively up to a level of US$10 per barrel-equivalent. Exemptions were proposed for energy-intensive industries. The proposal met strong opposition from most of the Member countries, and was abandoned in 1996. A new proposal by the Commission in 1997 included a minimum level of taxation instead of minimum tax rates *(i.e.* the tax base could be chosen by each country), and the taxation of electricity as a final product. Member states could choose the way the tax would be calculated and energy-intensive industries could be exempted (Bill, 1999). Compared with the current Finnish tax levels, the minimum tax levels are low, except for diesel. Overall, the Commission proposal is quite similar to the Finnish system, and would suffer from the same drawbacks.

The "competitiveness arguments" should be treated with caution. First, the CO_2 tax revenues could be used to reduce other distorting taxes (such as labour taxes), which would overall favour the competitiveness of non energy-intensive industries (Barker, 1999). Second, it is well known that the effects of a CO_2 tax on the competitiveness of energy-intensive sectors are small in comparison with other factors, such as the exchange rate or the wage rate. Given its natural resource endowment, Finland has definitely a comparative advantage in the forest industry, and would probably still have one even with increased energy costs. METLA(1998) discusses the implication of increased energy costs for the forest industry, concluding that it is not obvious that it would hamper the competitiveness of the industry. In general, the approach to competitiveness is too narrow. What matters in the end is the dynamic comparative advantage and, in the long run, as witnessed by the development of the electronic equipment industry, it may not be in metal or chemicals production, which account for around 20 per cent of manufacturing output in Finland. It is, indeed, paradoxical that energy taxation, introduced with the objective to reduce environmentally harmful

(continued on next page)

(continued)

effects and greenhouse gas emissions in particular, ends up providing incentives in the opposite direction, *i.e.* favouring energy-intensive sectors in relative terms.

In addition, the evidence for significant leakage effects (*i.e.* the displacement of emissions from countries applying environmental regulation to countries with no or less stringent regulation) is quite weak (Adams, 1997). It is true that the uncoordinated introduction of a CO_2 tax could lead to an undesirable switching of productive capacity, which would be costly to reverse, if other countries were to introduce a CO_2 tax later on. However, a gradual phasing-in of the tax (including, for instance, a temporary compensation scheme not related to emissions) could largely avoid such unnecessary costs by giving economic actors due time to adjust and provide sufficient time for the various national authorities to co-ordinate their climate change strategy.

Alternative ways to reduce greenhouse gas emissions by economic instruments are currently discussed in international fora. An alternative to EU-wide energy taxation would be the creation of an international market for greenhouse gas emission permits. It would reduce the overall costs of meeting the Kyoto target by allowing the emission reductions to take place where abatement costs are lowest. To the extent that emission quotas can be partly "grandfathered" (*i.e.* allocated in proportion to past emissions) a trading system would reduce the cost for the industries compared with a pure CO_2 tax, therefore making it more acceptable, while at the same time providing "correct" abatement incentives at the margin. The same could be achieved with a harmonised EU-wide CO_2 tax combined with a compensation scheme or tax credit which does not modify the incentives to reduce emissions. The burden sharing already takes into account differences in marginal abatement costs, but the harmonised tax would not guarantee that individual countries' targets would be reached.

(up to 50 per cent). Although this has not been evaluated, there is a risk that part of these subsidies finance investments that would have been undertaken even without subsidies, and that they represent to some extent a waste of taxpayer's money. Finally, the state is supporting R&D for energy conservation.

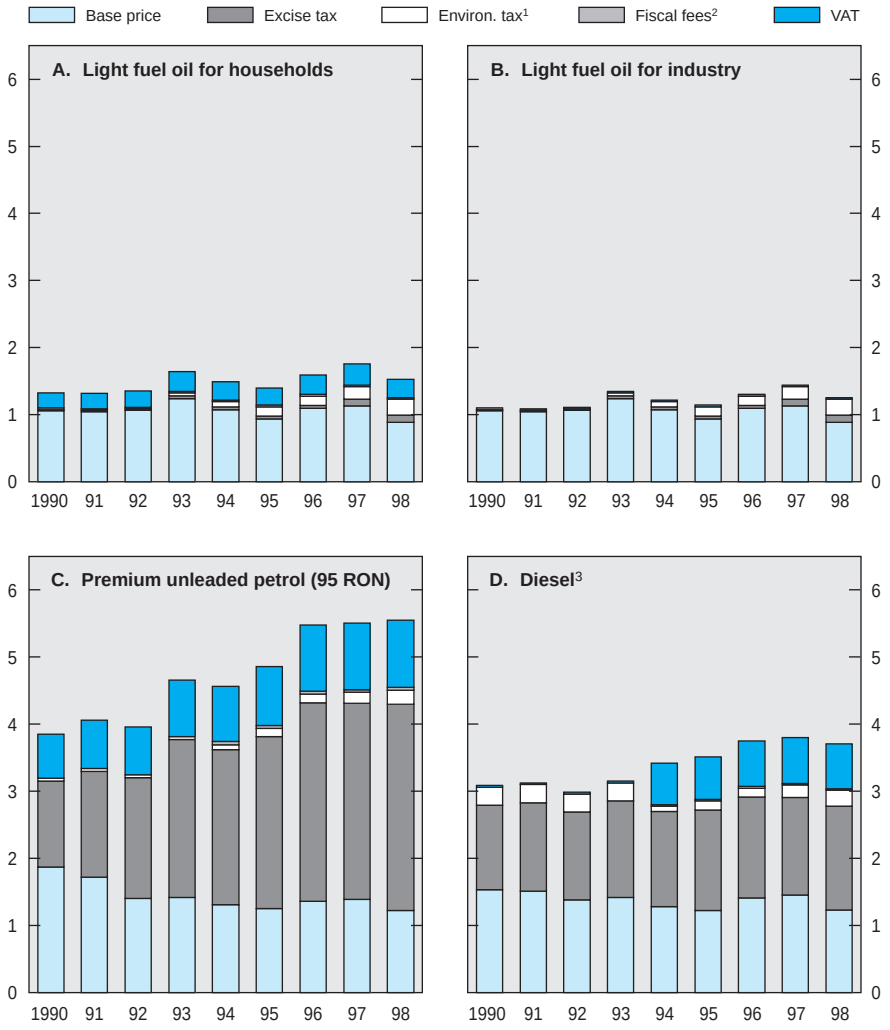
– To promote the development of indigenous energy sources, the government has also systematically supported the development of peat (which, as noted above, is not particularly "clean") and renewable energy sources, mainly wood. Wood combustion releases carbon, but if the wood is replaced, the CO_2 balance is neutral. CO_2 emissions from wood combustion are in fact not included in the Kyoto target. The following instruments are used: subsidies to investments in energy production based on these energy sources (up to 30 per cent of investment costs) and for the harvesting of energy wood; exemptions from the electricity tax; and funding of technological development related to these energy sources.[77] As for policies to improve energy efficiency, it is difficult to assess the effectiveness of this support.

– For transport, taxation has been the main instrument used. Transport fuels have been subject to the same CO_2 tax as other fuels. In addition, excise taxes, although they are generally motivated by fiscal rather than environmental objectives, are also an important steering factor. Taxes on gasoline have been raised in the 1990s, while those on diesel oil have been left almost unchanged (Figure 26). To date, the tax advantage for diesel compared with unleaded gasoline is the largest in the OECD (Figure 27). This is not in line with their relative pollution effect since CO_2 release is the same per litre of diesel and gasoline, while other environmental costs associated with diesel are higher than those from gasoline. A relatively high annual tax on diesel cars has prevented an increase in the number of private diesel cars. Overall, more weight is put on households (*i.e.* private cars) than on the transport of goods, reflecting, as in the case of energy taxes, concerns about competitiveness (Box 7). Taxes on the acquisition of vehicles and annual taxes are also important, amounting together to about 45 per cent of the revenues generated by the energy taxes in 1997, but they do not provide well-targeted incentives for emission reduction.
– Waste policy measures, which are mostly command and control, are also of importance for reducing CH_4 emissions. Increasing volumes of waste had been the main factor driving the increase in the past two decades. However, as a result of new regulation passed in 1993, the number of landfills has dropped and landfill management has improved. Recently, landfill gas recovery has also been made mandatory for new landfills, a move which – when combined with the use of the recovered gas for energy production – was found to be the most cost-effective waste treatment option to reduce GHG emissions (Pipatti and Wihersaari, 1998).
– Finally, as a result of agronomic measures to improve nitrogen fertilisation techniques and of reduced fertiliser use, N_2O emissions from agriculture have started to decrease (see below).[78]

Up to now, as in most other OECD countries, no radical decisions have been taken to mitigate GHGs. As noted in Box 5, the government is considering two main options for fuel switching (*i.e.* building an additional gas pipeline or a new nuclear power plant). Apart from that, the climate change strategy consists mainly in "doing more of what already exists". In the transport sector to control road traffic volumes, the government envisages adjusting taxation. One possibility is to grade the acquisition and/or annual tax on the environmental characteristics of the vehicles. However, to control CO_2 emissions an increase in taxes on fuels – the CO_2 tax as first best – would be more appropriate since it is directly linked to car use and emissions.

Figure 26. **Prices and taxes for fuel oil and gasoline**

FIM per litre

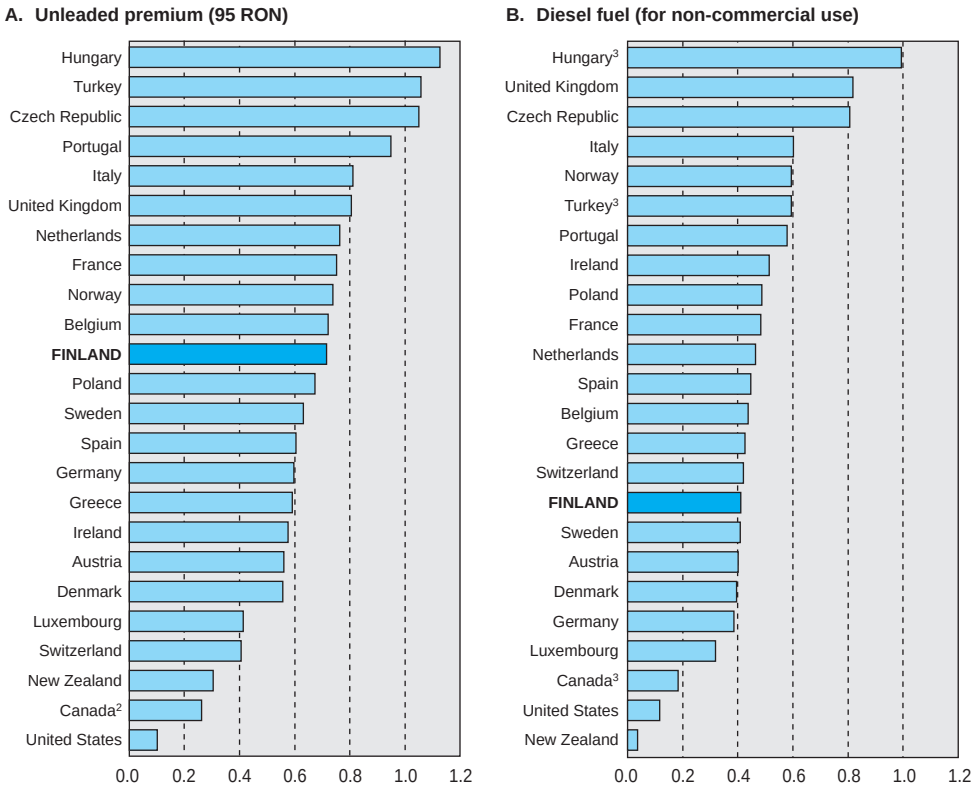


1. Environmental tax up to 1993 which was replaced by the energy/carbon dioxide tax from 1994 onwards.
2. Fiscal charges and fees covering a precautionary stock fee and an oil pollution fee.
3. Automotive diesel for non commercial use, sulphur-free quality as from mid 1994. The excise tax includes a sales tax (at the same rate as VAT) up to mid 1994, when the VAT rate was applied.

Source: IEA, *Energy Prices and Taxes.*

Figure 27. **Petrol taxes in international comparison**

Total taxes levied in 1998, US$ per litre[1]



1. Using current purchasing power parities.
2. 98 RON.
3. For commercial use.

Source: IEA, *Energy Prices and Taxes* and OECD, *Main Economic Indicators.*

One way of reducing carbon emissions is to store carbon in forests. Hence, land-use changes and forestry policies may be a complement to other mitigation policies. However, the Kyoto Protocol allows human-induced changes in take-up of GHGs by sinks to be treated as a credit only in a limited way. In practice, only afforestation (*i.e.* the human-induced extension of forest surfaces) since 1990 would be considered as a sink, and not the increment of the current growing wood stock (METLA, 1998). This is hardly an option for Finland since forests already cover 86 per cent of the country's surface. Moreover, given the

Box 7. Transport is emission-intensive

Transport volume has been growing considerably in Finland since 1970, spurred mainly by road traffic (passenger and freight). The economic recession in the first half of the 1990s has temporarily halted the rise, but the upward trend has resumed since 1996. On the other hand, public transport, mostly rail, had remained stable in the 1980s, but a substantial decline in rail freight traffic during the recession led to cuts in services, especially in low-density areas.[1] Rail, nevertheless, accounts for a market share of freight transport that is about double the European average of 14 per cent, and public transport has been recovering since 1996.

Transport is emission-intensive. Most difficult to tackle are CO_2 emissions. Although CO_2 emissions from transport still account for a smaller share of total CO_2 emissions than in other OECD countries, their importance has been growing quickly. Road traffic is the main culprit, accounting for 90 per cent of CO_2 emissions from transport, but emissions from air transport have been growing most rapidly. Emissions of acidifying gases, primarily NO_x for which transport is the main source, are causing important adverse environmental effects. They have been considerably reduced since 1990, however, with the adoption of emission standards and the progressive renewal of the vehicle stock. Local air pollution in urban areas from other exhaust gases such as carbon monoxide and of particulates is not an acute problem in Finland. Yet, threshold air quality values are exceeded on some occasions in the large towns. Finally, transport also contributes to other environmental problems such as local noise pollution, the fragmentation of ecosystems and landscapes, and waste. Official estimates are not yet available for emissions of the various gases per passenger kilometre or ton kilometre for the different transport modes, nor are abatement cost estimates.

Since 1994, Finland is implementing the Action Programme for Reducing the Adverse Effects of Transport on the Environment which covers all transport modes. Such a comprehensive approach is still quite rare in OECD countries. The plan fixes specific emission reduction targets for the year 2000, such as stabilising greenhouse gas emissions from the transport sector at their 1990 level and decreasing NO_x emissions by 30 per cent compared to 1980. These targets correspond to the national targets set for reducing these emissions, and do not reflect results of cost-benefit analysis. CO_2 emissions were still below the target in 1997, but they are projected to be 4 per cent higher in 2000 due to strong traffic growth. NO_x emissions from transport were 8 per cent lower in 1997 compared with 1980 and, according to official projections, the target should be reached with some delay around 2003.

Improving energy efficiency in the transport sector is one way of reducing emissions, but the Finnish authorities also aim at controlling traffic growth, at least for private cars. No CO_2 target has been adopted yet for 2010. Under current policies, road traffic is expected to increase by 27 per cent between 1997 and 2010, freight traffic being the component growing most rapidly, and CO_2 emissions would be 9 per cent higher than in 1990. There is no evidence on the relative abatement costs in the transport sector compared to other sectors of the economy, but unless they are much higher for transport, meeting the Kyoto target will require taking significant measures to reduce traffic growth. In this case, taxation will have to be increased and focused to a greater extent on fuel input. A radical shift from vehicle to fuel taxation both for cars

(continued on next page)

(continued)

and trucks would help in this respect. Vehicle taxes are very high in Finland compared with most other European countries, but they are not a well-targeted instrument and are probably slowing down the renewal of the vehicle park.

In line with the EU's approach, Finland has endorsed the principles of marginal cost pricing and full coverage of marginal social costs in the long run as part of its transport policy. Pricing transport at the marginal social costs, including the environmental costs, is necessary for the transport system to develop in a way which minimises the negative environmental effects. In principle, users should be charged operating costs, an efficient share of infrastructure costs, plus external costs (congestion, pollution and accidents). Ideally, the charges should vary according to time, place and condition. Environmental and other externality costs tend to be systematically under-covered in transport prices in most countries (ECMT, 1998).

Table 18. **Cost recovery in road and rail transport**

Per cent of total cost

	Marginal		Total	
	1990	1997	1990	1997
Road				
Lorries	42	93	33	63
Buses	38	96	28	69
Passenger cars and vans	62	163	82	162
Total	62	142	66	131
Rail				
Passenger	..	75	..	14
Freight	..	100	..	

Source: Ministry of Transport and Communication.

Cost recovery estimates including external costs (except for noise and congestion) are available for road and rail transport. They are calculated as the ratio of taxes and charges related to a given transport mode to costs and expenditure related to the same mode. Given the difficulty to value external – in particular, environmental – costs, the levels obtained in this type of calculation are uncertain, but in the Finnish case, the use of a common methodology allows interesting comparisons among the modes. The estimates show that marginal cost recovery for road transport has been substantially increased during the 1990s, and that overall, the charges imposed on road traffic are higher than marginal costs (Table 18). This tends to point to an inconsistency between the valuation of environmental costs used here and the one implicit in the Kyoto target. In any event, there is a sizeable difference in the treatment of private cars on the one hand and lorries and buses on the other, and this gap is increasing over time, largely reflecting the lower taxation of diesel fuel compared to

(continued on next page)

(continued)

gasoline.[2] In any case from an environmental point of view, lorries should be given the same incentives as private cars. This would help reducing the negative environmental impacts from transport.[3] Indeed, most of the projected increase in CO_2 emissions under current policies is coming from lorries and vans. Their share in NO_x emissions from road traffic is also increasing.

Concerning rail transport, relative prices are skewed in the opposite direction, since the marginal cost recovery ratio was estimated at 75 per cent for passenger transport and 100 per cent for freight in 1997. In the same year, with a 93 per cent marginal cost recovery ratio, road freight was charged slightly less than rail freight. Total cost recovery for rail is estimated at only 14 per cent. For shipping and air transport, cost recovery ratios including pollution costs have not been estimated. However, the fact that fuel for aviation is not taxed, as in most OECD countries, introduces a major distortion in favour of this transport mode. CO_2 emissions from international aviation are not included in the Kyoto Protocol. International negotiations to control these emissions are in progress.

Regarding rail, deregulation, which is lagging behind road freight liberalisation, may reduce rail prices and allow an increase in market share of rail freight, and favour the development of intermodal systems.

1. Support to public transport has been reduced in the first half of the 1990s (slightly more than one-third for urban transport), although much less than spending on transport infrastructure. Compared to other EU countries, the level of support is relatively low. The railway (operation) company, for instance, is profitable.
2. This gap between the cost recovery ratio for cars and trucks shows that the tax imposed on vehicles using diesel does not compensate for the lower taxation on diesel fuel for lorries. There is, however, uncertainty over cost allocation between cars and trucks.
3. A recent report from the Transport Ministry refers to a 1998 study by Räsänen which provides an elasticity of freight volume to diesel price of about 0.05. This figure is very low by international standards. If it is correct, this could imply that a command and control approach could be more cost-effective.

time profile of carbon sequestration in forests, afforestation is not relevant for the 2010 horizon. In any case, sinks have not been included in the EU burden-sharing mechanism.

The Finnish government has established scenarios for the effects of the two fuel switching options (gas and nuclear power) on CO_2 emissions in 2025. The two scenarios include a number of additional restraining factors, including higher world market energy prices, higher taxation, more stringent regulatory measures, and faster penetration of new, energy-saving technologies.[79] In both cases, however, CO_2 emissions in 2010 surpass the 1990 level (by 12 per cent with the gas option and 2 per cent with nuclear power). In an update by the Ministry of Trade and Industry, the growth in CO_2 emissions in the gas scenario has been revised upwards (by 6 percentage points to 18 per cent above the 1990 level). Hence, a strategy consisting of a combination of fuel switching with "doing more of what

exists already" is probably not sufficient for Finland to meet its Kyoto commitment.

According to a recent study, there is some evidence that abatement costs for CH_4 and N_2O emissions are lower than for CO_2 emissions, so that an integrated abatement strategy would allow substantial cost savings (Lehtilä and Tuhkanen, 1999).[80] In any case, the marginal abatement cost of meeting the Kyoto target in 2010 would be very high, in the range of FIM 200-300 per ton of CO_2 equivalent, according to official estimates.[81] There are no official estimates of the overall costs of meeting the Kyoto target nor of the potential ancillary benefits for other environmental issues associated with it. One option to reduce the costs would be to rely more heavily on imports of electricity than is the case today, but this is not considered by the government. With the liberalisation of the electricity market, the security margins for electricity supply have diminished, and they should continue to do so in the long run with the access to a wider liberalised European electricity market.[82] Given the high level of estimated marginal abatement costs, Finland would also strongly benefit from using the flexibility mechanisms included in the Kyoto Protocol. Some projects to reduce GHG emissions in other countries (joint implementation or clean development mechanisms) are already under study, but wider gains could be achieved by buying CO_2 emission quotas abroad, if that were possible. It depends, however, on decisions made within the EU and internationally.

It has been decided to appoint a committee to study the feasibility of a national trading system for emission quotas as well as the use of Kyoto flexibility mechanisms. This option would have some advantages. First, a partial grandfathering of emission quotas could make it more acceptable to industry than a pure CO_2 tax, whilst giving "correct" abatement incentives at the margin. Second, it would allow for a better control of the amount of emissions. Finally, it would make it easier to participate in an international (EU or broader) trading system as soon as it is created. It might also facilitate cutting domestic emissions whenever it becomes clear that the demand for quotas on the international market exceeds the supply. On the other hand, in a small country like Finland, there is a risk that unless they are linked to an international trading system, emission quota prices could be influenced by market power.

Transborder and local pollution problems

The Finnish territory is affected by a number of transborder problems, most prominently acidification of inland waters and soils and eutrophication in the Baltic Sea[83] (Box 8). These problems are dealt with within the framework of international treaties, namely the Sofia and Oslo Protocol for acidification and the Convention on the Protection of the Marine Environment in the Baltic Sea for

Box 8. Transborder pollution problems and co-operation with neighbouring countries

During 1991-97, the Finnish government spent FIM 440 million (about 2.5 per cent of overall government spending on the environment) on reducing pollution in neighbouring countries, covering 232 investment and 560 technical assistance projects. Co-operation on environmental issues between Finland and central and eastern European countries began in the mid-1970s under the Baltic Sea Convention and has been intensified since then through several bilateral and multilateral agreements, and through EU activities such as the Baltic Sea Region Initiative. As a follow-up, in 1998, the Baltic 21 Action Programme on Sustainable Development was adopted. In 1991, the Finnish Ministry of Environment set up the East Europe Project, with local and regional partners in neighbouring countries, particularly in north-western Russia, the Baltic states and Poland. The primary objective of the project is to reduce emissions in neighbouring areas of Finland and its focus has been on air and water protection as well as on waste management. Given the size of the problems, Finland does not have the financial capacity to resolve them alone.

Transborder air pollution, mainly SO_2 and particulates originating from smelters in the Russian Kola Peninsula, has caused environmental damage in the northern part of Finland. Due to the economic slump, capacity utilisation of the smelters has been low in recent years, resulting in a drop in SO_2 emissions from 660 000 tons in 1992 to 300 000 tons in 1994 which is still more than double the total Finnish emissions (140 000 tons in 1994). Finland has subsidised environmental investment projects to reduce SO_2 emissions of the Kostamuksha mining complex in Carelia, which emitted 48 000 tons in 1994, and affects neighbouring areas in eastern Finland. It also subsidised environmental investment projects at the Narva power plant in Estonia that had SO_2 emissions equal to the Finnish total.

The St. Petersburg area is responsible for 70 per cent of the nitrate and phosphorus load to the Gulf of Finland. The gulf receives large amounts of untreated waste water from St. Petersburg, as 30 per cent of municipal waste water from this area with 5 million inhabitants is untreated, while waste water from the Tallinn area in Estonia (0.5 million inhabitants) and from Helsinki (1 million inhabitants) is biologically and chemically treated. The Finnish government is supporting the renovation of the St. Petersburg sewerage system. Joint projects are also set up to reduce the pollution of the Gulf of Finland by the industry in the St. Petersburg area (industrial waste water and hazardous waste), by pulp and paper plants in Carelia, and by industrial facilities in Estonia.

In Finland, there is concern about the safety of nuclear power plants in adjacent areas. At Sosnovy Bor (Russia) and at Ignalina (Lithuania), Chernobyl-type reactors are used. Co-operation projects are under way to improve the reliability of these and other reactors in neighbouring countries. There is also concern about radioactive pollution by Soviet-Union nuclear submarines dumped in the Barents Sea. Finland has provided an advanced facility for treatment of liquid nuclear waste in this area.

eutrophication (Table 19). In addition, Finland is co-operating bilaterally with neighbouring countries, in particular Russia and Estonia. These agreements generally fix national targets, but the associated pollution problems are of a regional

Table 19. **Finnish policy targets for transborder environmental problems**[1]

Type of emission/ inputs	International targets	National targets
SO_2	Oslo Protocol (1994) • 80% reduction by 2000, base year 1980.	Council of State Decision-in-Principle (1991) • 80% reduction by the year 2000, base year 1980. EU directive on large combustion plants • 40% reduction in emissions by 1998 and 60% by 2003, base year 1980.
NO_x	Sofia Protocol (1988) • Stabilisation by 1994, base year 1987. Sofia Declaration • 30% reduction by 1998, base year 1980.	EU directive on large combustion plants • 30% reduction in emissions by 1998, base year 1980. Action Programme for transport • 30% reduction in emissions from transport in 2000, base year 1980.
VOCs	Geneva Protocol (1991) • 30% reduction by 1999, base year 1988.	
Nitrogen and phosphorus	Ministerial Declaration on Protection of the Marine Environment in the Baltic Sea (1988) • 50% reduction, base year 1987.	Water Protection Programme • 50% reduction in 2005 in discharges from agriculture and forestry compared with the average level in the first half of the 1990s. • 50% reduction in 2005 in discharges from industry, base year 1995. • 35% reduction in phosphorus discharges from waste water in urban areas compared with the average level in the first half of the 1990s, and 60% in nitrogen when it is the factor regulating eutrophication.

1. Only quantified targets are considered.
Source: OECD Secretariat.

nature. The design of policies is thus rather complex, since instruments have to be tailored to the damage caused in the different parts of the country.

Acidification

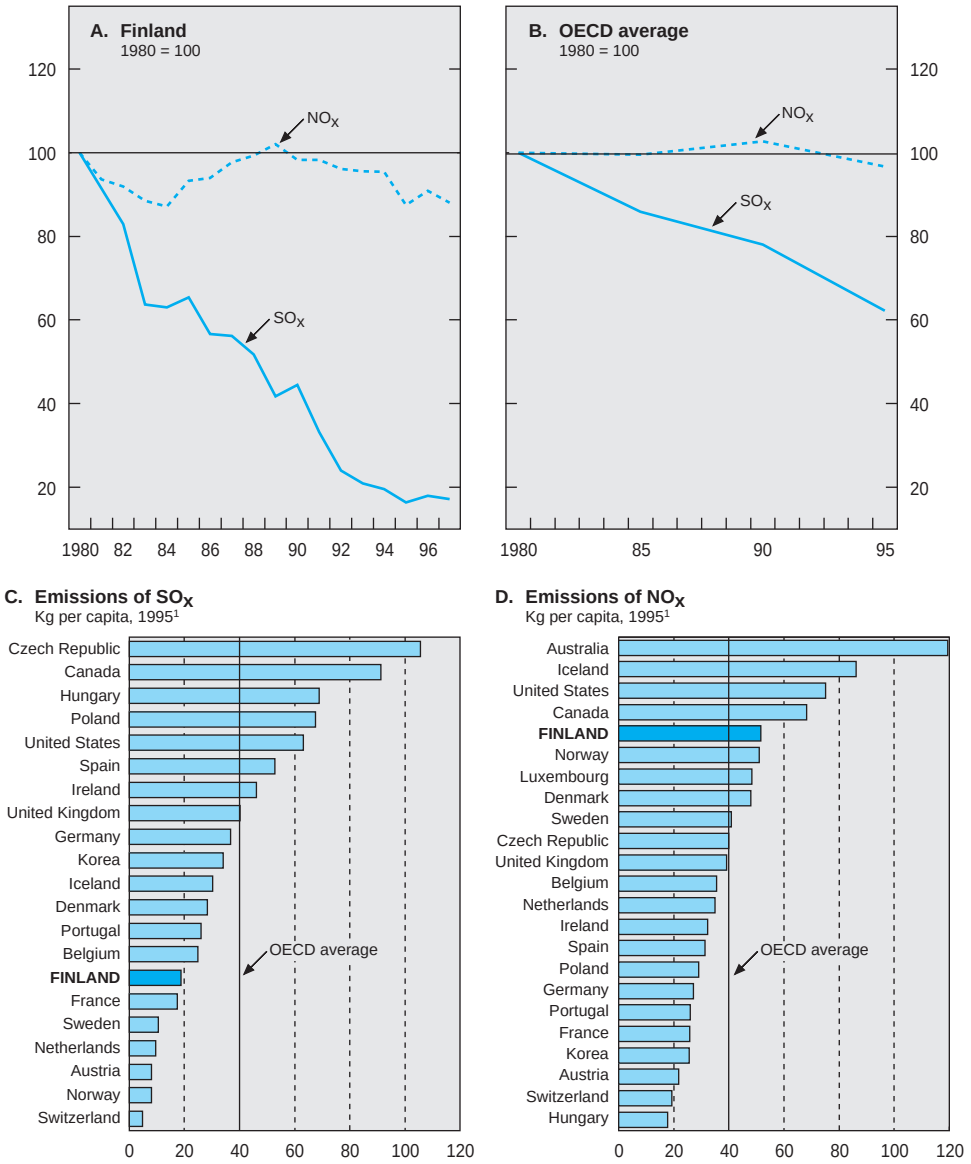
In the second half of the 1980s, the southern and central parts of Finland were suffering from acute acidification problems. Caused by atmospheric emissions of SO_2 and NO_x which through "acid rain", lead to deposition of sulphur and nitrogen in the soil and water surfaces, acidification triggered significant losses in fish populations in some lakes. The damage has built up at relatively low levels of pollution because Finnish soils and water have a low neutralising capacity (*i.e.* low

"critical loads"). The bulk of the acidifying depositions stem from long-range air transport, mainly from central Europe and Russia, which means that the improvement of the situation has come to a large extent from emission reductions abroad. Part of it results from the Russian crisis, and may therefore be reversed. Indeed, some 12 per cent of the sulphur deposition and 16 per cent of nitrogen deposition occurring in Finland comes from domestic sources. Finland is a net importer of SO_2, but a net exporter of NO_x.

For its part, Finland has been rather successful in meeting its international targets, in particular regarding SO_2 emissions. Indeed, the 80 per cent reduction target of the Oslo Protocol for the year 2000 was already reached in 1994, and in 1997 SO_2 emissions stood 83 per cent below their 1980 level (Figure 28). This reduction resulted from various factors including a change in the structure of energy production, a shift away from the use of heavy fuel oil in industry, a fall in the sulphur content of fuels and improvements in process technology. These were brought about mainly by command and control measures, in particular (non-tradable) emission permits. Permits have been granted at the provincial or municipal level based on the best available technology for the activity concerned and taking into account the local dimension.[84] Emission standards have also been established for some stationary sources such as coal-fired power plants, and fuel quality regulations specifying the maximum sulphur content of diesel and heavy fuel oil have been adopted. The only economic instrument targeted at reducing emissions from transport (*i.e.* less than 3 per cent of total emissions) is a lower rate of basic excise tax on sulphur-free diesel and reformulated gasoline.

It is difficult to assess the extent to which the reductions were achieved in a cost-effective manner. For large stationary sources, such as power stations, pulp and paper plants and refineries, emission reductions and necessary investments have been planned and abatement costs calculated by "Sulphur committees" set up in 1985 and 1991 by the Ministry of the Environment. Marginal abatement costs were not estimated, but the average abatement costs per ton of SO_2 vary between FIM 3 800 and FIM 26 000 among the various sectors, with a majority comprised between FIM 5 000 and 6 000. Given that there is a direct link between the sulphur content of fossil fuels and the amount of emissions generated, a tax on the sulphur content of fossil fuels, as implemented in Denmark, Norway and Sweden, would have been an appropriate instrument to enhance the cost-effectiveness of abatement.[85] Since SO_2 can be removed after combustion, the tax could have been made refundable on proof that SO_2 emissions had been reduced through the use of any technique. Emissions from industrial processes could have been monitored and also taxed, and the local dimension addressed by instruments such as regionally differentiated rates on the sulphur content of fuels. Another option would have been to establish a market for SO_2 emission quotas, as successfully implemented in the United States.

Figure 28. **Emissions of acidifying gases**



1. Or latest year available.

Source: Statistics Finland, *OECD Environmental Indicators 1997* and OECD Secretariat.

On the NO_x side, Finland has met its commitment in the Sofia Protocol to stabilise emissions in 1994 at their 1987 level, but did not meet the national target to reduce emissions by 30 per cent by 1998 compared with 1980, as emissions were only 8 per cent lower in 1997. Mainly due to high transport intensity, NO_x emission intensities have remained high in Finland, about 70 per cent higher than in the other European OECD countries (Figure 28). In contrast to SO_2 emissions, mobile sources account for a substantial share of total NO_x emissions (64 per cent in 1997) which makes the design of a cost-effective policy quite complex. Indeed, NO_x emission intensities depend on a combination of various factors and there is only a weak link between the amount of fuel consumed and the amount of emissions, so that a fuel tax would not be an efficient instrument. Moreover, the very high monitoring costs associated with a tax on emissions from many small mobile sources exclude this option. This means that a variety of instruments has to be used, adapted to the various processes underlying the NO_x emissions, seeking to equalise marginal abatement costs among the various sectors and sources.[86] As in most other countries, the main instrument for NO_x abatement has been vehicle emission standards for mobile sources, which have made catalytic converters necessary for new cars since 1991. To accelerate the renewal of the stock of cars, it has been complemented by a tax incentive – a tax allowance in the car tax for vehicles complying with those standards. Regarding emissions from stationary sources, the same instruments as for SO_2 have been used, namely permit procedures and emission standards for power plants.[87]

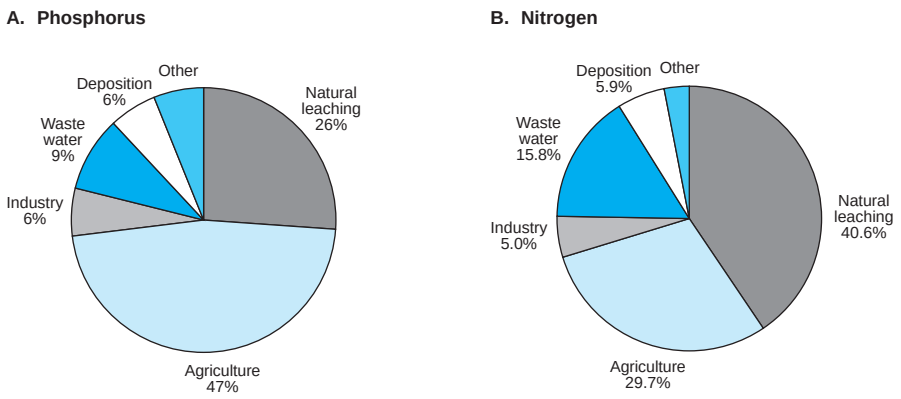
The situation regarding acidification has significantly improved in recent years, but problems remain in the southern part of the country where critical loads are still often exceeded. Hence, even if current policy measures are sufficient for Finland to meet the international targets, further reductions in acid depositions are required. According to the Acidification Committee (Ministry of the Environment, 1998), the cheapest domestic options for reducing SO_2 and NO_x emissions have already been used, and marginal abatement costs will increase sharply. To the extent that most depositions originate from outside Finland, the most cost-effective option would be for Finland to encourage abatement in neighbouring countries. Apart from the UN-ECE agreements, bilateral co-operation is taking place with Russia and Estonia (Box 8). However, reductions in domestic emissions in the south of the country will also probably be required. Decisions taken in connection with climate change mitigation policies regarding future energy sources will be of importance, since a shift from coal to gas (or nuclear power) will have ancillary benefits in terms of reduced SO_2 emissions.[88] Removing the favourable tax treatment for peat may also help reducing SO_2 (and to a lesser extent NO_x) emissions since emissions from peat have been growing quickly.[89] To contain NO_x emissions from power plants, a NO_x charge could be introduced to improve the cost-effectiveness of abatement. If the distributional effects of this charge should be mitigated, a compensation scheme could be

introduced. Such a system has proved quite effective in Sweden. The elimination of distortions in the transport sector should also help reducing NO_x emissions further (Box 7).

Eutrophication

Eutrophication is an important environmental problem for Finland, in particular in the Baltic Sea where intensive algae bloom episodes are frequent.[90] Resulting from nutrient enrichment (*i.e.* release of phosphorus and nitrogen), the excessive growth of plants damages aquatic life by reducing light penetration and consuming the oxygen in water when the algae die. It is, therefore, affecting aquatic biodiversity. The Finnish contribution to the nutrient runoff in the Baltic Sea is only about 10 per cent. In the Gulf of Finland, the most eutrophic part of the Baltic Sea, 80 per cent of the load is coming from the St. Petersburg area. Domestic emissions have nevertheless a major impact on coastal areas. Agriculture is the single largest source of anthropogenic nutrient discharges, with municipal and industrial waste water accounting for most of the rest (Figure 29). In the Convention for the Protection of the Baltic Sea, Finland had committed itself to reduce its nutrient loading by 50 per cent in 1995 compared to 1987 levels, a target which was reached for phosphorus but not for nitrogen, due mainly to an insufficient reduction in loading from agriculture. Since nitrogen is the controlling

Figure 29. **Sources of nutrient loading**
1995



Source: Statistics Finland.

factor in marine water eutrophication, this is where policy effort should focus in the coming years.[91]

Up to now, policy has been rather successful in reducing pollution from point sources using two main instruments: emission permits issued by the Water Courts based on the best available technology, taking into account the state of receiving waters, and water charges. Water is an abundant resource in Finland, and industries extract and discharge almost all the water they use directly rather than through the public system. The industrial sector, especially the pulp and paper industry – by far the main water polluting industry in Finland – has achieved significant reductions in nutrient discharges, well above the targets set by the Water Protection Programme for 1989-95.[92] However, it is estimated that that there is no room left to realise reductions at low cost, and that further increases in production will be associated with rising emissions (Statistics Finland, 1998*a*). Investments for water protection have been growing sharply in recent years, accounting for about half of total investments in the pulp and paper industry.[93] Water protection charges are imposed on heavy polluters on a case-by-case basis, but the amounts at stake are very small (FIM 2 million per year), so that they provide little incentive for further reductions.[94] In municipal waste water treatment, substantial decreases in phosphorus discharges have been achieved, but little improvement has been made in this sector regarding nitrogen removal (OECD, 1997*b*).[95] Municipal water charges are covering the economic (both investment and current) costs of water supply and sewage treatment, but there has been no attempt to cover the environmental costs associated with water pollution.[96]

Policies have been less successful in reducing nutrient discharges from diffuse sources, mainly agriculture.[97] Until 1994, a fertiliser tax based on the phosphorus and nitrogen content was applied to limit use.[98] The purpose of the tax was not strictly environmental since its revenues were earmarked to finance export subsidies for agricultural products. Together with specific agronomic measures, the tax helped to decrease significantly the use of phosphorus fertilisers and to a lesser extent nitrogen fertilisers, but not enough to meet the targets of the Water Protection Programme.[99] With EU membership in 1995, export subsidies were eliminated and, in the face of the deep adjustment that the sector had to undergo, there was strong pressure to reduce costs and the tax was removed (Box 9). At the same time, the agri-environmental aid system – half financed by the EU – was introduced, which provides subsidies conditional on compliance with certain environmental requirements including limits on the use of fertilisers and manure, and land management measures to reduce nutrient run-off. From the point of view of public finances, however, replacing a tax by a subsidy to achieve

Box 9. Finnish agriculture and the environment

Agricultural land covers 8 per cent of Finland's area. It is mainly concentrated in the southern and western parts of the country where the soils are richest, but dairy farming, which accounts for about half of total agricultural production, is also practised in central Finland. Farmers are generally involved in forestry activities, but have also "non-primary" activities. Only one-third of farms are run on a full-time basis. As in other OECD countries, the rural population has been declining over the last decades, and the share of primary production in rural employment has been falling. However, agriculture is still very significant in the economic activity of many rural areas, and forms an important element of regional policy.

The Finnish agricultural sector has undergone significant adjustment in recent years due to the adoption of the EU's Common Agricultural Policy. In 1995, the total producer support estimate stood at 69 per cent, exceeding the already high EU average by 20 percentage points. With EU membership, trade quotas were removed and replaced by EU tariffs. Exposure to EU competition resulted in an instantaneous halving of agricultural producer prices in 1995. To compensate for the loss in indirect support as a result of lower prices, "transitional" national support was implemented, but there was nevertheless a significant fall in farm income. Since 1995, the number of farms and employment has fallen significantly (by 10 per cent between 1995 and 1997).

Agriculture negatively affects the environment in many ways, but the most obvious direct adverse environmental effect is the runoff of nutrients, *inter alia* from fertilisers into lakes, rivers, aquifers and the sea. In 1995, agriculture was responsible for more than 60 per cent of anthropogenic phosphorus load, and about 50 per cent of the nitrogen load, and its share is increasing as pollution from point sources is reduced. The main resulting problem is eutrophication and the associated loss in aquatic biodiversity, but also impaired water quality in general. Fur farming, concentrated mainly in one western region (Ostrobothnia), is also causing elevated groundwater nitrate content. Pesticides do not pose a problem for water quality in Finland since they were never used intensively. Agriculture also causes emissions of gases contributing to acidification (mainly ammonia) and to the greenhouse effect (30 per cent of anthropogenic methane emissions come from agriculture). Finally, agriculture affects terrestrial biodiversity through the reduction in the number of habitats for wild species, and because of a specialisation of production and more uniform production practices. On the other hand, there are also some positive environmental effects of agriculture, like preserving a variety of flora and fauna, creating scenic landscapes, favouring land conservation including landslide and flood prevention (OECD, 1997*c*).

Some progress has been made over the last years in reducing these adverse environmental impacts. First, the agricultural support has progressively shifted away from market support towards direct income support (market price support represented 54 per cent of total support for EU countries in 1997, against 71 per cent in Finland in 1994). Second, an increasing share of the support (20 per cent in 1997) is tied to environmental requirements. About 90 per cent of the farmers have joined voluntary agri-environmental programmes, whereby they get financial support and advisory services in exchange for respecting limits on fertiliser use, establishing protective zones between arable land and water, etc. Support for organic farming has also been growing, and the government also provides support to farm households that

(continued on next page)

(continued)

diversify their earnings through other rural activities. As a result, fertiliser use has been reduced and associated agronomic practices improved. Between 1986-88 and 1994-96, the nitrogen balance (*i.e.* the nitrogen not taken up by the plants that results in runoff) has decreased by 20 per cent. However, in specific regions, levels have remained high and damage significant. Methane and ammonia emissions have been decreasing over the last years due to a reduction in the number of animals.

Most agricultural subsidisation has remained in the form of "coupled" support measures, which stimulates inputs and production. In any event, despite a strong decrease in support to agriculture (by almost 50 per cent between 1995 and 1997), the level of national support is still very high (amounting to more than 50 per cent of total agricultural value added). In addition to the climatic conditions which imply a short growing season and low yields, Finnish agriculture suffers from structural problems – mainly the small size of farms. Finland, as a country with a relatively clean environment, may have some comparative advantage in high quality agricultural products, in particular organic farming. Shifting some support to these activities would probably be a less costly and distorting way to integrate economic, regional and environmental objectives. In general, however, there is a conflict between agricultural, regional and environmental objectives: lower support would lead to a decline in the agricultural sector, and of the associated pollution. In this respect, the reform of the EU common agricultural policy (Agenda 2000), if it were to reduce the overall level of support substantially, could be expected to reduce environmental problems.

the same objective is quite inefficient. The amounts involved in environmentally related support are high, FIM 1.6 billion in 1997 (or 40 per cent of total government expenditure on environmental protection). It is estimated that about half of the support compensates for losses in the income associated with environmental management (Statistics Finland, 1998*a*). As a result, the use of fertiliser decreased between 1995 and 1997, while output grew at the same time. However, it is not yet possible to assess the environmental effects.[100]

New discharge reduction targets by sector have been established for the 1999-2005 period, ranging from 30 to 50 per cent. Investments are planned to install nitrogen removal facilities for municipal waste water treatment and the government is financing technical research in this domain. In agriculture, a reform of the subsidy eligibility criteria is being considered. Command and control measures have been rather effective in reducing point discharges to date, but they are probably less suited for the next stage of pollution control. The measures providing relatively "easy" reductions in point discharges are probably exhausted, and the clean-up or prevention costs per unit of pollution are probably rising. Hence, uniform targets are becoming increasingly inefficient compared to economic instruments such as water pollution charges. In addition, the conflict

between support to agriculture and water protection poses a major problem for realising cost-effectiveness.

Assessment and scope for further action

Finland has developed an extensive institutional framework for environmental management, with the Ministry of the Environment responsible for a large part of the environmental issues and various regional bodies implementing environmental policies and providing information to the public. The National Commission for Sustainable Development, chaired by the Prime Minister, has played a useful role as a forum of debate and in increasing the awareness of sustainable development issues. However, a weakness in the government's approach to sustainable development is the lack of systematic economic assessment. Such assessments by the Ministry of the Environment, accompanied by co-ordination with other key ministries, especially the Ministry of Finance, would be beneficial for the implementation of a set of comprehensive sustainable policies, especially where traditional economic instruments, like taxes or subsidies, are, or could be, involved. Moreover, environmental matters will need to be further integrated in sectoral policies and in the overall budgetary framework.

While overharvesting was an issue until the 1960s, the forest area has grown since then. Forest activities are no longer solely focused on sustainable wood production as practices are influenced now by wider sustainability objectives. However, protecting biodiversity and safeguarding old forests, especially in southern Finland, is on the policy agenda. The recent reform and extension of the legislative framework for forestry management is important for improving the performance in the coming years. Furthermore, the certification system is a tool that could lead to internalisation of customers' preferences for "sound" ecological products through the market. Some financial support of the government may be appropriate to start up such a system. Insofar as certification does not provide a solution, direct regulation, subsidies and public purchases of forests should be used, but in a cost-effective way. At present, however, economic assessment of the various measures for preserving biodiversity is practically absent and priorities are not clearly established. More systematic and comprehensive valuation of the costs and benefits of these measures would allow a better prioritisation in this field. The system of biodiversity indicators currently being developed could play a useful role in this process.

In many fields, pollution has been reduced. The extension and reform of legislation was important in this context. Air and water pollution, by the pulp and paper industry for instance, diminished substantially in the past decade due to stricter legislation and voluntary efforts by the industry. Finland has been successful in particular in reducing emissions contributing to acidification and

eutrophication. Since a substantial part of harmful depositions is imported from abroad, Finland's active involvement in the negotiation and implementation of related international agreements was clearly in the country's best interest. Finland is in a very good position to meet its international commitments regarding SO_2 and NO_x emissions. Regarding nutrient discharges, the date to meet the reduction target has been postponed internationally. To some extent, the reductions in the acidification problem in the southern parts of the country, and to a lesser extent in eutrophication in the Baltic Sea, result from the fall in output over the 1990s in Russia and other countries formerly belonging to the Soviet Union, and could, therefore, be reversed. In the future, further reductions in acid depositions and in nutrient discharges into the Baltic Sea are required. In Finland, there is not much room left for realising such reductions at low cost. Therefore, measures to reduce emissions of acidifying and eutrophying substances abroad, as already undertaken mainly in Russia and Estonia, should be pursued as they are more cost-effective.

However, measures are also needed to restrain domestic emissions. Up to now, command and control measures have been used successfully, especially in the case of SO_2. However, the regulatory authorities lack information about the abatement costs faced by the various sectors and firms and, given the steeply rising marginal abatement costs, command and control measures become increasingly inefficient compared to economic instruments. Regarding NO_x emissions, a charge on emissions from power plants, combined with a transitional compensation scheme, could be a cost-effective instrument, as experience elsewhere shows. Charges reflecting the environmental costs associated with nutrient release should also be applied to industries and municipal waste water treatment plants. Concerning water pollution, agricultural emissions may be more difficult to deal with. The agri-environmental measures taken so far have reduced nutrient balances, but at a high cost and not sufficiently. Further decoupling of agricultural support from production would help to reduce discharges and should therefore be pursued nationally as far as possible and promoted in EU negotiations on agricultural policy.

Combating climate change by reducing GHGs as agreed in the Kyoto Protocol is probably the most challenging issue. New policy measures will have to be taken in order to meet the target. Marginal abatement costs are high in Finland. This makes it even more difficult than elsewhere to decouple CO_2 emissions from GDP developments. Important energy policy decisions will have to be taken soon about fuel switching options, which could allow for substantial reductions in CO_2 emissions compared with the reference energy scenario. Nevertheless, even with a decision on a second gas pipeline or a fifth nuclear power plant, reaching the Kyoto target will probably require additional policy initiatives. Hence, to fulfil its commitment, Finland would greatly benefit from the cost-effective flexibility mechanisms allowed for in the Kyoto Protocol. Investments

reducing GHG emissions in neighbouring countries like Russia would help. Finland could also benefit greatly from buying emission quotas abroad, as soon as this becomes operational.

Finland was the first country in the 1990s to introduce a carbon tax aimed at diminishing CO_2 emissions while contributing to the necessary fiscal consolidation. However, after many changes, the current energy tax structure is far from being cost-effective since it provides only a very loose link between taxation and the carbon content. The favourable treatment granted to the most energy-intensive industries for competitiveness reasons increases the overall costs of CO_2 emission abatement, since it obliges others to pay higher taxes to achieve the emission reductions. Moreover, it is tilting incentives towards energy-intensive activities in the long run. Refocusing the CO_2 tax so as to equalise marginal abatement costs would greatly enhance the cost-effectiveness of abatement. However, in practice, the Finnish authorities consider that EU regulations limiting the possibilities to tax imported electricity are an obstacle to applying a "pure" CO_2 tax. One solution could be to soften the consequences for industry by introducing a compensation scheme that would allow firms to adapt to the tax gradually and would avoid unnecessary adjustment costs. Harmonisation of energy taxes at the EU level would also help in this respect. At the domestic level, reducing and progressively eliminating the favourable treatment of diesel compared to gasoline could enhance cost-effectiveness. Moreover, it would have ancillary benefits on NO_x emissions. Removing the favourable treatment for peat may also reduce acidifying emissions.

Another possibility to improve cost-effectiveness, while bringing emission levels closer to the target, would be to create a national trading system for emission quotas that would replace, at least partly, energy taxation. This would have some advantages. First, a partial grandfathering of emission quotas could make it more acceptable for industry than a pure CO_2 tax, while providing "correct" abatement incentives at the margin. Second, it would allow for a better control of the amount of emissions. Finally, it would make it easier to participate in an international (EU or broader) trading system as soon as it is created. It would be in the Finnish interest to support EU-wide measures, be it tradable quotas or an EU-wide carbon tax. Efforts to improve cost-effectiveness and to comply with the Kyoto Protocol should not solely focus on CO_2 emissions as abatement costs of other GHGs, especially methane, are estimated to be lower.

Initiatives should not be restricted to realising cost-effective environmental policies. It is also very important to improve the consistency of the various sectoral policies with respect to their environmental consequences. In many instances, it is difficult to reconcile the various objectives of agricultural, transport and energy policy with environmental goals. Concerning energy policy, peat production can be seen as an example where regional policy objectives are achieved

at an environmental cost. Regarding the options considered to increase the energy supply in the future, a greater use of gas from Russia would greatly reduce the diversity of energy supply and therefore clash with an energy policy objective. The interconnection to the European grid is under discussion, but it requires various decisions in neighbouring countries as well. Furthermore, the building of a fifth nuclear power plant, although it would help in containing CO_2 emissions, has the drawback of raising other environmental issues.

Emissions from the transport sector, in particular CO_2 emissions, have been growing strongly in the past decades. Although some gains in energy efficiency of vehicles can be expected, additional measures to control the growth in traffic volumes will be needed to contain emissions. Further efforts should be made to bring transport prices closer to marginal social cost. Given that trucks are expected to be responsible for most of the increase in emissions, it would be important in particular to align the marginal cost coverage of trucks with that of private cars by raising for instance the taxation of diesel. The tax advantage for diesel compared with unleaded gasoline in Finland is the largest in the OECD. Given the high level of vehicle taxes, there is also room for a radical shift from vehicle to fuel taxation which would allow a better internalisation of externalities.

Finally, despite a decrease in its overall level and a partial decoupling from input and production, the support provided to agriculture is still among the highest in the OECD and agriculture remains a significant source of environmentally harmful emissions. The most important adverse environmental effect is the runoff of nutrients, which is the main (human-induced) cause of water eutrophication, but agriculture also produces greenhouse and acidifying gases. Since it still plays an important role in the economic activity of many rural areas, agricultural activities form an important element of regional policy. However, farming in Finland suffers from fundamental structural weaknesses, such as the harsh climatic conditions and the small size of farms. A reshaping of regional policy would raise efficiency, and a lowering of support would allow the reduction of the associated pollution and government spending.

Notes

1. Since February 1999, the national accounts published by Statistics Finland are based on a new calculation method. This new method, European System of Accounts 95 (ESA95), will be introduced this year by all EU countries, making the national accounts statistics more uniform and therefore more comparable. According to the new method, Finnish GDP is 2 per cent higher than previously calculated. This is mainly because of the inclusion of computer software and other intangible fixed assets in fixed capital formation. As a consequence, capital formation in the 1990s is now 6 to 11 per cent higher than previously calculated. The other major difference is the introduction of depreciation on civil engineering. This increases consumption of fixed capital, increasing GDP in 1995 by 0.9 per cent. ESA95 aims to take the grey economy better into account but in fact only led to a minor revision. At the same time, the base-year changed, from 1990 to 1995 with the result that the 1990-93 recession is somewhat less deep than previously calculated while the subsequent pick-up is somewhat less steep.
2. On a quarterly basis, real GDP was 6.8 per cent higher than a year earlier in the first quarter, dropping to 3.7 and 3.9 per cent in the third and fourth quarter, respectively.
3. Nokia is the leading company for mobile telephones with a world market share of around a quarter in 1998.
4. The numbers are consistent with the *Economic Outlook No. 65* (OECD, 1999*a*).
5. Goods exports to Australia (export share of 1.3 per cent) and to Ireland (export share of 0.8 per cent) dropped even more, by 25 per cent and 20 per cent, respectively.
6. Imports from Argentina (+152 per cent) and Romania (+107 per cent) grew even stronger, but from very low levels.
7. Based on the registration of passenger cars.
8. Finland is still largely a deposit-oriented society, despite the rapid changes in financial markets and the low nominal interest rates currently paid on bank deposits. At the end of 1998, bank deposits held by the public amounted to FIM 315 billion (46 per cent of GDP), 6½ per cent more than at the end of 1996. Resources held in mutual funds increased more sharply, by 140 per cent, but were still only FIM 28 billion (4 per cent of GDP) at the end of 1998. The most popular funds are those investing solely in equities, accounting for one-third of the total. Asset allocation funds and bond funds are accounting for one-fourth and one-fifth, respectively.
9. House price development of owner occupied flats.
10. Within M3, there was a 10 per cent contraction in certificates of deposits, after the exceptional growth of 71 per cent in 1997. The narrow monetary aggregate (M1) rose by 5 per cent, substantially below the 9 per cent rise for the euro area as a whole.

11. The unemployment rate of the Uusimaa province was 7½ per cent in 1998, compared with 12½ per cent for the rest of the country, with rates ranging from 11.4 per cent in western Finland to 19.8 per cent in Lapland.

12. Young job seekers usually benefit more than proportionally from strong labour demand while female job seekers benefit from a favourable situation in the service sector. Surprisingly, this was not the case in 1998. The share of youth unemployment (those younger than 25 years) in total unemployment increased from 22.9 per cent in 1997 to 24.9 per cent in 1998, while the unemployment rate of females dropped only 1.0 percentage point, to 12.0 per cent, while that of males fell 1.4 percentage points to 10.9 per cent.

13. The index measures developments in average wage and salary earnings for normal working time. It does not include for example overtime pay and bonuses. The index has fixed weights (at the moment 1995) and therefore is not influenced by shifts in the employment composition. This is in contrast with wage rates calculated on the basis of national account figures that include overtime pay, bonuses and shifts in the employment composition.

14. The surplus was FIM 56.9 billion (8.4 per cent of GDP) on a customs basis.

15. The capital flows in connection with the merger of the Finnish company, Enso, and the Swedish company, Stora, are recorded as equally-sized outward direct investment and inward portfolio investment. Those related to the merger of the Finnish Merita bank and the Swedish Nordbanken are recorded differently: as equally-sized inward and outward direct investments. As a consequence, this merger has no influence on net direct investment but has a substantial impact on the gross direct investment flows.

16. On the basis of quarterly data, the drop was 12.1 per cent. The recent recession in Korea led to a drop in real GDP of 5.8 per cent on the basis of annual data and a drop of 7.1 per cent based on quarterly data.

17. The stock of bank lending at the end of 1998 was also still below the peak level reached in 1991, with a completely different currency composition. At the end of 1998, markka-denominated lending was 7 per cent above its 1991 peak level while foreign currency lending was only 16 per cent of its record 1991 level. The change in the currency composition was mainly due to firms paying back foreign currency denominated loans at a rapid pace after having suffered severe capital losses due to the depreciation of the markka in 1991-93. In addition, lower domestic interest rates and the prospective membership in the EMU reduced the interest in foreign currency borrowing.

18. Employers' organisations and labour unions have two-thirds and one-third of the seats of the board, respectively.

19. However, legislation can be introduced allowing a temporary funding deficit. This was done in 1994 when a temporary deficit of 2 per cent of the wage bill was allowed to prevent a sharp increase in pension contributions having a negative effect on the recovery.

20. The expenditure ceiling for 2000 was set at €32.4 billion in 1999 prices, compared with expected expenditures of €32 billion in 1999. The ceilings in 1999 prices for 2001, 2002 and 2003 are €32.5, €32.2 and €32 billion, respectively.

21. Italy, which had higher rates, lowered its key rate to 3.5 per cent.

22. In April 1999, the Bank of Finland announced a gradual staff reduction due to the shift of responsibilities to the ECB.
23. The projections presented in this *Survey* are the same as those published in *Economic Outlook No. 65* (June 1999).
24. The Irish standardised unemployment rate dropped from a peak of 15.9 per cent in May 1993 to 6.9 per cent in March 1999.
25. These countries, along with New Zealand, achieved a significant fall in unemployment since 1990, accompanied by improvements in other labour market indicators (OECD, 1999*c*). Other countries kept low unemployment rates: the United States, Norway and Portugal maintained low unemployment, as well as high and/or rising employment rates; Austria, Japan and Switzerland showed some deterioration, but unemployment nevertheless remained low.
26. The 1999 projection in the 1998 Stability Programme is based on a projected 1998 unemployment rate of 10.5 per cent, clearly below the actual rate of 11.4 per cent.
27. For a summary of the evidence pointing to the necessity of comprehensive labour and product market reform to achieve better labour market conditions, see OECD (1999*d*).
28. In 1997, the average hours actually worked per person in dependent employment was 1 687 hours – 10 per cent more than in Germany and France, but 11 per cent less than in Japan and 14 per cent less than in the United States (OECD, 1998*a*).
29. Large firms have to pay 100 per cent in the case of a disability pension and 50 per cent in the case of an unemployment pension but there is no contribution by small firms. The threshold is less than 50 persons for the unemployment benefit and less than 300 for the disability pension. For the disability pension, the employers' contribution rises gradually to 100 per cent between 300 and 1 000 employees. For unemployment pensions, the employers' contribution rises gradually to 50 per cent between 50 and 300 employees.
30. During August-October, share prices on the Helsinki stock exchange plummeted by 30 per cent. By January 1999, prices had already recovered fully from this fall.
31. The most important companies fully or partially state-owned are: Finnair (airline), Fortum (energy), Kemira (chemical industry), Leonia (banking), Outokumpu (mining), Partek (wood products), Rautaruukki (metal), Sonera (telecommunication), Stora Enso (pulp and paper), Valmet-Rauma (machinery), VR group (railways). Full privatisation is currently allowed for Leonia Corporate Bank, Partek, Stora Enso and Valmet-Rauma.
32. Beneficial for SMEs was also the merger of Kera Corporation, primarily granting soft-loans and participating in venture capital projects, and the Finnish Guarantee Board (FGB) on 1 January 1999 to form Finnvera.
33. The only remaining restrictions on inward direct investment concern nuclear power and mining.
34. Germany, Sweden and the United Kingdom also went beyond the minimum requirements of the EU directive and extended choice to all consumers.
35. Small electricity consumers can become eligible based on load profiling. In load profiling, real measurement of the consumption patterns of typical consumers are used to develop estimated profiles for the whole consumer group. The ultimate consumers' invoices continue to be based on the regular meter, but reimbursements

between retailers are now based on load profiles and metered total energy (IEA, 1999).

36. Fortum has a share of 80 per cent of the wholesale market for refined petroleum products in Finland. Sonera has a 70 per cent market share for mobile communication services in Finland, a 40 per cent market share for long-distance calls and a 55 per cent market share for international calls. It is the leading provider of Internet services with a market share of more than 35 per cent.

37. In 1998, the European Commission imposed heavy fines for price fixing and bid-rigging by a district heating pipe cartel, including the Finnish company KWH Tech.

38. In June 1998, the Ministry of Justice set up a committee to investigate the possibility of converting the Competition Council into a special court or transferring its tasks to ordinary courts. At the same time, a working group was established to examine the position of the Market Court. In February 1999, the committee and the working group proposed that a new special court should be established which would handle all the issues now falling within the jurisdiction of the two organs.

39. The European Commission makes the same recommendation in its recently published Broad Policy Guidelines (European Commission, 1999).

40. An exception is the Skopbank which was particularly hard hit by the banking crises. A liquidity crisis in September 1991 prompted the Bank of Finland to take over the bank but in June 1992, the State Guarantee Fund acquired it. During the rescue operation Skopbank received FIM 15.2 billion in government support. In 1998 it was decided to wind down its operations through a liquidation starting in early 1999.

41. In 1998, in line with the recommendation in the previous *Survey*, it was decided to transfer the supervision of insurance and pension companies from the Ministry of Social Affairs and Health, the supervisory body for the insurance industry, to the new Insurance Supervision Authority to be established under the Ministry's branch of administration, notwithstanding certain exceptions concerning the statutory employment pension insurance. It is intended to enhance co-operation between the financial and insurance surveillance authorities by also including a representative from the Financial Supervisory Authority in the Board of the Insurance Supervision Authority. In 1997, the Bank of Finland recommended combining the surveillance of the insurance and banking sectors in one independent agency, in line with the development in some other OECD countries. The proposal was generally well received in the financial industry, but resisted by the Ministry of Social Affairs and Health.

42. In 1997 the merger between the largest commercial bank in Finland, Merita Bank, and one of the largest banks in Sweden, Nordbanken, created the new bank group MeritaNordbanken. During the same year state-owned Postipankki and state-owned Finnish Export Credit merged, creating Leonia, while the OKOBANK Group introduced a new form of joint operation between the central co-operative bank and the member banks. In 1998, the investment bank Mandatum Bank merged with the commercial bank Interbank forming the commercial bank Mandatum Bank. Small savings banks and a group of independent co-operative banks made an agreement with Aktia Bank, which acts as their central monetary institution.

43. In addition to a sharp increase in the volume of bank operations, the rise in operating costs in 1998 was probably due to the non-recurring costs of the switch to the euro and additional investment to solve the Year 2000 (Y2K) problem.

44. In December 1997, the Helsinki Securities and Derivatives Exchange (HEX) was formed by the merger of Helsinki Stock Exchange and the Finnish Derivatives Exchange and Clearing House (SOM). In December 1998, HEX and Finnish Central Securities Depository were brought together in the parent company Helsinki Exchanges Group. The Finnish Central Securities Depository offers processing services for instruments traded on the Finnish money and capital markets, including clearing and settlement services.
45. By law, transaction accounts with an annual interest rate of 2 per cent or less are exempted from capital income taxation. In 1998, tax-exempt deposits were 70 per cent of total deposits held by the public. As these are mainly current accounts, a large proportion carries a very low or no interest rate.
46. The methodology is described in OECD (1994) and Suyker (1998).
47. The government's estimate of total discretionary expenditure measures, based on budget numbers, is FIM 55 billion (8 per cent of GDP), roughly the same as the OECD Secretariat estimate. The government estimate of total tax cuts is FIM 13.3 billion (2 per cent of GDP), substantially smaller than the Secretariat estimate. The low government estimate may be due to too high medium-term tax projections.
48. The Stability Programmes replace the Convergence Programmes for the euro area countries. To enhance surveillance of public finances, the euro area participants are obliged to present a programme annually for a four-year horizon, outlining the projected fiscal stance and targets as well as the government's assessment of economic growth prospects. The Finnish programme was discussed and approved by the Economic and Financial Affairs Council (ECOFIN) in October 1998.
49. It is based on economic growth of 4.0 per cent in 1999 and 2.6 per cent on average in the years 2000-02. The unemployment rate is projected to drop from 9.0 per cent in 1999 to 7.8 per cent in 2002.
50. Only the non-euro area EU member Sweden shows a higher surplus of 3 per cent of GDP in 2002 in its Convergence Programme.
51. Within the EU, only Sweden is more sensitive to economic fluctuations.
52. See also Box 3 ("How large a surplus to comply with the Stability and Growth Pact?") of the 1998 OECD *Survey* of Finland (OECD, 1998*c*).
53. Due to the derogation, tax-free alcohol import from other EU countries is currently significantly less than in the other EU countries: 1 litre of strong spirits can be imported duty-free (10 litres under the EU regulation), 3 litres of aperitifs (20 litres), 5 litres of table wines (90 litres) and 15 litres of beer (110 litres).
54. The new government decided not to cut excise duties before the derogation expires in January 2004.
55. In order to keep the focus on major issues for Finland, the chapter does not cover fisheries, waste management and urban air and noise pollution.
56. The discussion of environmental and resource management issues draws on earlier work by the OECD's Environment Directorate (in particular, the 1997 *Environmental Performance Review* for Finland, OECD, 1997*b*). Compared with this work, which provides a detailed assessment of environmental performance with respect to domestic objectives and international commitments, this chapter focuses only on a few key policy issues, but seeks to put them into a broader context. The chapter also draws on work by other OECD Directorates and specialised agencies, in particular the Finland Review by the International Energy Agency (IEA, 1999), the Directorate for

Food, Agricultural and Fisheries (OECD, 1999*b*), and the European Conference of Ministers of Transport (ECMT, 1998).

57. Data on natural resource wealth are currently being developed. Consistent wealth estimates are available for Norway for the period 1970 to 2050 and include physical, human, foreign net financial assets and the oil wealth.

58. The largest share of local expenditures on environmental protection is dedicated to sewerage and waste water purification, covered mainly by user fees.

59. For a detailed description, see OECD (1997*b*).

60. Education and training; changing production and consumption patterns; financial issues and transfer of technology; socially sustainable development and local aspects of sustainable development.

61. In the same year, the government also adopted a decision-in-principle on environmental impact assessment guidelines to be applied when preparing government bills. To support this decision, an interministerial network was established.

62. In addition, several research projects have focused on the impact of forest practices on biodiversity. Information that helps in implementing forestry policy has been developed (*e.g.* National Criteria and Indicators for Sustainable Forest Management, since 1995; a proposal for a Certification Scheme for the Sustainable Management of Forests in Finland made in 1997).

63. The annual increment is the growth of wood stock, equal to annual felling minus annual drain. The increment in 1998 was 75 million cubic metres, which is proportionally distributed across the three main species (Scots pine, Norwegian spruce and birch). The area dedicated to birch is considerably smaller than that for coniferous species.

64. Although there is no clear definition of what constitutes an old forest, in the case of Finland it is usually considered as one which has not been cut for 100 years.

65. In Finland there are approximately 42 000 species of living organisms, a small number by international standards reflecting the extreme northern location of the country. Of these, 1 per cent are vertebrates, 39 per cent plants and 60 per cent invertebrates.

66. The existence value is related to the desire of people to know that some species do exist, even if they may never see them. The option value refers to the possibility that these species will be of some use in the future, even though the use is unknown for the moment. There are also other "use values" (apart from direct wood exploitation) not included in wood prices (*i.e.* recreational use or prevention of erosion).

67. Biodiversity issues also concern maritime life, but are not pursued here. Environmental policies geared at preventing or reducing water pollution are described below.

68. See OECD (1997*a*).

69. These habitats are streams, small lakes, eutrophic peatlands and herb-rich forests.

70. In this sense, it extends the scope of the old Forest Improvement Act, which supported investment in the forest sector from a purely profit-oriented point of view.

71. See OECD (1997*a*) for a thorough discussion of the alternative certification systems and of the difficulties to reach an agreement among the participants in the negotiations.

72. However, NGOs have recently withdrawn from these negotiations because of disagreement on the certification system to be used.

73. The commitment for the EU as a whole is to reduce GHG emissions by 8 per cent in 2010 from their 1990 level. Finland got a lower reduction target for structural reasons, and because 1990 was a year with an unusually mild winter and a wet year with ample hydropower in addition to a high level of electricity imports, resulting in lower CO_2 emissions.

74. Emissions of the three other GHGs included in the Kyoto Protocol are virtually non-existent in Finland.

75. Peat and natural gas have always had a favourable tax treatment (in the early 1990s by means of a special deduction scheme in the sales taxation or the VAT). Wood has always been exempted from the CO_2 tax (Box 5). In addition, fuels used in industrial production as a raw material or inputs in the manufacturing of goods (*i.e.* for non-energy purposes) have always been exempted.

76. Companies that have paid more than 3.7 per cent of their value added in energy excise duties get a 85 per cent refund, if the amount involved exceeds FIM 300 000.

77. In recent years, wood-based energy has received a substantial share of energy investment support. In many instances, the production costs of new renewable energies are too high for taxation alone to make them competitive.

78. Methane emissions from agriculture depend on the volume of livestock, which is particularly important in Finnish agriculture; N_2O emissions stem mainly from animal manure and nitrogen fertilisation. There, the extensive support provided to the sector is important (Box 9).

79. As in the Business As Usual (BAU) scenario, a 2.5 per cent average annual GDP growth is assumed, which is slightly less than the rate observed during the past two decades. A fall in energy intensity is also assumed in the three scenarios, partly explained by a change in the structure of industry. In the two variant scenarios, taxation of fuels is assumed to be about twice as high in 2025 as it is today. In the first variant, the assumption is that natural gas will replace coal in power generation, both in conventional condensing plants and in combined heat and power plants (CHP). The second variant is based on the assumption that an additional nuclear plant is built before 2010, with less natural gas used.

80. Using a comprehensive energy system model for Finland, the authors analyse the cost-effectiveness of different technical options (but not policy measures) for reducing the emissions of CO_2, CH_4 and N_2O in the energy sector, industry, waste and agriculture. Integrated abatement is found to allow cost savings of about 20-25 per cent compared to abatement of CO_2 emissions alone.

81. Lehtilä and Tuhkanen (1999) provide an estimate of FIM 230.

82. In fact, if all countries are to achieve their Kyoto target, applying a perfect CO_2 tax without taxing electricity imports could provide an efficient solution. Indeed, in a liberalised market, the electricity will end up being produced where abatement costs are lowest.

83. Some episodes of high concentration of tropospheric ozone due to transborder movements of nitrogen oxides and volatile organic compounds do occur in some areas. Finland has committed itself in the Geneva Protocol to reduce VOC emissions by 30 per cent by 1999 compared with their 1988 level, a target which will probably

be reached at a later date since emissions had only been reduced by 13 per cent in 1996. However, high tropospheric ozone, is not a major problem in Finland.

84. In the pulp and paper industry, for instance, the licensing procedure and consequent changes in production processes have reduced the substantial emissions by more than 95 per cent.

85. In 1994, the Environmental Economics Committee rejected the possibility of introducing such a SO_2 tax, on the grounds that it would not add much compared with fuel taxation. Given that there are several ways to decouple SO_2 emissions from fuel consumption, this argument seems rather weak.

86. Moreover, as NO_x emissions also contribute to other environmental problems – eutrophication, tropospheric ozone and local air pollution – the measures should be tailored to local conditions. The future NO_x Protocol, which should be concluded in 2000, will take all these dimensions into account.

87. Emission standards are differentiated for larger and smaller power plants, reflecting differences in available technologies.

88. By contrast, the gas switch scenario will have little impact on NO_x emissions, since natural gas (and wood) produces about the same amount of NO_x emissions as coal.

89. SO_2 emissions from peat have grown by 40 per cent between 1992 and 1997, while emissions from other sources decreased by 34 per cent in the same time. As a result, the share of peat in total SO_2 emissions has risen from 7 to 14 per cent.

90. The Gulf of Finland experienced two intensive algae bloom periods recently, in 1995 and 1997.

91. For eutrophication of inland waters by contrast, phosphorus loading is the main culprit.

92. The Water Protection Programme for 1989-95 set quantitative targets for reductions in phosphorus discharges in 1995 compared to 1986 (25 per cent for the pulp and paper industry, 14 per cent for municipal waste water and 50 per cent for agriculture). The objective for nitrogen was qualitative, the aim being to "substantially" reduce total nitrogen discharges (OECD, 1997*b*). Over the period 1987-96, of relevance for the targets of the Baltic Sea Convention, industries have reduced their phosphorus discharges by more than 60 per cent and their nitrogen discharges by almost 40 per cent (Statistics Finland, 1998*a*).

93. In 1995 and 1996, investment for water protection accounted for about 55 per cent of total environmental protection investment (Statistics Finland, 1998*b*).

94. The proceeds are earmarked for water protection activities. A fish management charge (about FIM 4 million in 1993) is also levied on polluters whose action might have an adverse impact on fish stocks. This change is earmarked for fish conservation expenditure.

95. Nitrogen removal efficiency is still only at 30 per cent (OECD, 1997*b*), compared to 93 per cent for phosphorus (Statistics Finland, 1998*a*).

96. Some of the investments in water management (mostly in rural areas) are financed by state subsidies, amounting to FIM 44 million in 1997 or only 1 per cent of total water expenditure.

97. Forestry is another diffuse source, but changes in forestry practices and a very strong reduction in fertilisation have substantially brought down the contribution of this sector to eutrophication.

98. Introduced in 1979, the tax was initially a general tax on fertilisers, to which an environmental tax on phosphorus fertiliser was added in 1990. In 1992, the two were combined and an environmental tax on nitrogen fertilisers was also introduced.

99. Phosphorus fertilisers' use was reduced by half between 1989 and 1994, and that of nitrogen fertiliser by about a quarter (OECD, 1997*b*).

100. Compared to 1995, nitrogen loading from agriculture is estimated to have fallen by 3 to 14 per cent in various watersheds (Grönroos *et al.*, 1998).

Glossary of acronyms and terms

Acidification	Caused by atmospheric emissions of sulphur dioxide and nitrogen oxides which, through "acid rain", lead to deposition of sulphur and nitrogen in the soil and water surfaces (causing damage to aquatic life and forests)
Acid rain	Describes a type of pollution which washes out of the atmosphere as dilute sulphuric and nitric acids
Anthropogenic	Describes an object or disturbance to the environment which is man-made or which follows from the emissions of harmful substances as a result of human activities
ATM	Automatic Teller Machine
BAU	Business As Usual
Biomass	The total quantity or weight of organisms in a given area or volume
Biotope	The smallest subdivision of a habitat, characterised by a high degree of uniformity in its environmental conditions and in its plant and animal life
CEEC	Central and eastern European countries
CH_4	Methane
CHP	Combined heat and power plants
CIS	Community of Independent States
CO_2	Carbon dioxide
ECB	European Central Bank
ECOFIN	Economic and Financial Affairs Council
E-commerce	Electronic commerce
ESCB	European System of Central Banks
EMU	Economic and Monetary Union
ESA95	European System of Accounts 95
ERM	Exchange Rate Mechanism
EU	European Union
Eutrophication	A process of pollution that occurs when a lake or stream becomes over-rich in algae or aquatic plants which kill animal life by depriving it of oxygen
FIM	Finnish markka
GDP	Gross domestic product
GHG	Greenhouse gas
ha	Hectare
HEX	Helsinki Securities and Derivatives Exchange
IVO	Imatran Voima
kWh	Kilowatt hour
MWh	Megawatt hour
NFP	National Forest Programme

NGO	Non-Governmental Organisation
N_2O	Nitrous oxide
NO_x	Nitrogen oxides
Particulates	Tiny flecks of soot and similar substances in the atmosphere. A principal source of particulates is black exhaust from road vehicles
PPP	Purchasing Power Parity
R&D	Research and Development
sa	Seasonally adjusted
saar	Seasonally adjusted, annual rate
SDP	Social Democratic Party
Sink	In this context, a place where, or a process by which, greenhouse gases are removed from the atmosphere
SITC	Standard Industrial Trade Classification
SME	Small and Medium-sized Enterprises
SO_2	Sulphur dioxide
SO_x	Sulphur oxides
SOM	Finnish Derivatives Exchange and Clearing House
TEKES	Technology Development Centre
UN-ECE	United Nations - Economic Commission for Europe
VAT	Value Added Tax
VOC	Volatile Organic Compound
Y2K	Year 2000

Bibliography

Adams J. (1997),
"Environmental Policy and Competitiveness in a Globalised Economy: Conceptual Issues and a Review of the Empirical Evidence", draft paper prepared for a Workshop on Economic Globalisation and the Environment, Vienna.

Ali-Yrkkö, J. and Ylä-Anttila P. (1999),
"Omistus kansainvälistyy – johtamis- ja valvontajärjestelmät muuttuvat" (Internationalisation of Ownership – Management and Corporate Governance Culture in Change), Report to the Economic Council, Etlatieto Oy, Helsinki.

Bank of Finland (1998),
"Financial stability in Finland", *Bank of Finland Bulletin*, December 1998, pp. 9-14, Helsinki. <http://www.bof.fi/env/eng/ju/aik–buen.stm>

Barker, T. (1999),
"Limits of the Tax Approach for Mitigating Global Warming", in J. Hacker and A. Pelchen (eds.), *Goals and Economic Instruments for the Achievement of Global Warming Mitigation in Europe*, Kluwer Academic Publishers.

Bill S. (1999),
"European Commission's Experience in Designing Environmental Taxation for Energy Products", in J. Hacker and A. Pelchen (eds.), *Goals and Economic Instruments for the Achievement of Global Warming Mitigation in Europe*, Kluwer Academic Publishers.

Blöndal, S. and S. Scarpetta (1998),
"Falling Participation Rates Among Older Workers in OECD countries", *OECD Economics Department Working Paper*, No. 202, June 1998, Paris. <http://www.oecd.org/eco/wp/onlinewp.htm#1998>

Economic Council (1998),
"Labour Market Challenges and Improving Functioning of Labour Market", *Summary of the Expert Group Report*, Prime Minister's Office Publication Series 1998/3.

Economic Council (1999),
"Public Finances in the Twenty-first Century: Limitations, Challenges and Direction of Reforms", *Working Group Report*, Prime Minister's Office Publication Series 1999/1.

Environmental Economics Committee (1994),
"Interim Report — Environment Related Energy Taxation", *Working Group Report 4*, Ministry of the Environment, Helsinki.

ETLA, Research Institute of the Finnish Economy (1998),
"The Effects of the Electronics Industry and Nokia on Finnish GDP Growth", *The Finnish Economy*, 4/1998, pp. 72-73, Helsinki.

European Central Bank (ECB) (1999*a*),
Possible Effects of EMU on the EU Banking System in the Medium to Long-term, February 1999, Frankfurt. <http://www.ecb.int/pub/pdf/emubnk.pdf>

ECB (1999*b*),
"Banking in the Euro Area, Structural Features and Trends", *ECB Monthly Bulletin*, April 1999, Frankfurt. <http://www.ecb.int/pub/pdf/mb199904en.pdf>

European Conference of Ministers of Transport (ECMT) (1998),
Ecient Transport for Europe: Policies for Internalisation of External Costs, OECD, Paris.

European Commission (1999),
"1999 Broad Economic Policy Guidelines", March 1999, Brussels. <http://europa.eu.int/en/record/broad/index.htm>

Grönroos, J. *et al.* (1998),
"Maatalouden ympäristötuki", Finnish Environment Institute, Finnish Environment No. 239.

International Energy Agency (IEA) (1998),
"Implementing Environmental Commitments in Liberalised Energy Markets", Paris. <http://www.iea.org/clim/cop4/index.htm>

IEA (1999),
Energy Policies of IEA Countries: Finland 1999 Review, Paris.

Kam, F. de, and C. Bronchi (1999),
"The Income Taxes People Really Pay", *OECD Observer*, March 1999, pp. 13-16, Paris.

Keso, I. (1999),
"Kuinka yleisiä ovat tuloloukut? Eräitä arvioita efektiivisten rajaveroasteiden suuruudesta", (How Common are Income Traps? Some Estimations of Marginal Effective Tax Rates), Ministry of Social Affairs and Health, Helsinki.

Lehtilä A. and S. Tuhkanen (1999),
Integrated Cost-effectiveness Analysis of Greenhouse Gas Emission Abatement – The case of Finland, VTT Technical Research Centre of Finland, Publication No. 374.

METLA, Finnish Forest Research Institute (1998),
Finnish Forest Sector Economic Outlook 1998-1999, November, Helsinki.

Ministry of the Environment (1998),
"Happamoitumis-toimikunnan mietintö", (Report of the Acidification Committee), The Finnish Environment 219, May, Helsinki.

Ministry of Finance (1998*a*),
"Finland's Stability Programme", September 1998, Helsinki. <http://www.vn.fi/vm/english/va98etek.htm>

Ministry of Finance (1998*b*),
"Prospects and Challenges in Public Finances", Economic and Economic Policy Surveys, 4/98, Helsinki. <http://www.vn.fi/vm/english/national–economy/proschal.htm>

Ministry of Finance (1999),
Kannustinloukkutyöryhmän ehdotusten toteutumisen arviointia, (Evaluation on Practical Functioning of the Proposed Measures by the Incentive Trap Working Group), Helsinki.

Ministry of Trade and Industry (1997),
"The Finnish Energy Strategy — The Council of State's Report on Energy Policy", 11/1997, Helsinki.

Ministry of Trade and Industry (1998),
"Privatisation of Finnish State-owned Companies", September 1998, Helsinki.

Ministry of Interior Affairs (1999),
"Backmanin neuvotteluryhmän linjaukset kuntien peruspalveluiden turvaamisesta" (Broad Guidelines of Backman's Working Group on Securing Municipalities' Basic Public Services), 19 March 1999, Helsinki.

OECD (1994),
"Estimating Potential Output, Output Gaps and Structural Budget Balances", *OECD Economic Outlook 56*, pp. 31-37, Paris.

OECD (1997*a*),
"Finnish Case Study: The Act on the Financing of Sustainable Forestry and the Development of Forest Certification", Environment Directorate, forthcoming as a General Distribution document, Paris.

OECD (1997*b*),
Environmental Performance Reviews: Finland, Paris.

OECD (1997*c*),
Environmental Benefits from Agriculture, Issues and Policies, The Helsinki Seminar, Paris.

OECD (1998*a*),
Employment Outlook, Paris.

OECD (1998*b*),
"The retirement decision", *Economic Outlook 63*, pp. 179-192, June 1998, Paris.

OECD (1998*c*),
Economic Survey of Finland 1998, Paris.

OECD (1999*a*),
Economic Outlook 65, June 1999, Paris.

OECD (1999*b*),
"The Interim report of the OECD Three-Year Project on Sustainable Development", PAC/AFF(99)1, May 1999, Paris. <http://www.oecd.org/subject/mcm/1999/pdf/totrev4.pdf>

OECD (1999*c*),
"Policies to Create Jobs", May 1999, Paris. <http://www.oecd.org/subject/mcm/1999/pdf/employ.pdf>

OECD (1999*d*),
The OECD Jobs Strategy: Assessing Performance and Policy, forthcoming, Paris.

Pipatti, R. and M. Wihersaari (1998),
"Cost-effectiveness of Alternative Strategies in Mitigating the Greenhouse Impact of Waste Management in Three Communities of Different Size", in Mitigation and Adaptation Strategies for Global Change, Kluwer Academic Publishers.

Statistics Finland (1998*a*),
Finland's Natural Resources and the Environment 1998, in collaboration with the Ministry of the Environment, Helsinki.

Statistics Finland (1998*b*),
Environmental Protection Expenditure by Finnish Industry 1996, Helsinki.

Suyker, W. (1998),
"Structural budget balances: The Method Applied by the OECD", paper prepared for the Bank of Italy workshop on structural budget indicators, November 1998, Paris.

Vartianen, J. (1998),
"The Labour Market in Finland: Institutions and Outcomes", Prime Minister's Office Publication Series, 1998/2, June 1998. <http://www.vn.fi/vn/english/publicat/vnk19982.pdf>

Annex I

Labour market reform measures in the 1990s

Transfers

Only minor changes were made to the unemployment benefit level (the earnings-related benefit is still 90 per cent of the previous wage for most unemployed). In 1994, the indexation of benefits was temporarily suspended and, since 1998, indexation has been scaled down from wage to consumer price indexation. The waiting period was increased by two workdays in 1998. Currently, no benefits are paid on the first seven working days of unemployment.

Benefits in case of re-entering unemployment after a temporary (subsidised or normal) job were lowered somewhat in 1997 as the average earnings in the last 43 weeks (the qualifying period) became the basis for the earnings-related allowance. Before, the (higher) pre-unemployment earnings were the basis of the unemployment benefit if the interim period of the (temporary) job did not exceed 27 weeks.[1] The eligibility requirement for the unemployment insurance and allowance schemes was tightened in 1997 to an employment record of 43 weeks during the previous two years, from 26 weeks during 8 months. Since 1997, the unemployed can no longer "requalify" for full benefits through participation in subsidised job programmes, which usually last for six months. In 1998, the labour market support scheme, under which long-term unemployed and persons with an insufficient work history receive a means-tested benefit, was made conditional on participation in some form of labour market training for under 25-year-olds without vocational education. A comparable measure was already taken in 1996 for those aged 17 to 19 years. The availability requirement was somewhat tightened. The length of the period, during which benefits are suspended, in case an unemployed person stops looking for work, or refuses to accept a suitable job or training offer, was increased in 1998. There was, however, also some easing: since 1997, unemployed persons are allowed to make short, occasional trips abroad without losing their right to unemployment benefits.

Concerning early-retirement, in 1994, the minimum age for disability and unemployment pensions was raised from 55 to 58 years. In 1997, the minimum age of the unemployment pension was further raised, to 60 years. In 1998, the minimum age for part-time retirement was lowered temporarily for two years from 58 to 56 years. Employers are obliged to provide a part-time job for part-time retired persons. No major changes were made in the maximum duration of unemployment benefits (still no link with length of period worked; still 500 days for earnings-related benefits, followed by an unlimited period of a means-tested basic benefit), in the use of suspension of benefits (still very rare), in work incentives for older workers (still minimal for people aged 55 and above[2]), the standard-age of entitlement to old-age pension (65 years), access to invalidity schemes (still "soft" medical check for disability pensions) and sick-pay schemes (still 60 to 70 per cent of regular earnings after a nine-day waiting period for a maximum period of one year

with, in some collective agreements, the benefit topped up to 100 per cent for the first one or two months). No initiatives were taken regarding medical re-evaluation of disability beneficiaries.

Taxes

The tax wedge on labour has remained basically unchanged. All marginal tax rates of the central government income tax were lowered by 1 percentage point in 1997 and, with the exception of the top rate, by 0.5 percentage point in 1999. However, this was almost completely offset by a rise in the average local income tax rate of 1 percentage point since 1990. Since 1997, the tax burden for low-wage workers has been reduced by a substantial increase in the earnings-related deduction in the local income tax and by increases in the deduction for work-related expenses in the central government income tax.[3] In 1996, to attract foreign high-skilled employees and executives, a special 35 per cent flat tax rate for a maximum of 24 months was introduced for wage earners arriving from abroad. This special tax was extended for another two years in 1998.

Active labour market policy

The focus of active labour market programmes has shifted somewhat from providing subsidised jobs to various forms of labour market training and to job-search assistance. Since 1996, to promote training, participants in voluntary vocational training are receiving an allowance, equal to the basic unemployment allowance. The number of job counsellors at employment offices has increased substantially in 1998 to intensify counselling of the unemployed. Since 1998, in-depth interviews with unemployed are carried out after 5, 11 and 23 months of unemployment,[4] and individual job-seeking plans are set up. Moreover, a target was introduced of contacting every unemployed once per month. However, the employment offices have not yet met this goal. In 1998, the combined job subsidy was introduced, increasing the subsidy already received by private employers for a maximum period of 12 months who hire a long-term unemployed (unemployed for more than 500 workdays).[5] The increased targeting of labour market programmes has increased spending per unemployed person but the total budgetary costs have declined because unemployment has dropped. The public monopoly on job placement services was eliminated in 1994.

Employment protection legislation

Employment protection is comparable with the average European practice (although there are no legal severance payment obligations) and therefore fairly restrictive. In 1996, the notice period for a permanent employee having held a job for less than one year was reduced from two to one months, and the delay for the start of the notice period was reduced. The notice period was not adjusted for employees with a longer job history and increases gradually to six months for those employed by the firm for 15 years or longer. Reasons for dismissal are the work performance of the individual and the financial situation of the enterprise. There are detailed rules concerning unfair dismissals. No changes were made to trial periods (most contracts have a trial period of four months) and to lay-off procedures. The laid-off employee has to be informed ten working days in advance. Temporary and permanent lay-offs are allowed for economic reasons but in case of enter-

prises with more than 30 employees, there is the obligation to negotiate the lay-off conditions with staff representatives. Staff representatives and the labour market services have to be informed if the lay-off affects more than 10 persons. The employer is required to give priority to the affected workers when re-hiring. The regulation on fixed-term contracts was eased somewhat in 1997, as an enterprise in the service sector can conclude a fixed-term contract if demand for its services is not stable. Previously, such contracts were only allowed for temporary and seasonal work and in the case of hiring first-time entrants to the labour market. Restrictions on temporary work agencies and other private placement services were abolished in 1994. However, their role is still relatively limited, accounting for less than 1 per cent of job turnover, compared with 2 to 3 per cent in the other EU countries.

Working-time flexibility

Since 1996, following EU regulations, the maximum amount of overtime is 138 hours within a period of four months, and equal to 414 hours in a year. Finnish legislation became tighter than EU legislation later in 1996, reducing overtime to 330 hours per year (480 hours before 1996), but provisions were introduced permitting more flexible working time arrangements at the local level. In 1996, regulations restricting unusual working hours were eased. In 1997, the Employment Contracts Act was amended to grant part-time employment, and other atypical work, equal status to permanent employment. When an employee wishes to work part-time, the employer must attempt to provide part-time work for a maximum period of 26 weeks. Rotation leave was introduced in 1996. A person who has held a job for more than a year, can be off work for a period between 2 and 12 months and will receive 70 per cent of the earnings-related unemployment benefit. The employer has to agree with the rotation leave and has to hire an unemployed person for the leave period.

Wage formation

The wage formation process did not change during the 1990s. The core of the wage bargaining process has remained the central wage negotiation round, in which a general level of wage increases is agreed upon by the labour unions and the employer's federations, with government heavily involved and offering incentives to influence outcomes. Only in 1993 and 1994, negotiations were decentralised to the industry level, leading to an acceleration in wage inflation as faster wage developments in the export sector spread rapidly to the sheltered sector. Dispersion of earnings has remained among the lowest in the OECD area, although performance-related pay has increased in recent years. In 1997, a third of the white-collar workers and a fifth of the blue-collar workers were receiving some form of results-based pay. No major changes occurred to minimum wages, which have remained contractual and are not legislated, with administrative extension to non-organised workers in those industries where the union coverage exceeds 50 per cent. The only change, which took place in 1993, was the agreed waiving by social partners of the minimum wage provisions in certain cases for new, young entrants to the labour market with reductions in the required pay in the range of 10 to 50 per cent.

Notes

1. During the recession period, around 80 per cent of the unemployed used this option, with the assistance of local government. They were in a subsidised job for exactly 26 weeks to requalify with an unchanged unemployment benefit.
2. If, at the age of 57 years, the 500 working day period of earnings-related benefits is not fully used, the benefit is extended until one reaches the minimum age of the unemployment pension (60 years), while the obligation to look actively for work is abolished. This means that those becoming unemployed at the age of 55 years plus one month can get an earnings-related unemployment benefit till they receive, at the age of 65 years, the earnings-related old-age pension.
3. The earnings-related deduction was introduced in 1991.
4. After the 23rd month, in-depth interviews are held once a year.
5. The government expects to pay the combined subsidy in 1999 for the hiring of 10 000 long-term unemployed.

Annex II

Calendar of main economic events

1998

January

The 1998 Budget is adopted by Parliament. Expenditure amounts to FIM 188.6 billion and revenues to FIM 173.2 billion. Relative to the 1997 Budget, expenditure will decline by 0.9 per cent, while revenue will grow by 6.9 per cent. The net borrowing requirement is projected at FIM 15.4 billion.

February

The Cabinet presents Parliament with a statement on Finnish participation in the euro area. The effects of the different alternatives have been examined and the government believes that the interests of the Finnish nation can be best served by participation in the euro area from the outset.

March

The Cabinet reaches a decision on the budget guidelines for 1999-2002. The overall expenditure ceiling at 1999 prices amounts to FIM 191, 190, 193 and 191 billion for 1999 to 2002, respectively.

The Bank of Finland raises its tender rate from 3.25 per cent to 3.40 per cent. In addition, the interest rate on banks' excess reserves is raised from 1.25 per cent to 1.40 per cent.

April

Parliament votes on the Government's Statement on Finnish participation in the euro area with 135 votes in favour and 61 against.

The first supplementary budget for 1998 is approved by Parliament. The supplementary budget includes a FIM 1.1 billion increase in appropriations, of which about a half is related to the incomes policy agreement. The revenue side records an additional FIM 1.1 billion. Including the supplementary budget, central government budgeted expenditure totals FIM 189.6 billion, with revenues of FIM 174.3 billion and a net borrowing requirement of FIM 15.4 billion.

Parliament abrogates the stamp tax on lending and mortgages to ensure equal treatment of domestic and foreign financial institutions.

May

The new Act on the Bank of Finland is approved by Parliament including all revisions stipulated for the Bank of Finland to become part of the European System of Central Banks once the country joins the euro area. The powers of the Parliamentary Trustees of the Bank of Finland to obtain and relate information to Parliament on monetary policy implementation and other bank activities are increased. The Currency Act is repealed.

Eleven EU countries including Finland decide to introduce a single currency, the euro, at the beginning of 1999.

June

According to Parliament's decision, income brackets for all state income tax scales will be raised by 2 per cent at the beginning of 1999, and all marginal income tax percentages – except the top bracket – will be lowered by 0.5 percentage point.

The merger between two energy companies, IVO and Neste, forms Fortum. Another major merger takes place over the summer, combining partly state-owned company Enso and the Swedish company Stora to create the world's second biggest forest industry company.

July

Parliament agrees that the additional social security contribution rate of 0.45 percentage point collected on all income exceeding FIM 80 000 should be abolished at the beginning of 1999. In addition, a 0.30 percentage point reduction is to be made in the social security contributions collected from pensioners. All other social security contributions collected from both the insured and employers will remain unchanged.

Parliament authorises the privatisation of part of the telecommunication company Sonera.

August

Finland's Stability Programme for the EU is approved by the government. The programme establishes the basic premises and objectives in Finnish economic and fiscal policy, and gives an evaluation of their implementation up to the year 2002. Under the programme Finland will meet the budgetary criteria set out in the Maastricht Treaty. Despite a budgetary surplus for general government, central government is projected to remain in deficit over the entire period reviewed in the programme.

September

The 1999 Budget proposal is presented to Parliament. Expenditure amounts to FIM 186.8 billion and revenues to FIM 187.1 billion; the budget shows a FIM 0.3 billion surplus. Budgetary outlays are projected to fall by over 4 per cent compared to 1998 estimates. The budget proposal is based on guidelines drawn up in March 1998. The net borrowing requirement was estimated in March at FIM 10 billion, but net borrowing is now expected to be unnecessary due to additional tax revenues from the better than expected economic developments, spending cuts, and the use of the cumulative surplus in the budget year's final accounts. Owing to extra savings and faster than expected economic growth, the final expenditure proposal is almost FIM 4 billion less than in the original guidelines.

October

Government presents the second supplementary budget for 1998 to Parliament.

November

The state shareholding in telecommunication company Sonera is partly sold.

The partly state-owned company Valmet Ltd. and private-owned Rauma, both in the engineering and machinery industries, are merged. The turnover of the two companies amounted to over FIM 23 billion in 1997 and they employed over 23 000 persons.

December

The state-owned company Fortum (the energy company formed by the summer 1998 merger of IVO and Neste) is partly privatised. Due to the partial privatisations of Sonera and Fortum, the privatisation proceeds reach a record high of over FIM 10 billion in 1998.

Following a co-ordinated action of the euro area central banks, the Bank of Finland lowers the tender rate from 3.4 per cent to 3.0 per cent. In addition, it lowers the interest rate on excess reserves from 1.4 per cent to 1.0 per cent and the base rate from 4.0 per cent to 3.5 per cent.

Parliament approves the second supplementary budget for 1998. The upward revision in revenues amounts to FIM 17.5 billion, bringing expected total revenues to FIM 192 billion. Tax revenues have been boosted substantially by improved employment conditions, the positive trend in company profitability, and continued strong growth in private consumption. Income from asset sales exceeded the previous projections by a wide margin, owing *inter alia* to the privatisation of the telecommunication company Sonera and energy company Fortum. The positive trend in companies' financial results is also reflected in the central government's dividend receipts. On the other hand, projected revenues from the stamp tax and the mortgage tax are slightly revised downwards. The supplementary budget includes additional net expenditures of FIM 7.5 billion, bringing the expected total expenditures to FIM 197 billion. The central government net borrowing requirement is lowered by FIM 10.5 billion to total FIM 5 billion for 1998.

1999

January

Finland becomes a part of the euro area as of 1 January 1999. State-owned markka-denominated serial bonds and treasury bills are denominated in euro on 1 January 1999. The base rate calculated and quoted by the Bank of Finland is replaced by the new base rates confirmed by the Ministry of Finance twice a year for the next 6 months. The base rate is based on the 12-month market interest rate average for the three months preceding the confirmation. The Ministry determines that Euribor rates will replace the existing Bank of Finland Helibor rates.

The Budget for 1999 is adopted. Expenditure amounts to FIM 188.5 billion which is somewhat lower than in 1998, while revenues are projected to be 8.8 per cent higher. The income brackets in all tax scales for 1999 are raised by 2 per cent and all marginal tax rates except the highest one are lowered by 0.5 percentage point.

February

Government adopts expenditure guidelines for 2000 amounting to a total of FIM 191.5 billion (at 1999 prices).

March

In the Parliamentary elections of 21 March, the Social Democratic Party (SDP) of Prime Minister Paavo Lipponen remains the biggest party despite a substantial loss (23 per cent of votes in 1999 compared to 28 per cent in 1995). The other main party in government, the Conservatives has its best result ever (21 per cent of votes in 1999 compared to 18 per cent in 1995). The main opposition party, the Centre Party, also made gains (22 per cent of votes in 1999 compared to 20 per cent in 1995).

A government decision on Finland's National Forest Programme (NFP) is adopted. NFP 2010 covers Finland's forest policy objectives, to be implemented in collaboration with forest organisations.

April

The "rainbow" coalition government stays on with Paavo Lipponen as Prime Minister and Sauli Niinistö as Minister of Finance. Both parties have six ministerial posts in the new government (compared to seven SDP and five Conservative ministers in the previous government). The Left Alliance has two ministerial posts, while the Greens and the Swedish People's Party have one and a half each. Mr Kalevi Hemilä continues as Minister of Agriculture and Forestry, without party affiliation.

A new Insurance Supervision Authority is set up. The function of the new authority is to supervise and audit insurance and pension institutions as well as other agents operating in the insurance field.

May

Government agrees on the size of expenditure during the incoming four year term of parliament with a programme setting spending in real terms at the level of the 1999 Budget of FIM 191 billion expressed in the money of the year 2000. However, it was decided in February 1999 that expenditure next year would be FIM 2 billion higher at FIM 193 billion. During the previous government, expenditure ceilings at 1999 prices were set at FIM 193, 191 and 190 billion for the years 2001 to 2003, respectively.

OECD PUBLICATIONS, 2, rue André-Pascal, 75775 PARIS CEDEX 16
PRINTED IN FRANCE
(10 1999 31 1 P) ISBN 92-64-16989-X – No. 50755 1999
ISSN 0376-6438